SuperResearcher

The Map Is Wrong

Disputed borders, phantom islands, and the strangest lines on earth — a cartographer's tour of geopolitical weirdness

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Every map is a claim, and this report is about what happens when the claims fail. It opens with places that do not exist: islands copied from chart to database for more than a century until a research vessel sailed through one in 2012, and how such phantoms are born — myth, hope, strategy, deliberate traps for plagiarists, honest mistakes no one checks. Fictional land turns out to have real prices, because modern law converts shores into baselines and baselines into territorial seas, exclusive economic zones and continental shelves, so the accuracy of the geography decides who may fish, drill and police enormous areas of ocean. The middle of the report tests that machinery against the world's hardest coastlines. Equidistance lines are constructed and then corrected for equity; historic titles collide with modern rules in the South China Sea, where an arbitral tribunal extinguished a famous dashed claim that its maker continues to assert; straits regimes sort passing ships by cargo and propulsion; and enclaves trapped inside other countries' territory were finally dissolved by a negotiated exchange of land and people. Courts recur throughout — a world court working through a decades-long African boundary case, judges dissenting over whether precision finer than a millimetre can mean anything on the ground — alongside the disputes no court has settled: Kashmir's line recognised by everyone and accepted by no one it encloses, a Western Sahara referendum deferred for decades, and hundreds of checkpoints turning one small territory into a lattice of permissions. The final movement turns from the lines to the instruments that draw them. No flat map can preserve area, shape and distance at once, so projections are ranked by what they sacrifice, and the honest ones declare their invariants outright. Plotting a boundary on a chart is shown to be a craft of tolerances — distances borrowed from the latitude scale, bearings good to a quarter of a degree — while national surveys industrialised first the quadrangle and then the database. The report closes where mapping now lives: in satellite services, digital charts and global gravity models, still negotiating a single world height datum. Its conclusion is practical rather than cynical: treat every map as an argument with an error budget, and check what the argument costs when it is wrong.

Phantom Islands and Places That Are Not There

Obituary: Sandy Island (1876–2012)

Sandy Island died as it had lived: on a map, in public, and to general astonishment. In October 2012 the research vessel Southern Surveyor, working the eastern Coral Sea on the 25-day ECOSAT tectonics voyage, transited toward a feature that appeared in numerous scientific data sets and in Google Earth, labelled “Sandy Island” — a landmass some 25 kilometres long and 5 kilometres wide, between the Chesterfield Islands and Nereus Reef. It was absent, however, from the hydrographic charts the ship’s master used for navigation, and when the crew looked out over the expanse of open ocean where the island was supposed to be, they confirmed what they had suspected: the data sets showing the island were wrong [1] [1]. The “undiscovery” became the most-read story of 2012 on the Sydney Morning Herald website, and the mystery — how well do we really know our own planet? — drew global attention from traditional and social media alike [1].

The deceased was born in 1876, reportedly sighted by the whaling ship Velocity, and first plotted on a British Admiralty chart of 1908: an island roughly 500 kilometres northwest of New Caledonia, at around 19°S, 160°E, whose shape, size, and location match what later appeared in scientific databases [1]. Its chart life was always doubtful — hydrographic charts eventually annotated it ED, “existence doubtful,” the official abbreviation for a reported island that later searches failed to find [1]. The professional verdict came early: seafloor mapping by the Australian Hydrographic Service (AHS) found minimum depths of 1,488 to 2,353 metres over and around the site, and on the basis of no island and no shallow reef, the French Hydrographic Service removed Sandy Island from official charts in 1974 and the AHS followed in 1985 [1]. As far as the hydrographic world was concerned, Sandy Island was a phantom, and word was passed to other national services [1].

Yet it refused burial, because the scientific community's maps were no longer drawn from charts but compiled from databases. The vector of resurrection was the World Vector Shoreline (WVS), a global coastline data set built by the U.S. National Imagery and Mapping Agency (now the National Geospatial-Intelligence Agency) during the conversion from hard-copy charts to digital formats, carrying both human digitizing errors and errors inherited from the original maps — worst in the least-explored parts of the planet [1]. From WVS the island propagated into the widely used GSHHS shoreline database shipped with Generic Mapping Tools, and beyond: the GEBCO bathymetry and the Smith and Sandwell satellite-derived topography reported an elevation of 1 metre above sea level over Sandy Island, ETOPO reported 2 metres, and some satellite products showed gaps in sea-surface temperature exactly where the island sat [1]. None of this was independent evidence: a land mask — a preprocessing layer distinguishing land from water so different algorithms can be applied to each — had been applied to those data sets, so the island's appearances in bathymetry and imagery were the same spurious digitized geometry, echoing from product to product [1]. Google Earth rendered the sum of these echoes as an elongated black polygon, because the satellite imagery it used over land contained no land there to show [1].

Against the echoes, the physical evidence was unanimous. The Southern Surveyor ran 5 kilometres south of the purported island, directly over the bathymetric high in the legal charts, and its swath bathymetry (Simrad EM-300) found a minimum depth of about 1,300 metres — consistent with the hydrographic surveys and discrepant with the global data sets by up to 2,500 metres along the track [1]. Landsat and MODIS imagery indicated deep or at best shallow water; satellite chlorophyll showed none of the biological signature seen over real nearby reefs such as Bampton and Nereus; the absence of active volcanism nearby rules out an emergent island that subsided; and the regional gravity high of 80 milligals sits west of Sandy Island, not over it [1]. Within a week of the story breaking, crowdsourced analysis of historical maps and data sets had sunk the island once and for all; it was removed from GSHHS (version 2.2.2) and the latest satellite topography, slated for removal from GEBCO, and deleted — polygon and name — from Google Earth [1].

What, then, did the Velocity see in 1876? Erosion cannot explain it: the seafloor lies more than 1,400 metres down. The leading candidate is a pumice raft — floating volcanic rock from a submarine eruption. Pumice from the 2001–2002 Tofua Arc eruption off Tonga travelled more than 3,000 kilometres westward to Australia within a year, passing within 20 kilometres of Sandy Island's position about 200 days after the eruption, along a current-driven "pumice raft superhighway" funnelling between Fiji and New Caledonia [1]. Scale is no objection: the July 2012 Havre eruption produced a raft estimated at 463 kilometres long and 55 kilometres wide, so a 25-by-5 kilometre "island" of pumice is entirely reasonable [1]. The obituary's moral, as its authors draw it, is about provenance: global data sets are invaluable but inherit the errors of their raw inputs; interdependent products give a single error many apparent confirmations; and satellite compilations still require ships to go and look [1]. Sandy Island's 136 years were lived entirely inside the archive — which is why it could be killed in a week once someone checked the ocean instead.

How a false island is born

Sandy Island was a data error, but the species is ancient. A phantom island, in Christensen's definition, is a purported island that was included on maps for a period of time and later found not to exist; most appeared on maps from the 1200s to the early 1800s, and some survived into the present century [2]. Surveying the genus, Christensen sorts phantom islands into six birth categories — myth and legend, mystical lands of hope and treasure, strategic islands, cartographers' copyright islands, and islands made by weather and nature among them — and the taxonomy is the real answer to how a false island is born: a report enters the chart, and the chart's copying chain does the rest, because for centuries no falsification pass ever ran [2] [2].

The oldest births are legends with coordinates. Antillia, the "Fore-Island" at 33°N, 54°W, grows from the Iberian legend of 711, when seven Catholic bishops fled the Moorish conquest of Spain with their treasures to an Isle of Seven Cities; revived around 1424 by the Seven Cities of Gold story, it drew searchers — Columbus among the disappointed — before vanishing from maps by the end of the seventeenth century [2]. Hy-Brasil (51°N, 17°W), on maps from 1325, was supposedly sighted from Ireland's west coast every seven years, complete with black rabbits and a druid's castle; sailors searched into the 1800s, British cartographers were still recognizing it in the 1870s and 1880s, and it has since been identified with Porcupine Bank, 200 kilometres west of Ireland — yet, as Freitag notes, neither Hy-Brasil nor St. Brendan's Island appears in the oldest Irish myths or earliest Irish literature at all, so the "ancient Irish belief" is itself a later accretion pasted onto the chart [2]. St. Brendan's Island (28°N, 23°W) descends from the monk Brendan's landfall with 14 monks in 512, thereafter hidden behind mist 200 kilometres west of the Canaries; Atlantis, from Solon and Plato around 600 BC, is the limiting case — the most famous phantom of all, searched for worldwide, with no evidence ever found, including underwater [2].

The second birth canal is hope. The Island of California was born in fiction — Garci Rodríguez de Montalvo's romance Las Sergas de Esplandián (1510) placed an island near the Terrestrial Paradise peopled by black women living in the manner of Amazons amid abundant resources — and the fiction hardened into geography: Vizcaíno acknowledged the island in 1602, and the Carmelite friar Antonio de la Ascensión, who had sailed Vizcaíno's routes, confirmed it, separated from the mainland by a "Mediterranean Sea of California" [2]. Killing it took a century of overland falsification: the Jesuit cartographer Kino, who had accepted California's insularity while studying in Europe, doubted it in Mexico and ran expeditions from Sonora toward the Colorado delta in 1698–1706; fellow Jesuits Ugarte (1721), Konščak (1746) and Linck (1766) pressed the point; Anza's journeys between Sonora and Alta California in 1774–1776 settled it, and Ferdinand VI had already declared in 1747 that California is not an island [2].

The third is strategy. A charted island has political mass — under the modern law of the sea it can anchor 200-nautical-mile exclusive economic rights over oil- and gold-bearing seabed — and states have drawn phantoms deliberately: at Louis XVI's behest, French maps from 1783 scattered Isle Phelipeaux, Isle Royale, Ile Maurepas and Ile Pontchartrain across Lake Superior, and the 1783 Treaty of Paris wrote the fiction into law, running the US–Canada boundary "northward of the Isles Royal and Phelipeaux to the Long Lake" [2]. When surveyors tried to stake the boundary in the 1820s, Isle Phelipeaux did not exist and no one could identify the "Long Lake," so the line followed the Pigeon River instead — a treaty boundary quietly rerouted around an island that was never there [2].

The fourth and fifth births are professional. With no copyright regime in the fifteenth to seventeenth centuries and map theft routine, cartographers planted phantom islands as watermarks — traps only the originator could explain: the island of Satananez, depicted by the Venetian Zuane Pizzigano in 1424, disappears after 1436, resurfaces in 1462 under Benincasa as "Salvaga," and wanders under a string of name-variants (Salvagio, Salvaje, Salirosa) that told every map user exactly whose work had been copied — "a watermark does not need a legend" [2]. Mercator's Rupes Nigra, a highly magnetic black rock placed at the North Pole itself, persisted until science separated the geographic from the magnetic pole, whereupon the island evaporated [2]. And outright fraud occurs: Robert Peary reported sighting Crocker Land from Ellesmere Island in 1906 and named it for his backer George Crocker, while his own diary recorded that no land was visible — an invention apparently meant to secure funding for his 1909 expedition (the money had gone to rebuilding San Francisco after the 1906 earthquake), which nonetheless launched a real Crocker Land Expedition in 1913 to investigate land that its inventor knew was not there [2].

The sixth birth is the sky itself. The Fata Morgana — a complex mirage, commonest in polar regions — projects detailed false coastlines that dissolve into heat haze as you approach; Fata Morgana Land, sighted between Svalbard and Greenland by the Danish explorers Koch and Bertelsen in 1907, was hunted for 30 years before being recognized as a mirage, probably of Tobias Island, whose own position was fixed accurately only in 1993 [2]. Sannikov Land, reported by Iakov Sannikov in 1809–1810 north of Kotelny Island, drew the Zarya expedition of 1901 (lost without describing it) and a 1937 expedition that found nothing; it is now read either as a real island of fossil ice and permafrost eroded into a submerged shoal, or as Sannikov's mirage of Bennett Island's coast [2]. Buss Island, recorded in September 1578 by sailors of Frobisher's third expedition aboard the Emmanuel of Bridgewater, shrank on successive charts, was renamed the Sunken Land of Buss by 1745, and is now attributed to a dead-reckoning error compounded by optical effects near Greenland; Dougherty Island (sighted 1841, confirmed by Keates and Stannard in the 1860s, still on international maps in the 1930s) was fog banks or icebergs all along [2]. The pattern across all six births is uniform: the island is cheap to report and expensive to kill, and it lives exactly as long as the interval between those two events — an interval the next sections show modern law and modern surveying redrawing, but never closing.

Mystical islands: land of hope and treasures

Christensen's third category of phantom islands is defined by a motive rather than an error: hope. Even in modern times, when the earth has been almost 100% geographically described, there lives in man a hope that something unexplored remains — something that gives hope of more wealth, or simply of areas not yet taken [2]. An island of hope is cheap to keep: it sits beyond the reliably surveyed routes, it promises gold, souls, harbours or a shorter passage, and every year it is not found can be charged to weather, currents or the searcher's incompetence. The category's two type specimens — the Island of California, born in a 1510 romance and killed only by a century of overland expeditions, and Sandy Island, this paper's opening obituary — were treated in the taxonomy section; the category's wider membership shows how many distinct mechanisms hope can power [2].

Frisland (63°N, 20°W) shows hope compounded by cartographic vanity. Depicted as a roughly rectangular island with promontories on its west coast, near Iceland, Frisland's leading explanation is that it is Iceland: when Nicolo

Zeno assembled his famous map collection in 1558, he did not simply copy the maps of his time — he moved Iceland further south, a displacement Christensen calls also a kind of watermark, since Zeno was the first and only cartographer to place Iceland at exactly that spot in the Atlantic [2]. The duplicate then lived a life of its own: just under 250 years passed before Frisland disappeared from maps, by which time it had wandered as far as the southeast coast of Greenland [2]. A real island, cloned by a copyist's flourish, became a phantom with its own geography.

Pepys Island (47°S, 58°W) shows hope working on a single sighting. In January 1684 the British journalist William Ambrose Cowley sighted an island some 400 kilometres north of the Falkland Islands; later in the 1680s it appeared on a map by William Hacke, known as the pirate's cartographer [2]. Nobody ever saw it again. By about 1840 Pepys Island was forgotten, and the modern scientific view is that Cowley's report was a misidentified account of the Falkland Islands themselves; tellingly, the island appears on different maps misspelled in some places and nameless in others — which Christensen reads as an obvious case of cartographers marking the island as their own [2]. One ambiguous sighting, one prestigious map, and a century and a half of copied existence.

Mayda — the Isle of Mam (52°N, 33°W), midway between Ireland and Canada — is the category's purest ghost: nobody, on Christensen's account, has ever sighted it at all. It simply appeared, under the name Brazir, on the Venetian brothers Domenico and Francesco Pizzigano's map of 1367, and several cartographers depicted it in the following centuries, under different names and different spellings [2]. After 1650 Mayda disappeared from all documentation; its plotted location had ranged from Brittany in the west to Labrador in the east, a wander that made it untrustworthy — and, Christensen concludes, the possibility that it was a watermark is high [2]. Hope here needed no sighting, no legend, no coastline: only the blank Atlantic and the professional incentive not to leave it blank.

The category's longest shadow falls across the Gulf of Mexico, where hope acquired a documentary pedigree early. Bermeja — whose full case occupies the next section — enters the written record with Alonso de Santa Cruz's El Yucatán e Islas Adyacentes, a list of the region's islands published in Madrid in 1539, and receives a precise location in Alonso de Chaves's Espejo de navegantes (Seville, around 1540), which records that from a distance the small island looks blondish or reddish — the observation behind its name [2]. Even its death was reported hopefully and vaguely: according to the French-Mexican cartographer Michel Antochiw Kolpa, several British maps from 1844 onward reported the island as having sunk some 60 fathoms — 360 feet, 110 metres — below the surface, a submergence that has never been proven [2]. An island that sinks is an island that cannot be falsified by an empty horizon, which is why Bermeja survived to become, four centuries later, a pawn in an oil dispute. The mystical category thus grades imperceptibly into the strategic one: hope draws the island, but interest decides how long it stays — and the next section follows Bermeja into the treaty era, where a phantom's coordinates could be worth an oil field.

Bermeja and the legendary islands that never were

Christensen's survey divides the births of phantom islands into two great causes: error and imagination. Error covers the honest failures — sailors mistaking mirages, low clouds and rock formations for land, deceived further by imprecise navigational and charting equipment; imagination covers the deliberate constructs, islands invented by cartographers guarding against copyright theft or by creative writers whose adventure tales claimed new and exciting lands [2]. Modernization has been steadily debunking both kinds, and the survey closes its framing with a lament for the result: comprehensive databases, free maps on every phone and satellites overhead leave a world that feels "flatter," "all-discovered," no longer extraordinary [2]. Bermeja is the standing rebuke to that lament — a phantom whose importance grew in the age of satellites and conventions, because what it could move was no longer a coastline but an oil field.

Bermeja (or Vermeja), at 22°N, 91°W in the Gulf of Mexico north of the Yucatán Peninsula, is presented in the survey as the most famous of the strategic islands; its name means "red" in Spanish, supposedly the colour of its rocks [2]. Its sixteenth-century documentary birth was covered in the preceding section; its afterlife is the subject here. One theory of that birth is itself strategic: Julio Zamora, president of the Mexican Geographical Society, theorized that Bermeja was an imaginary island purposely charted incorrectly in sixteenth- and seventeenth-century maps to dissuade enemies from traversing waters that were routes of navigational interest [2]. On that reading the phantom was born as a decoy — disinformation with coordinates.

What resurrected it was law. The 1982 UN Convention on the Law of the Sea left legal loopholes in the Gulf of Mexico — the so-called "Donut Holes," pockets of international water between the two countries' exclusive economic zones — whose ownership might have mattered little had the Gulf not been rich in oil reserves [2]. In the bid to determine control of those waters, ancient maps were brought out, and Bermeja was discovered to be the northern-

most Mexican land toward the area of contention: if the island existed, Mexico's maritime projection from it could claim the Donut Holes; Bermeja was therefore Mexico's best chance to claim ownership of the international waters [2]. A reddish speck in a 1539 island list had become a sovereignty instrument with a quantifiable value in hydrocarbons.

Then the state did what four centuries of sailors had not: it went and looked, twice. In 1997 Mexico deployed a vessel to prove the existence of Bermeja, but no sign of any island was found in the charted location [2]. In 2000, notwithstanding, Mexico signed the treaty it had been negotiating with the United States — settling the Donut Holes without its phantom — and in 2009 a team of expert researchers from the National Autonomous University of Mexico (UNAM) searched again, by both sea and air, finding no trace; analysis of the ocean floor led the team to conclude that there had not been an island at the given coordinates [2]. The falsification is as complete as Sandy Island's — and it changed nothing about the island's political career, because the treaty was already signed.

Falsification, moreover, failed to kill the island in the public mind; it merely changed its genre. Bermeja's disappearance from the charted location around the year 2000 fuelled a conspiracy theory that the CIA had blasted the island off the surface of the ocean, so that the United States would retain rights to the oil-rich areas of the Gulf — and resolutions were passed in the Mexican parliament to examine the blasting theory in more detail [2]. The episode is the taxonomy's categories collapsing into one another: an island possibly born as deliberate disinformation (Zamora's theory), kept alive by hope, promoted by strategy, and, once disproven, preserved by suspicion — because an island removed by a secret explosion is exactly as unfalsifiable as an island reported sunk 60 fathoms down. The legendary islands that never were thus end their history with a lesson the rest of this paper keeps re-learning at sea: a map is not only a record of coasts but an instrument of claim, and a phantom's lifespan is set not by hydrography but by the value of what its coordinates would command. From phantom land, the paper turns to the real rulebook that prices such claims — the Law of the Sea Convention, and the baselines from which all maritime zones are measured.

The Rulebook: The Law of the Sea Convention

Baselines: normal, straight, and combined methods

Every maritime zone a state claims — territorial sea, exclusive economic zone, continental shelf — is measured from a baseline, and UNCLOS builds that foundation in three forms. The **normal baseline** is the default: the low-water line along the coast, as marked on large-scale charts officially recognized by the coastal State (Article 5) [3]. Its full definition is composite, assembled from several articles: the low-water line along the mainland shore and around islands, the outer limits of permanent harbour works, the low-water line of certain low-tide elevations, and the seaward low-water line of atoll reefs and fringing reefs around islands (Articles 5, 6, 11, and 13) [3]. **Straight baselines** replace the waterline with straight segments joining selected points on the coast, under the criteria of Article 7, and they change the legal character of what lies behind them: waters landward of a straight-baseline system are internal waters, not territorial sea [3]. **Straight archipelagic baselines** apply the same geometric idea to an island group, joining the outermost islands in a succession of straight lines under Article 47 to define the periphery of the archipelago [3]. Real coastlines are served by combinations — normal-baseline reaches interrupted by straight closing lines where the Convention permits them: across the mouth of a river (Article 9: a straight line between points on the low-water line of its banks), across the mouth of a juridical or historical bay, as part of a straight-baseline system, or as archipelagic baselines [3].

Whatever the method, the result must be made legible to the world. UNCLOS requires that baselines — like the outer limits and delimitation lines of EEZs and continental shelves — be shown on charts of a scale adequate for ascertaining their position, or replaced by lists of geographical coordinates specifying the geodetic datum; the coastal State must give them due publicity and deposit a copy of each chart or list with the Secretary-General of the United Nations (Articles 16, 47, 75, 84) [3]. The chart, in other words, is not a picture of the legal order; it is an instrument of it. The TALOS Manual, the standard technical guide to these provisions, notes that the nautical chart — a special-purpose map designed for safe navigation, showing depths, seabed character, coastal configuration, dangers, and aids to navigation — is in almost all countries the only chart type that comes close to meeting the needs of the legal bodies and cartographers who carry out delimitation, even though its fitness for that purpose is incidental to its design [3].

And here the manual issues the warning on which this whole report turns. Hydrographic surveys are time-consuming and executed area by area over years, so the data underlying a single chart are of mixed vintage, and for some areas may be dated and below modern standards: “information shown on a chart can be different from the reality” [3]. The zone between the high-water line and the 5-metre isobath — precisely the strip from which the normal baseline is read — is in many parts of the world based on old data; on older charts the symbology may not clearly distinguish the low-water line from shallow-water contours, and the depiction of an isolated shoal or reef may indicate presence rather than specify nature, leaving it genuinely ambiguous whether a feature is a low-tide elevation (which can anchor a baseline) or something else — an ambiguity the manual resolves, in ascending order of authority, through Sailing Directions, modern land maps, aerial photography, satellite imagery, and, where serious doubt persists, a field survey [3]. Charts built on inadequate geodetic control can carry both overall positional error and distorted relative positions between features, curable only by resurvey; where negotiations cannot wait, the best supplementary sources — land maps, systematic aerial photography, high-resolution satellite imagery — may outperform the chart itself [3]. Layered over all of this are the cartographic fundamentals any delimitation must respect: projection (and its distortions), scale, horizontal datum, and vertical datum [3].

Read back against the phantom-island chapters, the logic closes. A baseline system inherits every error in the chart it is drawn from, and the Convention’s own technical manual concedes that charts are strata of surveys of different ages, sometimes wrong about the very line the law treats as foundational. Sandy Island is the limiting case — a feature that existed only in data — but the mundane version is everywhere: a low-water line last surveyed a century ago, a reef symbol copied forward for generations. The methods of this section are the law’s answer to geography; their weakness is that geography reaches the law only through the chart.

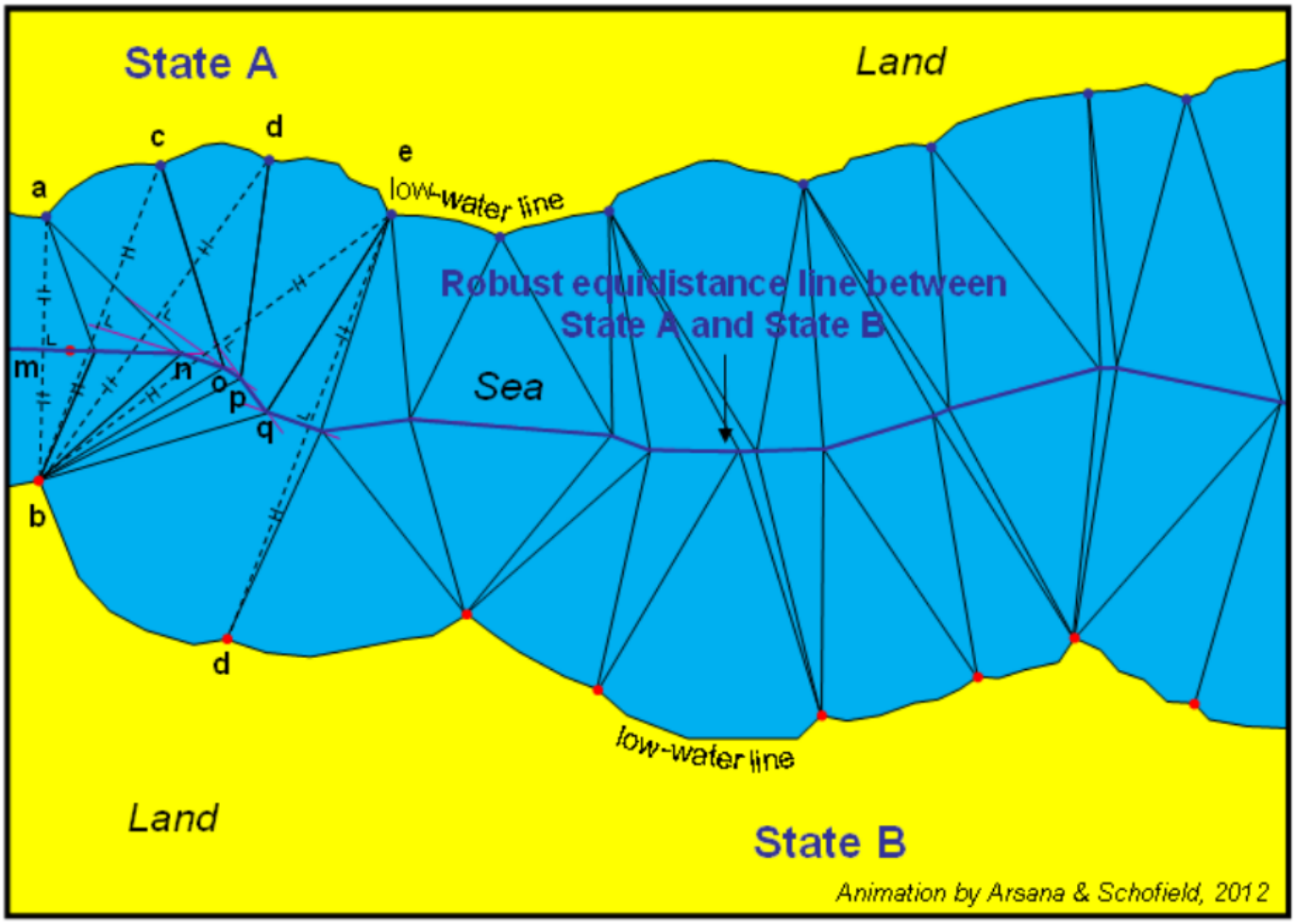


Exhibit. Constructing a robust equidistance line between two opposite coasts: basepoints on each low-water line generate the median line between the two States. Source: S0012.

The territorial sea and contiguous zone

The territorial sea is sovereignty with a hole in it. The coastal State is sovereign over the belt — its waters, air space, bed and subsoil — yet “ships of all States, whether coastal or land-locked, enjoy the right of innocent passage

through the territorial sea" [4]. Passage itself is tightly defined: navigation through the territorial sea either traversing it without entering internal waters or calling at a roadstead or port outside them, or proceeding to or from internal waters or such a call; it must be continuous and expeditious, and stopping or anchoring counts only where incidental to ordinary navigation or rendered necessary by force majeure, distress, or the rendering of assistance to persons, ships or aircraft in danger [4]. The zone's breadth — up to 12 nautical miles from the baselines — was set out in the previous section; this section's subject is what the coastal State may and may not do inside it.

The criminal law is deliberately blunted. The coastal State "should not" exercise criminal jurisdiction on board a foreign ship passing through the territorial sea to arrest any person or investigate a crime committed on board during passage, save in four cases: where the crime's consequences extend to the coastal State; where it is of a kind to disturb the peace of the country or the good order of the territorial sea; where the master, or a diplomatic agent or consular officer of the flag State, has requested local assistance; or where measures are necessary to suppress illicit traffic in narcotic drugs or psychotropic substances [4]. Where jurisdiction does lie — or where the ship is passing after leaving internal waters, a case the restraint does not protect — the coastal State must, if the master requests, notify the flag State's diplomatic or consular officer before taking steps, facilitate contact with the crew, and have "due regard to the interests of navigation" in deciding whether and how to arrest [4]. And a ship merely transiting from a foreign port, without entering internal waters, is immune from enforcement steps for crimes committed before it entered the territorial sea, Part XII (environment) and Part V (EEZ) violations excepted [4].

Civil jurisdiction is blunter still. The coastal State should not stop or divert a passing foreign ship to exercise civil jurisdiction over a person on board, and it may not levy execution against or arrest the ship in civil proceedings except for obligations or liabilities the ship itself assumed or incurred in the course or for the purpose of its voyage through the coastal State's waters; a ship lying in the territorial sea, or passing through it after leaving internal waters, enjoys no such shelter [4]. Warships occupy their own corner of the regime: the Convention defines them precisely — belonging to a State's armed forces, bearing distinguishing external marks of nationality, commanded by a duly commissioned officer on the service list, and manned by a crew under regular armed-forces discipline — and the coastal State's remedy against a warship that defies its passage laws and ignores a compliance request is not arrest but expulsion: it "may require it to leave the territorial sea immediately," while the flag State bears international responsibility for any loss or damage the non-compliance causes [4]. Sovereign immunity, in other words, is preserved even inside someone else's sovereignty; the balance is struck by responsibility rather than enforcement.

Beyond the territorial sea lies the Convention's first buffer: the contiguous zone. In a zone contiguous to its territorial sea, the coastal State may exercise "the control necessary to" do exactly two things — prevent infringement of its customs, fiscal, immigration or sanitary laws and regulations within its territory or territorial sea, and punish infringement of those same laws committed within its territory or territorial sea [4]. The drafting is surgical and worth dwelling on, because it is the template for every functional zone that follows. The contiguous zone confers no sovereignty and no general jurisdiction: the four enumerated subject matters are exhaustive; the preventive power reaches seaward only in service of laws that bite on land or in the territorial sea; and the punitive power attaches to violations already committed inside the line. The zone "may not extend beyond 24 nautical miles from the baselines" — exactly doubling the maximum territorial sea, a policeman's apron of 12 further miles [4]. Together the two zones display the Convention's basic grammar: full sovereignty qualified by passage rights in the first 12 miles, then a narrow, purpose-built control jurisdiction for 12 more — after which, as the next section shows, the grammar changes again, from sovereignty to "sovereign rights," in the exclusive economic zone.

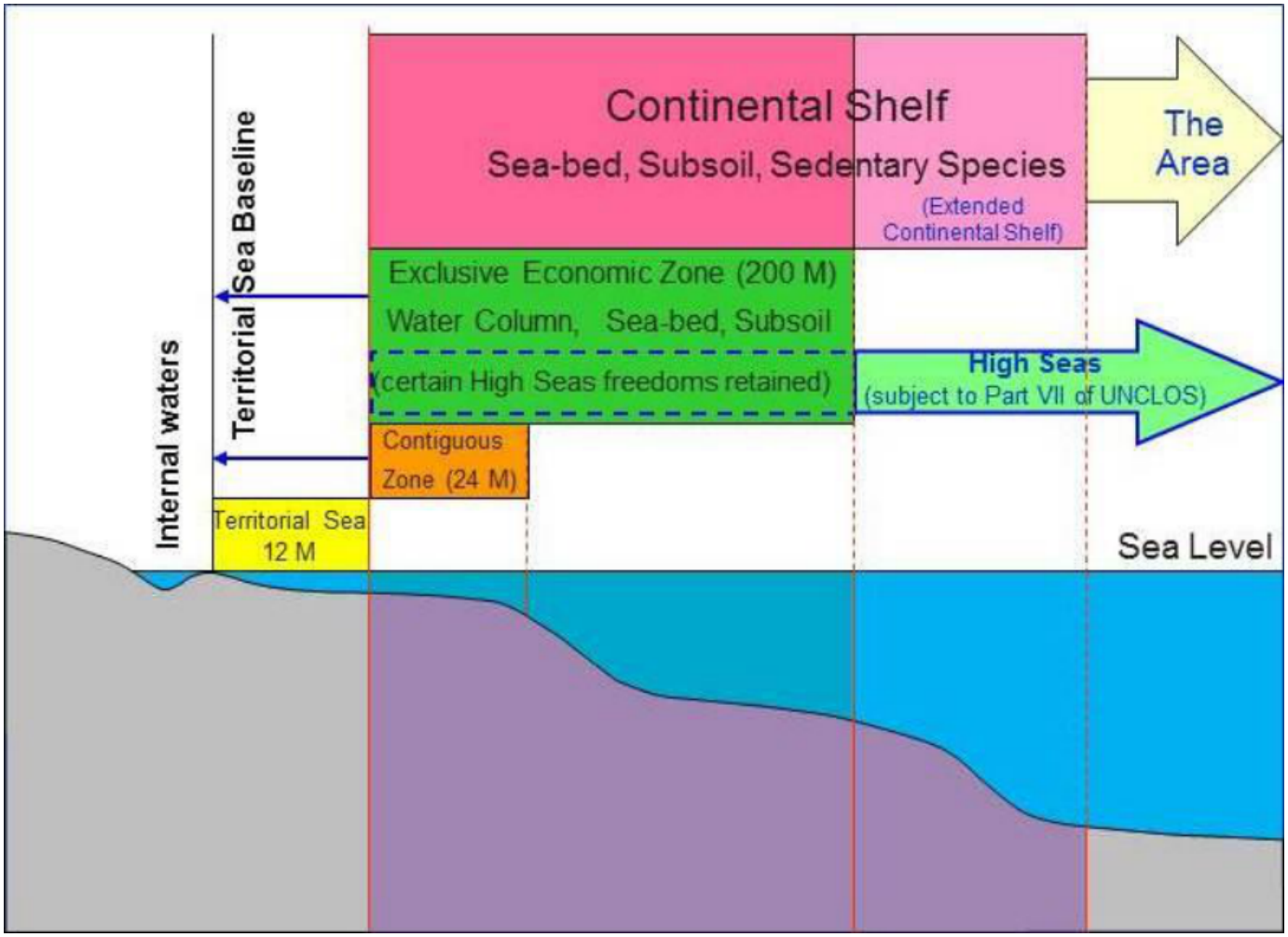


Exhibit. The maritime zones of a coastal State, from internal waters and the 12-mile territorial sea through the 200-mile exclusive economic zone to the continental shelf and the high seas. Source: S0012.

The exclusive economic zone

The exclusive economic zone is the Convention's most original invention: a belt that belongs to neither category the law previously knew. Article 55 defines it as "an area beyond and adjacent to the territorial sea, subject to the specific legal regime established in this Part, under which the rights and jurisdiction of the coastal State and the rights and freedoms of other States are governed by the relevant provisions of this Convention" [4]. Article 57 fixes its maximum extent: it "shall not extend beyond 200 nautical miles from the baselines from which the breadth of the territorial sea is measured" [4]. Two hundred nautical miles is the number that made Bermeja worth a parliamentary inquiry: a single qualifying island projects this zone across an ocean-sized disc of water, seabed and subsoil, which is why the existence or non-existence of one small island in the Gulf could price an oil province.

What the coastal State holds in the zone is deliberately not sovereignty. Article 56 grants, first, "sovereign rights for the purpose of exploring and exploiting, conserving and managing the natural resources, whether living or non-living, of the waters superjacent to the seabed and of the seabed and its subsoil" — and, notably, "with regard to other activities for the economic exploitation and exploration of the zone, such as the production of energy from the water, currents and winds" [4]. The phrase "sovereign rights," purpose-limited and resource-bound, is the Convention's central drafting distinction: the coastal State owns the fish, the oil and the wind farm, but not the water. Second, it has jurisdiction as provided in the Convention over, among the heads listed, "the establishment and use of artificial islands, installations and structures" — with seabed and subsoil rights exercised in accordance with Part VI, the continental shelf regime — and it must exercise everything "with due regard to the rights and duties of other States" [4].

What other States hold is equally deliberate. Under Article 58, in the EEZ "all States, whether coastal or land-locked, enjoy… the freedoms referred to in article 87 of navigation and overflight and of the laying of submarine cables and pipelines, and other internationally lawful uses of the sea related to these freedoms," with the high-seas rules of Articles 88 to 115 applying so far as they are not incompatible with Part V; in return, those States must have due regard

to the coastal State's rights and comply with its laws adopted in accordance with the Convention [4]. The EEZ is therefore a negotiated overlap, not a gradient: full navigational freedom of the high seas, minus the fish; full resource rights for the coastal State, minus the power to close the sea. Most of the disputes in this paper's later chapters are arguments about where, inside that overlap, a given activity — a warship's survey, a coast guard's boarding, a wind farm, a cable — actually sits.

Marine scientific research shows the balance under load. Coastal States have "the right to regulate, authorize and conduct marine scientific research in their exclusive economic zone and on their continental shelf," and research there "shall be conducted with the consent of the coastal State" [4]. But the consent regime is structured against caprice: in normal circumstances coastal States shall grant consent for projects carried out exclusively for peaceful purposes to increase scientific knowledge of the marine environment for the benefit of all mankind, and must maintain rules ensuring consent "will not be delayed or denied unreasonably" — with normal circumstances expressly capable of existing despite the absence of diplomatic relations [4]. The discretionary refusals are enumerated: a project of direct significance for the exploration or exploitation of natural resources; one involving drilling into the continental shelf, explosives, or harmful substances; one involving artificial islands, installations or structures; or one whose notified particulars are inaccurate or whose sponsor has outstanding obligations from a prior project [4]. Even that discretion is fenced on the outer continental shelf: beyond 200 nautical miles, consent may not be withheld on resource-significance grounds except in specific areas the coastal State has publicly designated as ones where exploitation or detailed exploratory operations are occurring or will occur within a reasonable period [4]. The procedural duties run the other way too — a researching State must supply, not less than six months in advance, a full description of the project's nature and objectives, its methods and vessels, its precise geographical areas, its expected dates, its sponsoring institution, and the extent to which the coastal State should be able to participate [4]. Knowledge, like fish, is a resource the zone allocates; the next section descends from the water column to the seabed itself, where the continental shelf regime gives the coastal State its most geology-bound rights.

The continental shelf and its outer limits

The continental shelf is where the Convention stops measuring from the coast and starts measuring from geology. Article 76(1) defines it as "the sea-bed and subsoil of the submarine areas that extend beyond its territorial sea throughout the natural prolongation of its land territory to the outer edge of the continental margin, or to a distance of 200 nautical miles from the baselines… where the outer edge of the continental margin does not extent up to that distance" [4]. The continental margin itself "comprises the submerged prolongation of the land mass of the coastal State, and consists of the sea-bed and subsoil of the shelf, the slope and the rise. It does not include the deep ocean floor with its oceanic ridges or the subsoil thereof" [4]. Every coastal State therefore gets at least 200 nautical miles of shelf, geology or not; States with broad geological margins may get more — and Article 76's remaining paragraphs are the machinery for deciding how much more, the most technically demanding delineation rules in the Convention, elaborated in the Commission's Scientific and Technical Guidelines and the TALOS manual on the Convention's technical aspects.

Where the margin extends beyond 200 miles, its outer edge is established by one of two formulae, applied at the coastal State's choice point by point. The sediment-thickness formula uses fixed points "at each of which the thickness of sedimentary rocks is at least 1 per cent of the shortest distance from such point to the foot of the continental slope"; the distance formula uses "fixed points not more than 60 nautical miles from the foot of the continental slope"; and the foot of the slope itself, "in the absence of evidence to the contrary, shall be determined as the point of maximum change in the gradient as its base" [4]. As the TALOS manual puts it, the outer edge of the margin is the outer envelope of the two formula lines [3]. Two constraint lines then cut the formulae back: the fixed points "either shall not exceed 350 nautical miles from the baselines… or shall not exceed 100 nautical miles from the 2,500 metre isobath" [4]. On submarine ridges only the 350-mile cap applies — but that ridge rule "does not apply to submarine elevations that are natural components of the continental margin, such as its plateaux, rises, caps, banks and spurs," a distinction several later chapters will find worth fighting over [4]. The final limit merges the two constructions: if the margin lies inside the combined constraint line, the margin defines the limit; if it extends beyond, the constraint truncates it — and the limit is then drawn as straight lines not exceeding 60 nautical miles connecting fixed points defined by coordinates of latitude and longitude, points that "may be arbitrarily selected, although they will normally be chosen to maximize the enclosed area" [3] [4].

Uniquely in the Convention, this delineation is peer-reviewed. Information on limits beyond 200 miles must be submitted to the Commission on the Limits of the Continental Shelf, a body set up under Annex II on the basis of equitable geographical representation; the Commission makes recommendations, and "the limits of the shelf established

by a coastal State on the basis of these recommendations shall be final and binding" [4]. The coastal State must then deposit charts and geodetic data "permanently describing" the outer limits with the UN Secretary-General, who gives them due publicity — the publication discipline of Article 16 carried to the ocean's edge [4]. None of this prejudices delimitation between opposite or adjacent States, which remains a separate, bilateral or judicial question [4].

The rights themselves are the strongest resource rights in the Convention. The coastal State exercises sovereign rights over the shelf "for the purpose of exploring it and exploiting its natural resources"; those rights are exclusive — if the coastal State does not exploit, "no one may undertake these activities without the express consent of the coastal State" — and, distinctively, they "do not depend on occupation, effective or notional, or on any express proclamation": shelf rights exist by virtue of geography alone [4]. The protected resources are "the mineral and other non-living resources of the sea-bed and subsoil together with living organisms belonging to sedentary species" — organisms which at the harvestable stage are immobile on or under the seabed or move only in constant contact with it [4]. Yet the shelf is a floor, not a ceiling: coastal rights "do not affect the legal status of the superjacent waters or of the air space above those waters," and their exercise must not unjustifiably interfere with navigation and other States' freedoms; all States may lay cables and pipelines on the shelf (pipeline routing, unlike cables, needs coastal consent), and the coastal State holds the exclusive right to authorize and regulate drilling "for all purposes" [4].

One provision prices the privilege of going beyond 200 miles. For exploitation of the shelf's non-living resources beyond that line, the coastal State must make payments or contributions in kind through the International Seabed Authority, which redistributes them to States Parties on equitable criteria favouring developing, least-developed and land-locked States: nothing for the first five years of production at a site, then 1 per cent of value or volume in the sixth year, rising 1 per cent annually to a permanent 7 per cent from the twelfth year, with developing States that are net importers of the mineral exempt [4]. The outer shelf is thus the one place where national resource rights are taxed for the common heritage — a bridge to the deep seabed regime examined later, and a standing reminder that in this Convention, every extension of the map's coloured area is negotiated, never free.

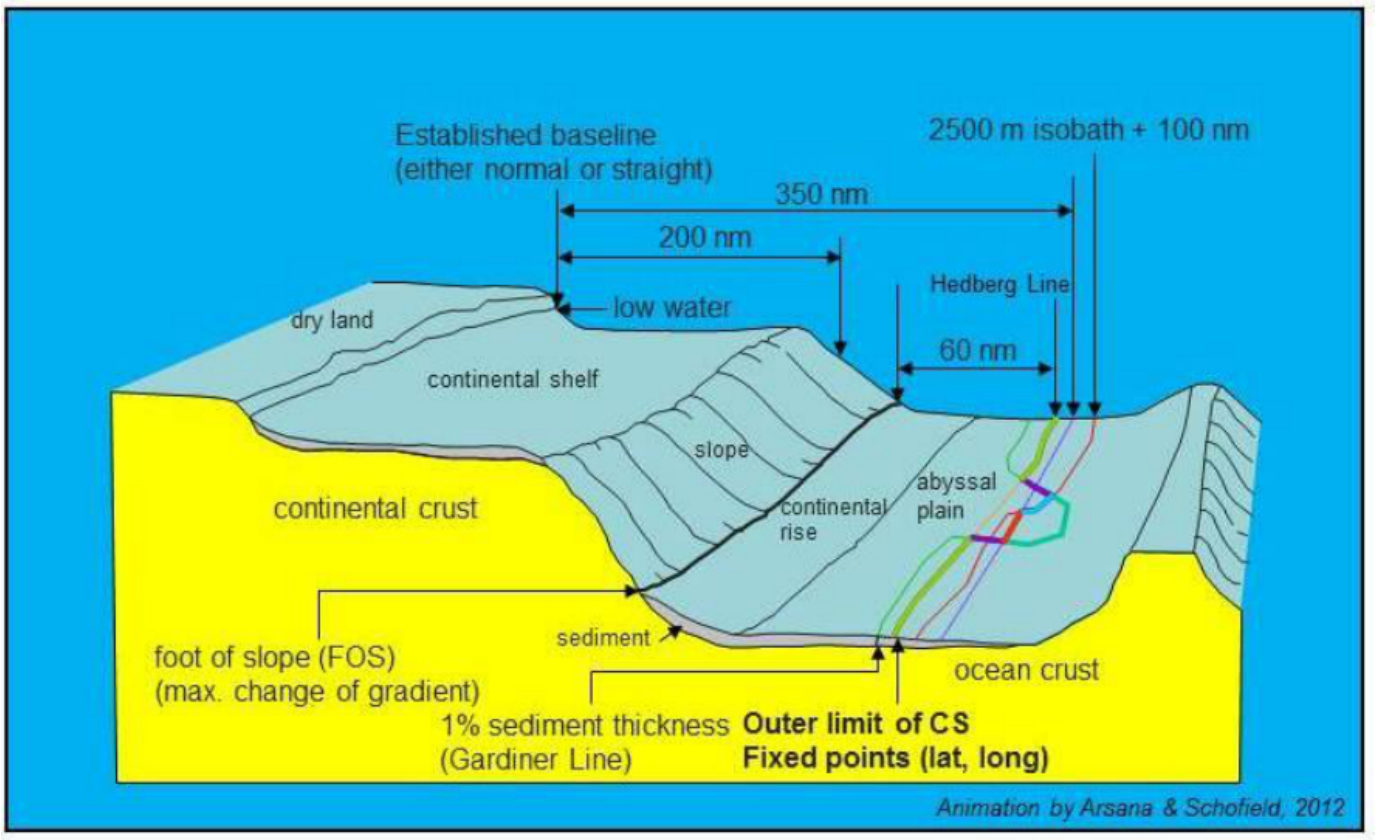


Exhibit. Establishing the outer limit of the continental shelf: the foot of the slope, the Gardiner and Hedberg lines, and the 2,500-metre isobath plus 100 nautical miles. Source: S0012.

The high seas and their freedoms

Every zone examined so far is defined by what a coastal State may do; the high seas are defined by what no State may do. Article 86 fixes the domain by exclusion: Part VII "applies to all parts of the sea that are not included in the

exclusive economic zone, in the territorial sea or in the internal waters of a State, or in the archipelagic waters of an archipelagic State," without abridging the freedoms other States enjoy inside the EEZ under Article 58 [4]. Article 89 then states the regime's founding prohibition: "No State may validly purport to subject any part of the high seas to its sovereignty" [4]. The high seas are not ungoverned; they are governed by a ban on appropriation.

Article 87(1) opens them affirmatively: "The high seas are open to all States, whether coastal or land-locked," and enumerates six freedoms — navigation; overflight; the laying of submarine cables and pipelines (subject to Part VI); the construction of artificial islands and other installations permitted under international law (subject to Part VI); fishing (subject to the conditions of section 2 of Part VII); and scientific research (subject to Parts VI and XIII) [4]. Each freedom arrives pre-qualified: fishing is conditioned, research is conditioned, construction defers to the shelf regime. And Article 87(2) wraps the whole list in the reciprocal discipline met throughout this report — the freedoms "shall be exercised by all States with due regard for the interests of other States in their exercise of the freedom of the high seas, and also with due regard for the rights under this Convention with respect to activities in the Area" [4]. Article 88 adds a purpose limitation: "The high seas shall be reserved for peaceful purposes" [4].

Order on this unowned ocean is supplied by nationality rather than territory. Every State, coastal or land-locked, "has the right to sail ships flying its flag on the high seas" (Article 90); each State fixes the conditions for granting nationality and registration to ships, and "there must exist a genuine link between the State and the ship" (Article 91) [4]. A ship "shall sail under the flag of one State only and, save in exceptional cases expressly provided for in international treaties or in this Convention, shall be subject to its exclusive jurisdiction on the high seas"; a ship that sails under two or more flags "using them according to convenience" may claim none of those nationalities and "may be assimilated to a ship without nationality" (Article 92) [4]. Jurisdiction follows the flag the way, in the zonal chapters, it followed the chart: a mobile, documentary fact — registry, papers, a genuine link — substitutes for the coastline.

For this report's argument, the high seas complete the gradient. Moving seaward from the baseline, sovereignty thins in stages — full sovereignty in the territorial sea, functional resource rights in the EEZ, seabed-only rights on the shelf — until on the high seas it vanishes altogether and only freedoms, exercised with due regard, remain. Every stage of that gradient is anchored to the same measured lines, and the final irony is symmetrical to the first chapters': a phantom island, if it were real and generated zones, would carve its exclusions out of this commons; being unreal, its 200-mile shadow on the chart was an appropriation of the high seas by a feature that never existed — exactly what Article 89 forbids a State to do on purpose.

Archipelagic states and archipelagic sea lanes passage

Part IV of the Convention solves a problem the baseline articles cannot: a state made of islands scattered across open ocean, whose "coast" is an abstraction. Article 47 lets an archipelagic State draw straight archipelagic baselines "joining the outermost points of the outermost islands and drying reefs of the archipelago," subject to conditions that keep the geometry honest: the enclosed area must include the main islands and satisfy a water-to-land ratio between 1 to 1 and 9 to 1 (land area may include waters within fringing reefs and certain enclosed oceanic plateau waters); no baseline segment may exceed 100 nautical miles, except that up to 3 per cent of segments may run to 125 nautical miles; the system "shall not depart to any appreciable extent from the general configuration of the archipelago"; and baselines may not be drawn to or from low-tide elevations unless a lighthouse or similar permanently above-sea-level installation stands on them, or the elevation lies within a territorial-sea breadth of the nearest island [4]. One external discipline matters most: the system "shall not be applied... in such a manner as to cut off from the high seas or the exclusive economic zone the territorial sea of another State" [4]. The usual machinery then applies — baselines shown on adequate charts or replaced by coordinate lists with geodetic datum, given due publicity, and deposited with the Secretary-General — and Article 48 measures everything from them: "The breadth of the territorial sea, the contiguous zone, the exclusive economic zone and the continental shelf shall be measured from archipelagic baselines drawn in accordance with article 47" [4].

Inside the lines, the waters are "archipelagic waters," over which the State's sovereignty extends "regardless of their depth or distance from the coast," together with the air space above, the bed and subsoil, and the resources therein (Article 49); within them the State may draw closing lines under Articles 9, 10, and 11 to delimit internal waters (Article 50) [4]. That sovereignty is expressly encumbered: the archipelagic State must respect existing agreements and recognize the traditional fishing rights and other legitimate activities of immediately adjacent neighbouring States in certain areas of its archipelagic waters, and must respect and permit maintenance of existing submarine cables laid through them (Article 51) [4]. And all ships enjoy innocent passage through archipelagic waters on the Part II model,

which the State may suspend temporarily in specified areas — without discrimination, after due publication — only where essential for its security (Article 52) [4].

Archipelagic sea lanes passage is the stronger, transit-passage-style right layered on top. Under Article 53, an archipelagic State may designate sea lanes and air routes "suitable for the continuous and expeditious passage of foreign ships and aircraft," in which all ships and aircraft enjoy passage "in the normal mode solely for the purpose of continuous, expeditious and unobstructed transit" between one part of the high seas or an EEZ and another [4]. The lanes must include all normal passage routes used for international navigation or overflight; they are defined "by a series of continuous axis lines from the entry points of passage routes to the exit points," and ships and aircraft in passage "shall not deviate more than 25 nautical miles to either side of such axis lines" nor "navigate closer to the coasts than 10 per cent of the distance between the nearest points on islands bordering the sea lane" [4]. Designation is procedural as well as geometric: proposals go to the competent international organization for adoption in agreement with the archipelagic State; the State "shall clearly indicate the axis of the sea lanes and the traffic separation schemes designated or prescribed by it on charts to which due publicity shall be given"; and if no lanes are designated, the right may be exercised through the routes normally used for international navigation [4].

The archipelagic regime is the Convention's most thoroughly drawn legal order: a sovereignty constituted by lines, hedged by numerical ratios and segment lengths, and threaded by charted corridors whose axes are legal objects in themselves. It is also, for this report, the fitting culmination of the zonal chapters — the farther the law reaches from a walkable coastline, the more completely it lives on the chart, and the more its validity depends on the fidelity of the features the lines connect. An archipelagic baseline anchored to a drying reef that is not there would not merely misdescribe the sea; it would misdraw a nation.

The regime of islands and low-tide elevations

If one article carries this report's whole argument, it is Article 121. An island, it provides, is "any naturally formed area of land that is surrounded by water and which is above water at high tide" — and the low-water line surrounding such a feature, "regardless of its size, may form the baseline or a part of the baseline from which to measure the maritime zones" [3]. Every element of that sentence is a factual predicate. Naturally formed excludes the artificial. Surrounded by water and above water at high tide fix the feature's relation to the tide. And the entitlement is size-blind: a speck that satisfies the predicates anchors the same 200-nautical-mile geometry as a continent's headland. Article 121 then grades downward: a feature that is a "rock, which cannot sustain human habitation or economic life of its own," has no exclusive economic zone and no continental shelf at all — though, the TALOS Manual notes, "UNCLOS does not explicitly define what a rock is, nor does it distinguish a rock from an island," leaving the most consequential classification in the article to inference [3].

Below the island sits the low-tide elevation, defined in Article 13 as a "naturally-formed area of land which is surrounded by water at low-tide but submerged at high-tide" [3]. Its baseline value is strictly conditional and strictly geographical: a naturally formed low-tide elevation "may be part of the 'normal' baseline only if all or a part of the elevation lies within the breadth of the territorial sea, measured from the mainland or an island"; if it "lies wholly outside the breadth of the territorial sea measured from the mainland or an island, it may not be used as part of the baseline" [3]. Human construction can upgrade an elevation's usability but not its nature: straight baselines may be drawn to low-tide elevations only where they carry "a lighthouse or similar installation permanently above sea level," or where baselines to and from such elevations have received general international recognition; archipelagic baselines may reach an elevation on the same distance criterion or with such an installation [3]. Artificial islands occupy the bottom of the hierarchy: "Such features, which include artificial islands, do not possess the status of islands and do not form a part of the baseline. They have no territorial sea of their own" — in pointed contrast to permanent harbour works, which Article 11 treats as part of the coast itself, jetties, breakwaters, and sea walls included, though not structures such as sewage outfalls that serve neither harbour nor coast protection [3].

The Manual's most telling admission comes first, as a practical warning: "Even on a detailed chart, it is not always possible to determine whether a small feature is a natural island, an artificial island, or a low-tide elevation with a structure built upon it," and the matter may have to be resolved by the Sailing Directions or, failing that, "a visual inspection or a hydrographic survey, to determine if the feature is a low-tide elevation or an island" [3]. Consider what that means. The legal categories that decide whether a feature generates no zone, a territorial sea, or a full EEZ-plus-shelf — categories carrying hundreds of thousands of square kilometres of ocean — turn on properties (natural or artificial? dry at high tide or only at low?) that the chart, the law's own instrument, frequently cannot certify. The classification must be verified in the world: by looking, and by sounding.

Sandy Island is this problem with the volume turned up: a feature charted as an island — naturally formed, dry, 25 kilometres long — that satisfied none of the predicates because it did not exist at all, yet whose low-water line would in principle have anchored baselines under the very article quoted here. The regime of islands is a machine for converting geography into jurisdiction, and it has no internal sensor for the geography's truth. Its only error-correction is the one TALOS prescribes and the Southern Surveyor performed: go and look.

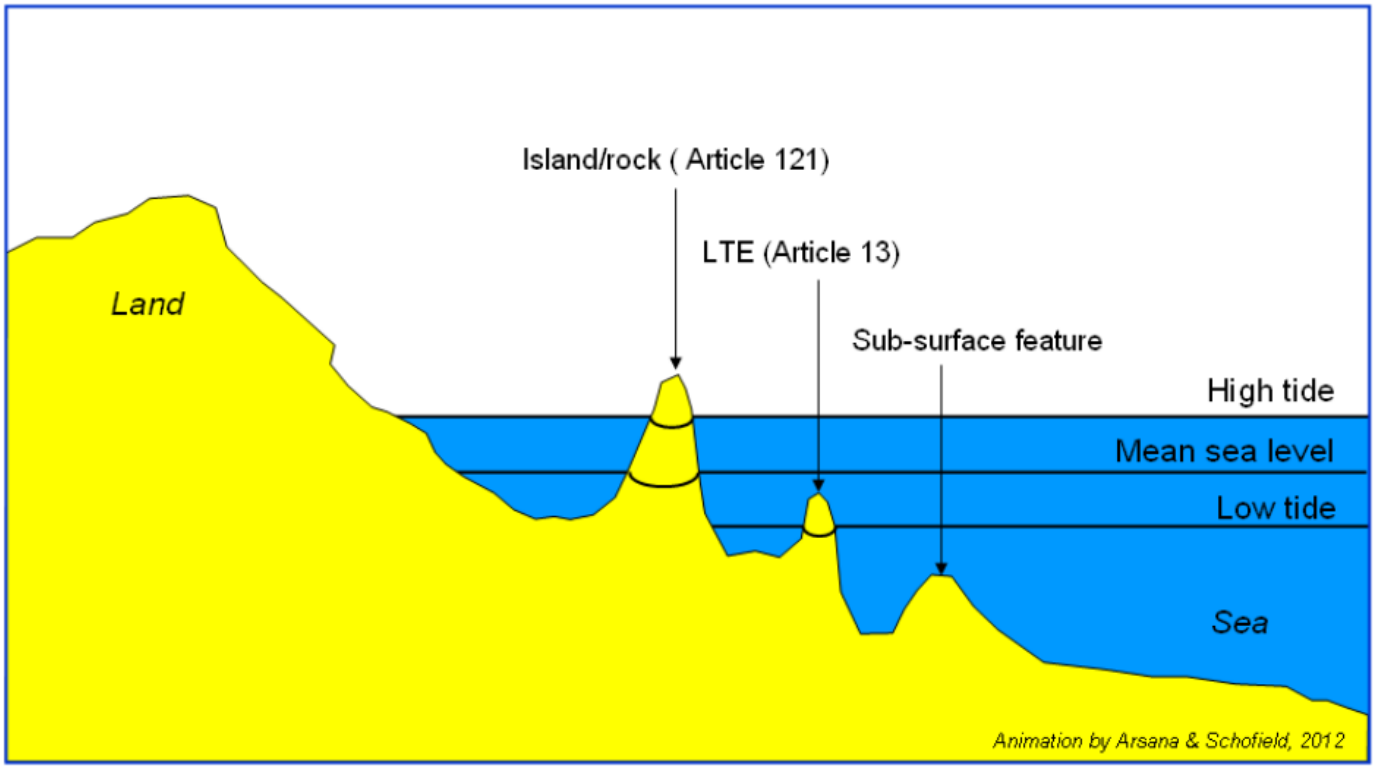


Exhibit. Under UNCLOS Articles 13 and 121, whether a feature is an island, a rock, or a low-tide elevation that dries at low water determines the maritime zones it can generate. Source: S0012.

Drawing the Line at Sea: Maritime Delimitation

The line of equidistance: construction, graphical and automated methods

Every delimitation in the chapters ahead is drawn, at least initially, by one construction, defined in the IHO's TALOS Manual with geometric exactness: an equidistance line is "a line every point of which is equidistant from the nearest points on the territorial sea baselines of two States" [3]. Article 15 calls it a median line; the technical literature often reserves median line for the equidistance line between opposite States and calls the line between adjacent States a lateral equidistance line — though opposite and adjacent are themselves often difficult to define and apply, and the construction method is identical either way [3]. The method's authority rests on two properties: it is the method that must be employed in the territorial sea in the absence of agreement or special circumstances, and it is "an objective geometric method which is relatively easy to apply, particularly using computer methods (if the baselines are clearly defined) and gives a unique line" [3]. The International Court of Justice has said as much repeatedly, including in the Romania–Ukraine judgment (paragraph 115), as has the Barbados–Trinidad and Tobago arbitral tribunal (paragraph 242), and the resulting three-stage approach to delimitation has since been applied by the ICJ, ITLOS and arbitral tribunals alike [3].

The construction itself is a walk along the baselines. Between opposite coasts with normal baselines, start with basepoint a on State A's baseline and basepoint b on State B's: the midpoint m of the line joining them fixes a point of the equidistance line, and the perpendicular through it (direction m–n) fixes the line's azimuth, which holds until a third basepoint c is reached that is equidistant from a and b; the perpendicular of the line joining b and c then sets the next leg, and so on, segment by segment, until the whole median line is derived — points equidistant from three basepoints being known as tri-points [3]. Where straight baselines replace the low-water line, the segments change species: equidistance between two points yields a straight line, between two lines a straight line, and between a line and a point a section of a parabola [3]. For adjacent coasts there is no essential difference, except the practical advice to begin the exercise seaward rather than at the land boundary terminal — starting offshore from a point t

equidistant from points in each State, drawing successive angular bisectors as new basepoints come into play, and running shoreward until the line terminates at the land boundary [3]. Only the salient — seaward-most — basepoints ever affect the line, and how many do so depends on the interplay of the two States' baseline segments, the coastal configuration and the line's distance from the nearest basepoints: the farther offshore, the fewer the controlling points, and the wider the spacing permissible along a smooth coast [3].

Two honest caveats frame the technique. First, the plane geometry is a fiction of exposition: "in practice they are made on the ellipsoid," so the point-versus-line locus called a parabola "is in fact a more complex curve" — not even the intersection of a paraboloid with the ellipsoid, which would hold only for chord rather than surface distances — and an equidistance line between two single basepoints, though unique, is "very nearly but not exactly coincident with a Geodesic," treated in practice as the geodesic between successive turning points [3]. Second, the older graphical method — chart, dividers and plotted turning points — is worth knowing but is expertise-hungry and time-consuming, and "will only achieve an approximate result, which may not be acceptable as a final solution" [3]. Its workflow nonetheless survives inside the automated one: examine the largest-scale nautical chart to decide which coastal points and sections may legitimately be used — whether a feature is a low-tide elevation or an island, whether an island carries a lighthouse and may anchor a straight baseline — keep every basepoint on the same horizontal and vertical datum, and list candidate coordinates in sequence, reading them from charts of larger scale than those used to identify them [3]. Where a feature that looked like one point resolves into several, or two features appear to become equidistant simultaneously, the graphical stage cannot decide: the coordinates are all carried forward, and "the further resolution of these problems must await the computation process, when accurate distance checks must be made from points on the equidistance line to discover whether other listed base-points are closer than those being used" [3].

That division of labour — human judgment to select the candidate geography, machine computation to find the line it implies — is why equidistance disciplines everything that follows in this part of the paper. A court or tribunal does not draw a boundary freehand; it provisionally constructs this line, then asks whether the coastal geography makes the line inequitable and must be adjusted. The line's objectivity is therefore both its strength and its provocation: because it is unique and reproducible, any departure from it must be justified — and the justifications, islands given half effect, concavities cut off, coastlines of wildly unequal length, are the subject of the delimitation case-law chapters ahead.

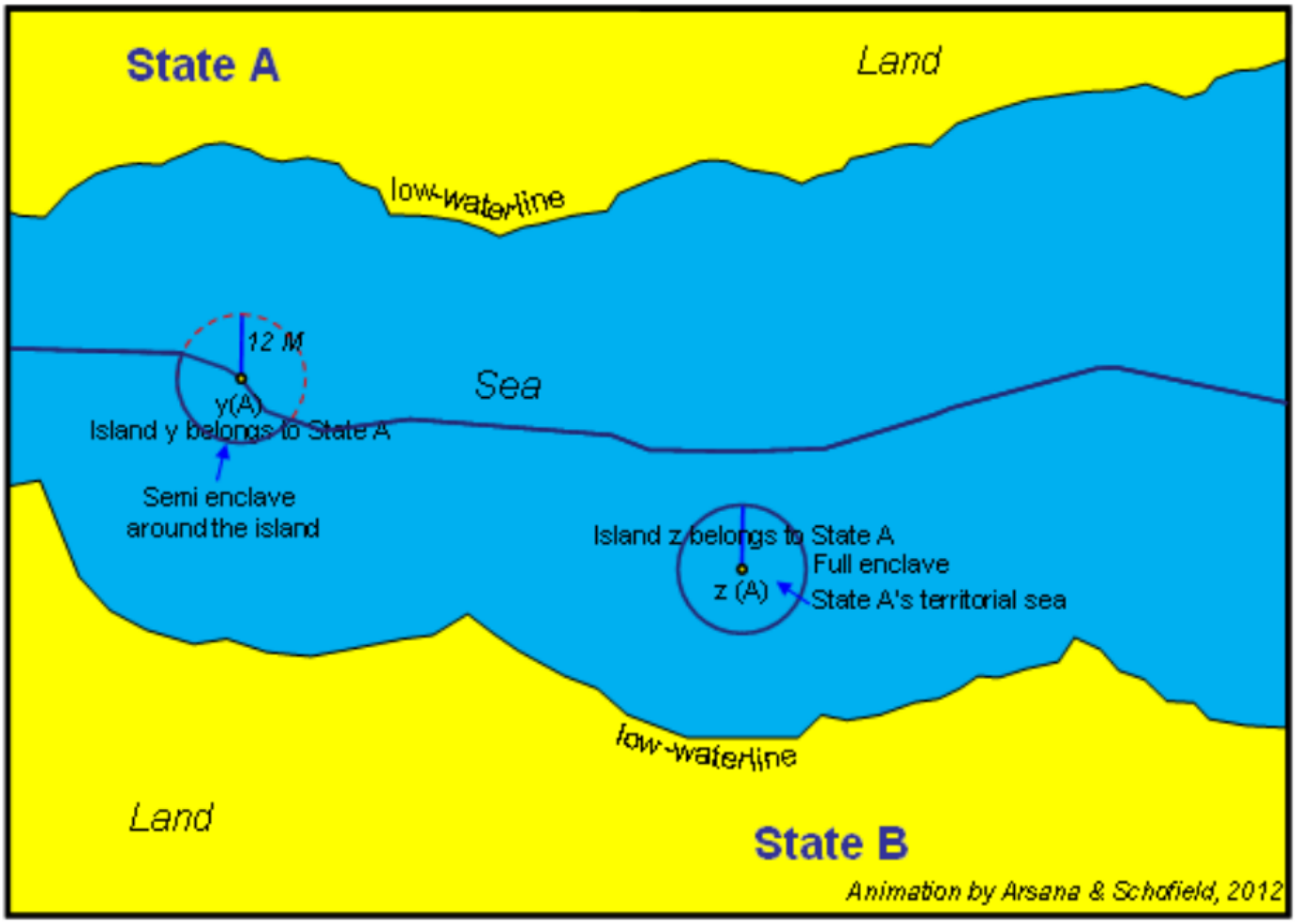


Exhibit. Semi-enclave and full-enclave effects: how an island belonging to one State is treated when an equidistance line is drawn between two opposite coasts. Source: S0012.

Hypothetical median lines in the Northern Gulf of Aqaba

The delimitation case studies begin where the geometry is kindest. The IBRU's Red Sea briefing (Parting the Red Sea, 2001) surveys a region in which only three of thirteen potential boundaries in the Red Sea and Gulf of Aqaba have actually been delimited — seven of the thirteen principally adjacent-type, six between opposite States — with a Saudi–Sudanese joint development agreement for the Red Sea brines standing in for a fourth [5]. The Gulf of Aqaba is the simple end of that inventory. It is less than 26 nautical miles wide, so "most boundaries within it will separate the territorial seas of the littoral states"; only in the southern portion, at its widest, are there small strips of continental shelf and EEZ for Egypt and Saudi Arabia to divide [5]. The coastline is generally smooth, with no significant islands, and "except for its southern entrance, where there is a latent island sovereignty dispute between Egypt and Saudi Arabia" — Tiran and Sanafir — "few features should complicate maritime delimitations"; given the simple geometry, "median line boundaries would appear to divide the Gulf equitably" [5]. The briefing's proportionality figure for the northern Gulf shows why: comparing each State's share of the area a median line would allocate against its share of the relevant coastline, the pairings are Egypt 23 per cent of area to 23 per cent of coast, Saudi Arabia 19 to 16, Jordan 5 to 8, and Israel 4 to 2 — the median line roughly pays each littoral in proportion to its coastal frontage [5].

Two baseline questions hang over the construction. Egypt defined its straight baseline system in 1990 and supplied the turning-point coordinates to the United Nations, but long stretches of its Gulf of Aqaba coastline "are not deeply indented or fringed with islands, which are the principal requirements before a country may claim straight baselines," and one analysis quoted in the briefing calls Egypt's Mediterranean system "illegitimate though moderate," a verdict the author extends to the Red Sea claims; US Department of State analysis concludes that in the Aqaba vicinity, where the coast is relatively free from indentations, "the normal baseline – the low-water line – must be used" [5] [5]. Saudi Arabia's position is worse for the cartographer: its 1958 decree claims straight baselines and appears to treat any feature not below lowest low-water as an island — "at variance with UNCLOS language" — but "no charts or coordinates of the Saudi baselines have been published to date," so their segments cannot even be located [5]. The briefing accordingly identifies "what effect, if any, to attribute to Egypt's straight baseline system and to the undelimited Saudi straight baseline system" as the most significant technical issue in dividing the Gulf — and

the greatest impediment as political: “Israeli-Arab relations” [5]. A further asymmetry: every Red Sea littoral claims a 12-mile territorial sea except Jordan, which maintains a 3-mile claim [5].

The one concluded boundary shows the median line doing treaty work. Article 3(7) of the 1994 Israel–Jordan Peace Treaty required the parties to negotiate their Gulf of Aqaba maritime boundary and conclude within nine months; they took longer, signing on 18 January 1996 an agreement under which the boundary “begins at Boundary Pillar 0 on the seashore and follows a straight line for 2.84 kilometres where it meets the median line of the Gulf,” then “follows the median line of the Gulf southwards until the last point of the maritime boundary between the two countries” [5]. The accord fixed no coordinates itself, calling for a joint team of experts to do so; the author knew of no published coordinates, and even the precise location of Pillar 0 was unclear — but a median-line estimate puts the final boundary at about 12 kilometres, dividing the two States’ territorial seas, with the treatment of the Egypt–Israel–Jordan tripoint left open [5]. The remaining northern segments are short and, technically, easy. Egypt–Israel have a potential territorial sea boundary of approximately 5 kilometres with “no intrinsic difficulty,” though its negotiation may be linked to the far harder Mediterranean boundaries with the Palestinian Authority [5]. Egypt–Jordan face some 10 kilometres (5.4 nautical miles) between opposite coasts less than 24 miles apart, where “a strict median line would seem both appropriate and equitable,” the only complications being their tripoints with Israel and Saudi Arabia [5]. Egypt–Saudi Arabia hold the Gulf’s longest boundary — territorial sea along most of its length, with the small southern shelf/EEZ strips at the widest part — and any delimitation would simultaneously confirm the status of Tiran and Sanafir at the entrance, so it will presumably be settled together with the pair’s longer, island-complicated northern Red Sea boundary [5].

The Aqaba exercise is the paper’s control case: where coastlines are smooth, islands absent and widths narrow, the hypothetical median line is nearly self-executing, and the obstacles migrate entirely off the chart — into unpublished baselines, an unlocated boundary pillar, a disputed island pair at the strait, and a peace process. The Red Sea proper, studded with mid-sea islands and rocks whose sovereignty and effect are contested, inverts every one of those conveniences; it is there, and in the Saudi–Yemeni and Eritrea–Yemen delimitations, that the median line meets its real tests.

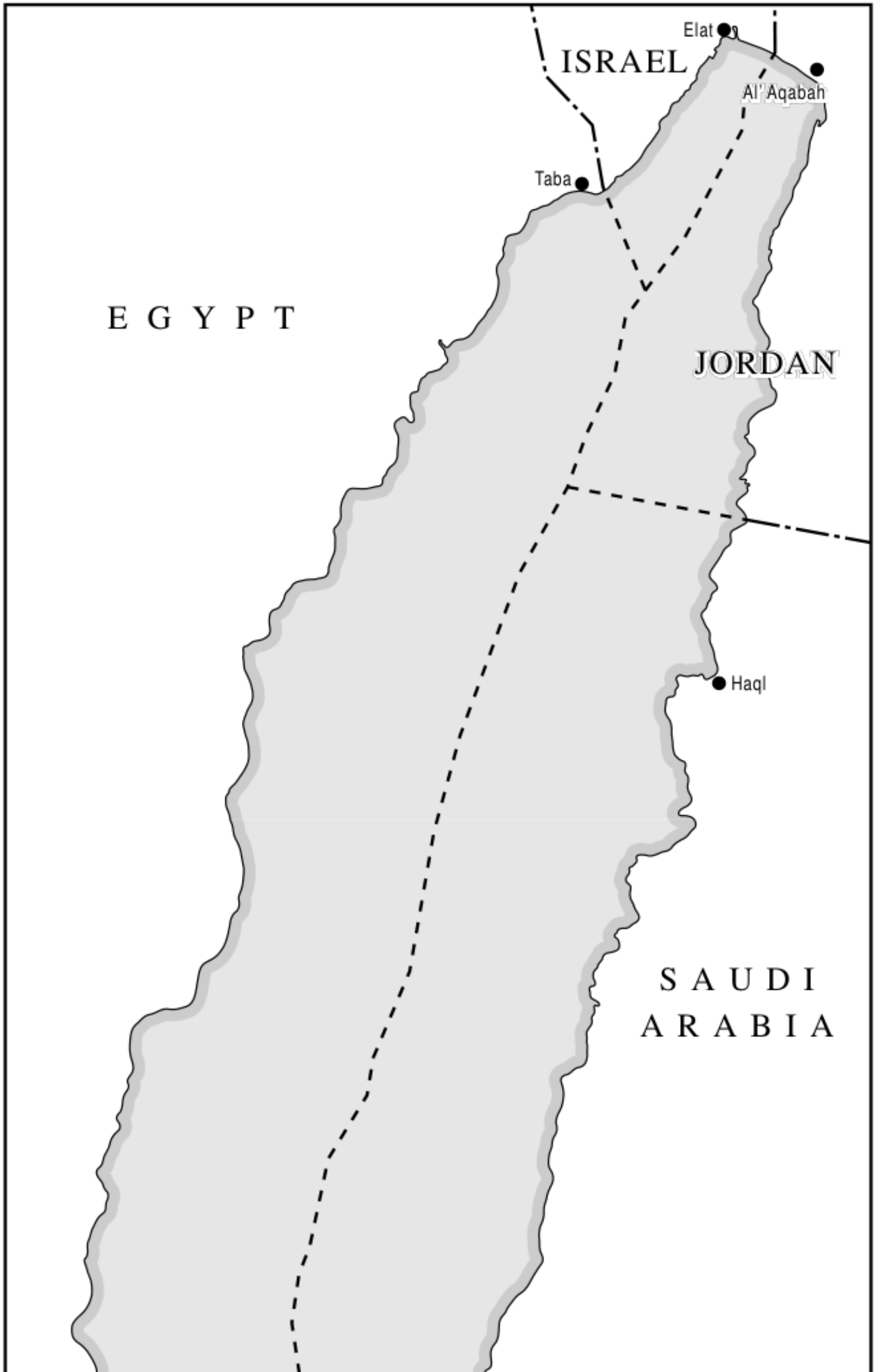

Elat
ISRAEL
Al' Aqabah
Taba
E G Y P T
JORDAN
Haql
S A U D I
A R A B I A

Exhibit. The Gulf of Aqaba and its approaches, where Egypt, Israel, Jordan, and Saudi Arabia meet at the head of the Red Sea. Source: S0009.

Egypt's straight baselines

Egypt is the Red Sea's most committed baseline claimant, and its system is the region's best test of how far Article 7's elasticity stretches. The claim has two birthdays: a 1951 decree proclaimed straight baselines along Egypt's coasts without delimiting the segments, and Presidential Decree No. 27 of 1990 finally defined the system and provided the turning-point coordinates to the United Nations — covering virtually the entire Egyptian coastline and its nearshore islands in both the Mediterranean and the Red Sea [5] [5]. The IBRU briefing's segment arithmetic gives the system its shape: within the Red Sea and the Gulf of Aqaba Egypt claims 55 straight baseline segments totalling 973.0 kilometres (525.4 nautical miles); the Aqaba segments (points 1–32) total 187.6 kilometres and average only 6.1 kilometres each, hugging the coast, while the Red Sea proper segments (points 32–56) total 785.4 kilometres and average 32.8 kilometres — five times longer, and correspondingly more ambitious [5]. The system's most eloquent feature is an omission: the 1990 claim excludes Tiran and Sanafir, the disputed islands at the Aqaba entrance that Egypt has occupied since 1949 — perhaps because Egypt no longer claims them, or perhaps because enclosing them might have closed the Strait of Tiran inside straight baselines and triggered a strong response from the maritime powers and Israel [5].

The claim carries a surveying defect that this paper's theme makes pointed. The 1990 decree omits the geodetic datum Article 16 requires, and its Article 2 instead specifies the coordinates "in accordance with geodetic datum (Mercator projection)" — but a Mercator projection is not a geodetic datum, and when the briefing's author plotted the Egyptian coordinates on a WGS-based nautical chart, "many of the turning points plotted 0.4km or more inland," with some segments appearing to cut through capes and promontories: "clearly, the Egyptian coordinates are not based on the WGS," and until Cairo identifies its geodetic system "a precise analysis of the straight baseline system will not be possible" [5]. A baseline claim is a list of coordinates offered as geography; Egypt's list, for a decade after publication, could not be placed on the earth to within half a kilometre — a paper coastline in the most literal sense, its segment lengths recoverable mathematically (the briefing computes them along great-circle arcs) but its position genuinely indeterminate [5].

On conformity, the verdict is split by segment. In the claim's favour, "the overall pattern of the Egyptian baseline segments closely follows the coastline, except for a few areas," the system does not incorporate vast areas as internal waters, and it does not appear to hamper navigation through the Strait of Tiran; enclosing the Gulf of Suez as internal waters should in any event leave a regime of innocent passage toward the Suez Canal [5]. Against it, many Red Sea turning points appear to be low-tide elevations without lighthouses or similar installations — an Article 7 violation built into the point list — and the two segments closing the Gulf of Suez, from Ras Muhammad (point 33) to a low-tide elevation off Giftun el-Saghir (point 35), run a combined 35.9 nautical miles, exceeding the 24-mile closing-line limit of Article 10, so Egypt "could not close the Gulf of Suez as a juridical bay with the proposed segments" [5]. In the Red Sea proper the baselines largely link low-tide elevations along fringing reefs, and the system notably does not reach Jazirat El-Ikhwan (The Brothers), twin islets 61 kilometres offshore [5].

The southern end is where the claim leaves the coastline behind. Approaching the land boundary disputed with Sudan, Egypt runs segments from Ras Banas (point 48) out to Rocky Island, an islet near Jazirat Zabarjad (point 49), and back to the reef Shi'b Abu Findirah (point 50) — "among the longest claimed in the Red Sea [67.6 and 75.2km] and digress from the general direction of the coast" [5]. Zabarjad itself — barren, waterless, rising 235 metres, reported inhabited in 1950 — is the weak link in the chain: "because Zabarjad is not a fringing island, there is no justification for its inclusion in Egypt's straight baseline system" [5]. The claim has not gone unchallenged: the United States lodged a detailed protest in 1991, holding in particular that segments 36–56 in the Red Sea fail the Article 7 criteria, the coastline there being neither deeply indented nor fringed with islands [5]. Egypt's baselines thus stand as the region's worked example of the straight-baseline bargain from the baselines chapter: modest, coast-hugging segments in the Gulf of Aqaba that a delimitation court would likely honour, shading southward into long oceanic jumps to isolated islets that it likely would not — and the difference matters, because every hypothetical median line in the northern Red Sea is computed from exactly these points.

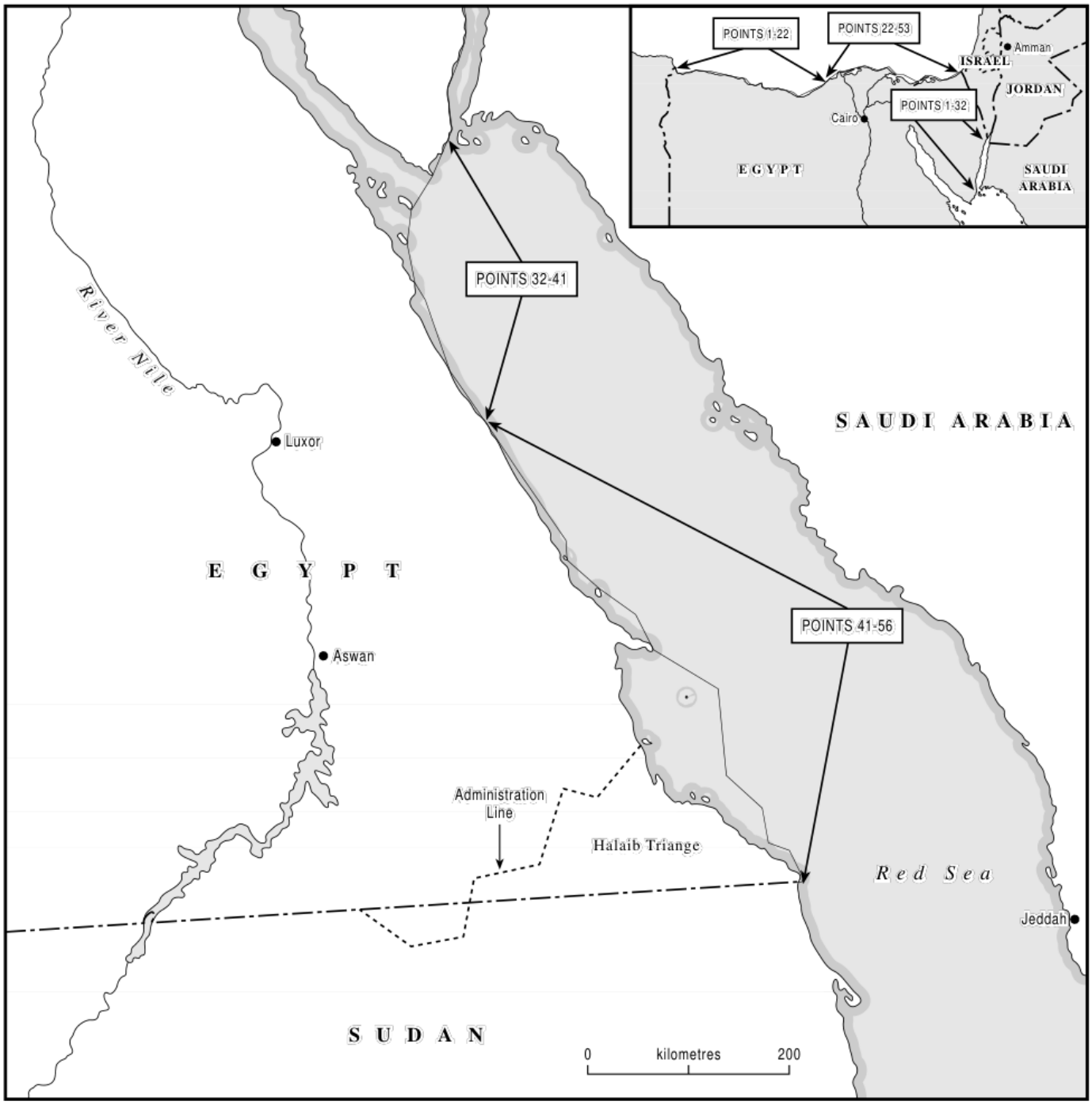


Exhibit. Egypt's straight baselines along its Red Sea coast, showing the baseline point groups and the Halaib Triangle administration line. Source: S0009.

Undelimited maritime boundaries of the Asian Rim

The IBRU's companion briefing, Undelimited Maritime Boundaries of the Asian Rim (2001), applies the equidistance instrument prospectively: for each neighbouring pair that has failed to agree a boundary, it constructs the line of equidistance on charts of appropriate scale, from all appropriate basepoints, and then audits the result for inequity. The choice of instrument is argued, not assumed. A median line is drawn, first, because every point on it is equidistant from the nearest baseline points of the two countries, so "providing there is agreement on the basepoints to be used the line of equidistance constructed by different cartographers will be identical" — "an impartial construction once the basepoints have been selected" — and second because it is the first line States themselves construct when preparing for negotiations, the area inside it being what most countries regard as their minimum acceptable share [6].

What the briefing refuses to build on is other States' straight baselines. They are excluded from the constructions because "some straight baselines have the potential to inequitably deflect the line of equidistance to the distinct advantage of the state that constructed them," because no party can be certain another will accept their validity, and

because a State without them might answer by drawing its own — Egypt's Red Sea system, examined in the previous chapter, is exactly the kind of artifact this rule guards against [6]. Archipelagic baselines are treated more gently, their rules being "more rigorous," so two archipelagic States such as Indonesia and the Philippines might well base a boundary on their long-established systems [6]. The live battleground is the basepoint itself, and above all Article 121(3): an objector may argue a feature is only a rock, unable to sustain human habitation or an economic life of its own, and so fit to generate a territorial sea and contiguous zone but not shelf or EEZ rights [6]. The briefing calls Article 121 "perhaps the most badly drafted article in UNCLOS," notes the two academic camps (wide "rock" versus narrow), and records the governmental reality with dry precision: States adopt the narrow-rock interpretation "for their own islands and rocks and the view of the first group of academics for the small islands and rocks of neighbours," so that "for every self-denying decision, such as that by Britain regarding Rockall, there are dozens of cases of countries defending the use of tiny rocks as basepoints" [6]. The study accordingly applies a deliberately generous interpretation of Article 121 — owners may treat sand cays and islets as islands, inhabited features generate full zones, barrenness does not disqualify — while flagging every feature a neighbour has queried or is likely to query [6]. Three structural blockers precede all geometry: Taiwan's unresolved status makes a Taiwan Strait boundary impossible and Taiwan-coast negotiations unlikely; no Spratly boundaries can be drawn until sovereignty there is settled (the briefing draws lines only between the outermost Spratlys and undisputed coasts, and excludes Scarborough Reef from the group); and the Philippines' colonial treaty-limits doctrine blocks its delimitations with Japan and Indonesia, while its latent North Sabah claim — pressed as recently as its March 2001 application to intervene in the Indonesia–Malaysia Sipadan and Ligitan case — clouds Malaysia talks [6].

The Cambodia–Vietnam analysis shows the method at full resolution. In the Gulf of Thailand, the offshore equidistance line is controlled by Cambodia's Poulo Wai group against Vietnam's Tho Chu group, diverted south by Cambodia's Îlots du Sud Est and pushed further south by Depond Reef — a low-tide elevation whose charted position is "almost exactly 12nm from the nearest feature in the Poulo Wai island group," so that its validity as a territorial-sea basepoint hangs on an approximate position being inside 12 miles; without it, the line retreats to Poulo Wai control [6]. Inshore, the controlling points are an almost absurd granularity of rocks and capes — Rocher Déchiré, Rocher Rosita, Hon Dung and a rock off a rock — against Mui Nai and the Pirate Islands, a reminder that at territorial-sea scale boundary lines are decided by features a chart-plotter must hunt for [6]. History then does what geometry cannot: the two States' 1971–72 shelf claims both annexed Phu Quoc, Poulo Wai and Tho Chu and overlapped wholesale; the 1982 settlement awarded Poulo Wai to Cambodia and the rest to Vietnam, but wrapped the nearshore in an oblong joint "historic waters" area whose seaward limit is the straight line linking the Tho Chu archipelago and Poulo Wai, the two States' baselines terminating at those groups and meeting at a floating "Point O" whose position still awaits mutual agreement [6]. Beneath even that lies a colonial artifact, the Brévié Line — Governor-General Jules Brévié's decision of 31 January 1939 in a Ha-tien–Kampot islands dispute — still charted as an alternative alignment to strict equidistance [6].

Two briefing-wide observations frame the region. Each analysis closes by rehearsing the grounds on which equidistance might be attacked as inequitable, including the economic-disparity plea — the poorer State asking for a compensating shift — which the ICJ has repeatedly rejected as irrelevant and unstable over time, yet which thrives in negotiation, where States may raise anything and economics is often the point of the exercise [6]. And where sovereignty over an island is disputed, the briefing simply draws both lines, one per owner, assessing neither claim: the map is offered as a measuring instrument, not a verdict [6]. Joint zones, several of which dot the region, receive the same restraint — once established, "the search for a single boundary becomes a low priority," and no replacement lines are drawn for them [6]. The Asian Rim thus emerges as Aqaba's opposite at continental scale: the geometric instrument works everywhere and settles almost nothing, because its inputs — rocks, reefs, treaty limits, floating points, unreconciled sovereignties — are precisely what is in dispute.

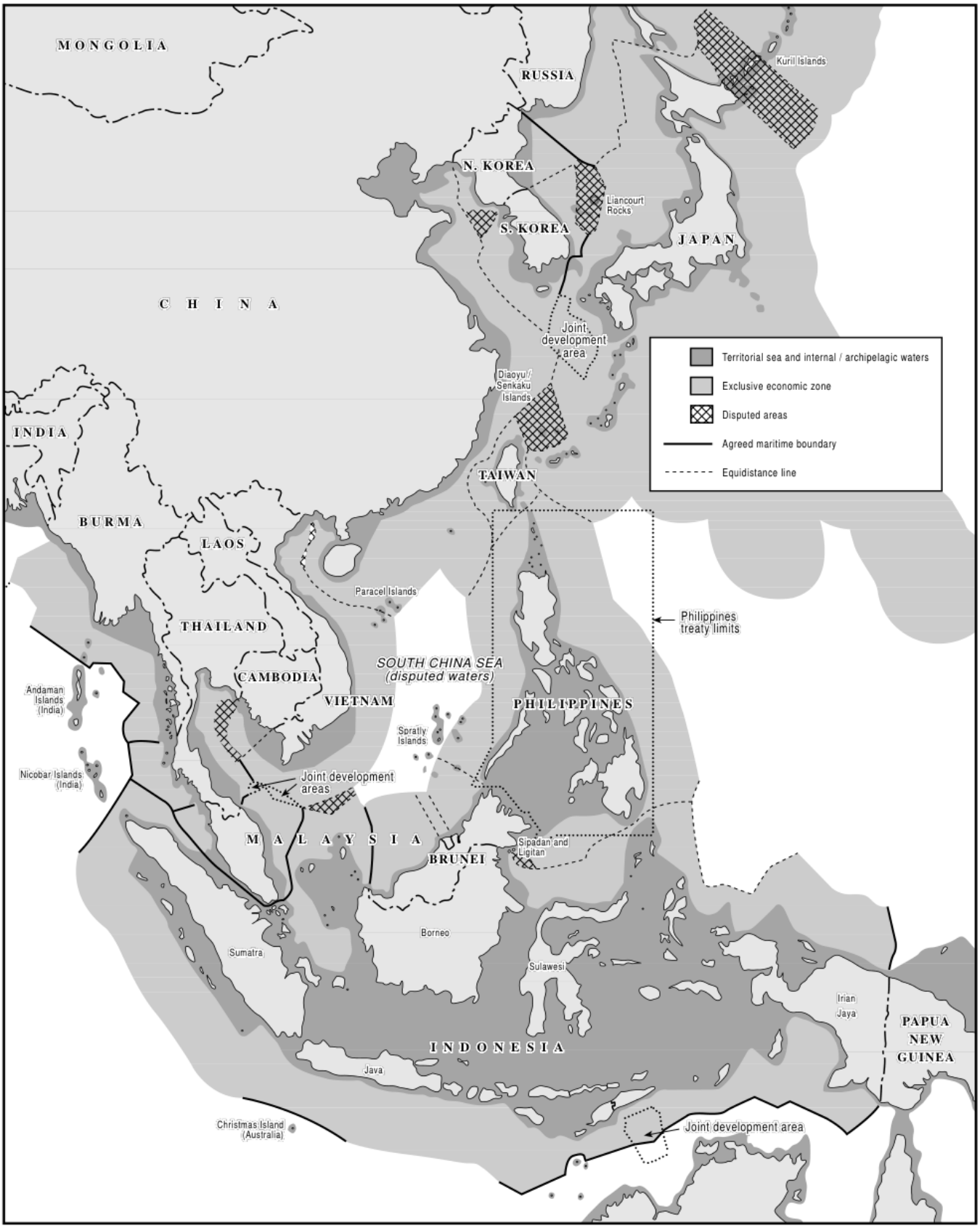


Exhibit. Maritime claims and archipelagic baselines across Southeast Asia and the South China Sea. Source: S0011.

Delimitation between opposite and adjacent coasts

All the zones of the preceding chapters radiate from one State's baselines; delimitation begins where two States' radiations intersect. The neighbouring States may be opposite — coasts facing each other across water, as in the Gulf of Aqaba — or adjacent — coasts side by side along a shared land boundary, as in the Gulf of Thailand — and the

TALOS Manual's delimitation chapter proceeds on two working assumptions worth stating because reality so often violates them: that in the adjacent case the coastal terminal point of the land boundary is agreed, and that "there is no dispute concerning baseline claims" [3]. The Convention then supplies three distinct legal regimes for the three kinds of space. For the territorial sea, Article 15 is a rule with an escape hatch: neither State is entitled, failing agreement to the contrary, to extend its territorial sea beyond the equidistance line between them — but the provision "is inapplicable if historic title or other special circumstances make it necessary to use another method" [3]. For the wider zones, Articles 74 (EEZ) and 83 (continental shelf) are textually identical and conspicuously rule-free: delimitation "shall be effected by agreement on the basis of international law, as referred to in Article 38 of the Statute of the International Court of Justice, in order to achieve an equitable solution"; if no agreement is reached within a reasonable period, the States resort to the Part XV dispute procedures; pending agreement they must make every effort to enter provisional arrangements of a practical nature and must not jeopardize or hamper the final agreement; and an existing agreement governs where one is in force [4].

The gap between those regimes is the delimitation problem in one sentence: the Convention mandates equidistance, subject to exceptions, in the narrowest zone, and mandates only an equitable result, by agreement, in the zones forty times wider. The manual is blunt about what that means for the practitioner: "unfortunately for the technical expert engaged on maritime delimitation, an equitable solution has no objective meaning, and there are many possible ways in which an equitable solution may be achieved" [3]. Practice has filled the vacuum with a standard choreography: "in any particular case... the approach to the problem is likely to follow a well-known method, such as equidistance, which is then adjusted in the last stages in order to achieve the required result" [3]. That is the provisional-equidistance-then-adjust sequence the courts later formalized as the three-stage approach — the construction audited in the equidistance chapter, whose output is a starting line with a unique claim to objectivity, and whose adjustment is where geography's hard cases (islands, concavities, disproportionate coasts) are litigated. The manual itself confines its ambition accordingly: not the catalogue of possible solutions, but "only a few basic techniques on which the expert charged with the technical work may build," with the key elements of a representative bilateral boundary assembled in its Figure 6.1 [3].

Three technical considerations, easily overlooked beside the law, govern whether a drawn line can actually serve as a boundary. First, accuracy and precision: "an appreciation of the appropriate accuracy and precision obtainable or to be achieved is as important in bilateral delimitation as in the determination of baselines or unilateral limits" — a line is only as defensible as the datums and coordinates beneath it, the lesson of Egypt's datum-free decree [3]. Second, administrability: where a dense selection of basepoints has produced the line, "the resulting line may be too complex for description in a treaty or for administration," and "some form of simplification may then be necessary" — the boundary as finally drawn is often a deliberately smoothed descendant of the mathematically exact equidistance line, trading fidelity for a turning-point list a treaty can hold and a coastguard can patrol [3]. Third, continuity and unity: it is possible, but inconvenient for maritime security and management, for bilateral boundaries not to form a continuous division of jurisdiction, so "in order to prevent 'gaps'... it is desirable to have a continuous delimitation line"; and while distinct lines for different zones are possible, "there is a clear preference in practice for maritime boundaries that separate all types of maritime jurisdiction" — the single all-purpose boundary over the layered alternative [3].

The framework now meets its cases. Opposite-coast delimitation across narrow seas (Aqaba) runs on Article 15's equidistance default inside territorial seas; adjacent-coast delimitation (the Asian Rim pairs) must first solve the land-boundary terminus the manual assumes away; and the wide-zone cases engage Articles 74 and 83, where the equitable-solution standard hands the advantage to whichever side better argues the adjustment. It is in that adjustment — the space between the provisional line and the equitable one — that the case law of the next chapters, from the North Sea to the Black Sea, is fought.

Asia's Contested Waters

The Spratly Islands

The Spratlys are the Article 121 problem at geopolitical scale. The "170-plus features collectively termed the Spratly Islands" stretch some 900 km across the southern South China Sea, but "[t]he majority of the Spratlys are in fact really submerged banks, reefs and low-tide elevations" — only 36 are known to rise above high tide to form islands at all, the largest of them, Itu Aba, a mere 1.4 km long and 400 m wide, and the total land area above the highest astronomical tide is estimated at less than 8 km² scattered over roughly 240,000 km² of sea [6]. Recall what the regime

of islands does with such facts: submerged banks generate nothing; low-tide elevations generate baseline rights only within a territorial-sea breadth of real land; rocks that cannot sustain habitation or economic life generate no EEZ or shelf. The contested maritime area, by contrast, "far exceed[s]" the land figure by orders of magnitude — which is precisely why the classification of each speck matters, and why six coastal states claim the archipelago [6].

The claimants are China, Taiwan (Republic of China), Vietnam, the Philippines, Malaysia, and Brunei, laying claim "to all or part of the Spratly and Paracel archipelagos and their surrounding maritime space"; of the six, "all save Brunei maintain a military presence on one or more islands" [6]. The occupation ledger, as compiled in the IBRU briefing from information supplied in November 2000, runs: Vietnam at least 19 features, the Philippines 8, China 7, Malaysia 5, Taiwan 1 — and Brunei claiming Louisa Reef without occupying it [6]. Occupation, however, cannot do here what equidistance did in the Gulf of Aqaba: the five occupying powers "also claim all or some of the other Spratly features that they do not occupy," and since none recognizes the claims of the others, the briefing concludes flatly that "it is not possible to draw lines of equidistance within the Spratly Islands to show the areas claimed by each country" [6]. The analysts had to exclude the Spratlys from their equidistance study altogether and define the surrounding region from British Admiralty chart 4508 instead — a delimitation study with a hole cut in its middle, the hole being the dispute [6].

The surrounding sections of the same briefing show what is at stake in feature classification once lines are drawn elsewhere. In the Gulf of Tonkin, Vietnam's Dao Bach Long Vi, lying 58 nautical miles off the mainland, "delivers an additional 5,680 km² to Vietnam" through its effect on the equidistance line — a single offshore island worth a small province of sea [6]. The briefing also dismantles a recurring cartographic temptation in that gulf: Vietnam's claim that the Sino-French Convention of 26 June 1887 had already fixed the boundary at meridian 108°03′13″E — the line had in fact been drawn only to allocate islands between the colonial powers, a common late-19th-century shorthand (Britain in Torres Strait, 1879; France and Portugal in West Africa, 1886; Spain and the United States for the Philippines, 1898), mentioning no division of seas or seabed and providing no southern terminus [6]. The Paracel Islands, the Spratlys' northern counterpart, complete the pattern of paper and possession diverging: occupied by China since 1974, claimed in Chinese place-name lists of 1935, 1947, and 1983, yet awarded the stronger historical title by Chemillier-Gendreau's survey, which found the indigenous kingdoms of Vietnam held the Paracels for at least two centuries before French annexation in the 1880s — against which stands North Vietnam's own message of 14 September 1958 approvingly recognizing China's 12-nautical-mile territorial-sea declaration that treated the Paracels as Chinese territory [6].

The Spratlys thus invert Sandy Island one final time. There, a feature that did not exist was treated as land, and the error was corrected by a single survey. Here, features that unquestionably exist — but barely, intermittently, a few of them above water only by metres — are asked to bear EEZ-scale entitlements, and no survey can settle the question, because the dispute is not about where the features are but about whose they are and what class they belong to. The chart can show all 170 features perfectly and the map of claims will still not converge: measurement was never the missing piece.


South China Sea
BRUNEI
BORNEO
MALAYSIA
INDONESIA
Kalimantan
Celebes

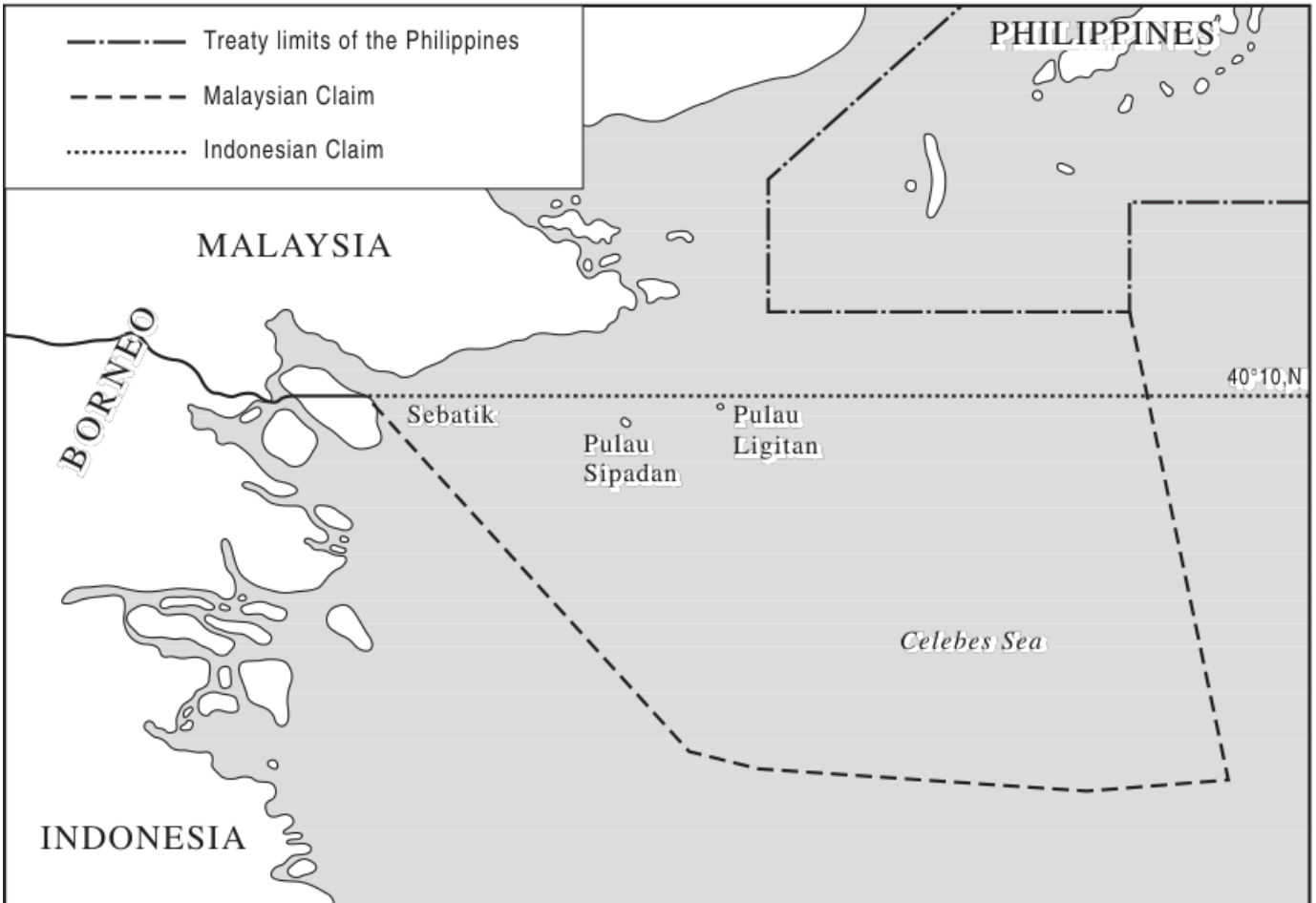

Treaty limits of the Philippines
Malaysian Claim
Indonesian Claim
PHILIPPINES
MALAYSIA
BORNEO
40°10,N
Sebatik
Pulau Sipadan
Pulau Ligitan
Celebes Sea
INDONESIA

Exhibit. The Sipadan and Ligitan area: Malaysia, Indonesia, and Brunei around northern Borneo and the Celebes Sea. Source: S0008.

The Nine-Dash Line

The Spratly chapters' legal climax is a line with no coordinates. The nine-dash line is, in the words of a close analysis of the 2016 South China Sea Arbitration, "a cartographic denotation that was developed, first by the Republic of China in the 1940s, and then by the People's Republic of China (PRC) in the 1950s, and subsequent years to affirm Chinese sovereignty over the islands and maritime areas of the South China Sea" [7]. Since the 1950s the PRC has consistently stated that the line demarcates Chinese sovereignty over the four island groups and the adjacent maritime areas, and two Chinese scholars close to the government describe it as "synonymous with a claim of sovereignty over island groups that have always belonged to China and with an additional Chinese claim of historical rights of fishing, navigation, and other marine activities… on the islands and in the adjacent waters" (Gao and Jia, 2013) [7]. The same scholars note the line was drawn, beginning in 1948, to function as "median lines between the [South China Sea] islands and the opposite coasts of neighboring states" — a delimitation purpose from the beginning [7]. What the line has never had is a published legal definition: China, which did not participate in the arbitration, "has never officially stated the precise legally intended effect of the nine-dash line, preferring to leave the assertion of this claim a vague political pronouncement" [7]. The US Department of State's 2014 analysis (Limits in the Seas No. 43) states the consequence: "unless China clarifies that the dashed-line claim reflects only a claim to islands within that line and any maritime zones that are generated from those land features…, its dashed-line claim does not accord with the international law of the sea" — an assessment with which virtually all law-of-the-sea experts agree [7].

The tribunal, in its Merits Award of 12 July 2016, first had to reach the question. Article 298(1)(a)(i) lets a State except from compulsory dispute settlement all disputes concerning "sea boundary delimitations or… historic bays or titles," and China activated that exception in 2006; a literal reading seemed to exclude the line's validity altogether, and the tribunal's October 2015 admissibility ruling reserved the point [7]. The Merits reasoning dissolved the obstacle by definition, in three moves: the dispute over the line is "a dispute over the source and existence of maritime entitlements" — and while all delimitation concerns entitlements, "the converse is not the case" — so it is not a delimitation dispute (paras. 204–206); the nature of China's claim is not "title" but historic "rights" to living and non-living resources (paras. 207–214); and the Article 298 exception extends only to historic "titles," not historic "rights" (paras. 215–229) — so, China's claim being "only a 'constellation of historic rights short of title,'" jurisdiction stood [7]. On the merits the answer was categorical: China's claim to historic rights to the living and non-living resources within the nine-dash line "is incompatible with the [UNCLOS] to the extent that it exceeds the limits of China's maritime zones as provided for by the [UNCLOS]"; any such rights were "superseded… by the limits of the maritime zones provided for by the [UNCLOS]"; and China has acquired no rights under the line since the Convention entered into force [7]. The Award's companion holdings — features judged by their natural condition, incapable of upgrade by construction, and every high-tide feature considered held to be a mere "rock" entitled to a 12-mile territorial sea only — stripped the line's contents as thoroughly as its perimeter [7].

The essay's author accepts neither the reasoning nor, entirely, the result. The jurisdictional route "does not quite ring true": the tribunal defined the claim until the obstacle "became a straw man, easily brushed aside" — though China abetted the process by absenting itself and leaving its claim legally undefined, so that "China's lack of legal acumen redounded to its detriment" [7]. On the substance, the author concedes the line fails as title: historic title in the modern law is confined to historic bays (Article 10(6)), straight-baseline contexts (Articles 7 and 10(6)) and territorial-sea delimitation (Article 15), and the customary test from the 1951 Fisheries Case — historic waters are waters treated as internal "but which would not have that character were it not for the existence of historic title," requiring open, effective, long-term and continuous exercise of authority plus foreign acquiescence — is one "the nine-dash line does not qualify even remotely" to meet [7]. But historic rights are a second, lesser category — non-exclusive rights other States must take into account — and Article 62(3) expressly preserves their core: a coastal State giving access to its EEZ "shall take into account… the need to minimize economic dislocation in States whose nationals have habitually fished in the zone" [7]. The tribunal's categorical statements — that "no article of [UNCLOS] expressly provides for… the continued existence of historic rights" in the EEZ, and that EEZ resource rights belong "to the coastal state alone" — are therefore "not entirely correct," and its position that traditional fishing rights survive in archipelagic waters and the territorial sea but are extinguished in the EEZ is an evidently result-driven distinction [7]. The wiser ruling, the essay argues, would have left the line definable as an assertion of non-exclusive,

historical/habitual fishing rights under Article 62(3), inviting negotiation among all the South China Sea riparians instead of a blanket extinction [7].

The nine-dash line is this paper's theme at maximum scale: a mark on a map — dashed, uncoordinated, officially unexplained — asserted for seventy years as if the drawing itself were the title. The Convention's system, which prices claims by baselines, features and formulae, had no cell in which to file it; the tribunal answered by filing it under nothing, and its critic answers that a smaller, older category — habitual use, short of title — was available and was refused. Between extinction and sovereignty lies the negotiation neither side has yet had.

The South China Sea arbitration decision and the merits award

On 12 July 2016, a five-member tribunal of the Permanent Court of Arbitration at The Hague, constituted under Annex VII of UNCLOS, handed down its Merits Award in the case the Philippines had brought against China in 2013 (PCA Case No. 2013-19) — a judgment Thomas Schoenbaum calls "surely the most important set of jurisprudential rulings in the modern history of the international law of the sea" [7]. The proceeding had two acts: an Award on Jurisdiction and Admissibility of 29 October 2015 (151 pages) and the Merits Award itself (479 pages) [7]. The tribunal "ruled in favor of the Philippines on virtually every issue of the dispute," and in doing so "clarified many murky issues and problems inherent in provisions of the UNCLOS" — among them, as the previous section detailed, the extinction of historic-rights claims beyond the Convention's zone limits and the classification of every high-tide Spratly feature it assessed as a mere rock [7] [7]. Its significance, Schoenbaum stresses, extends well beyond the South China Sea to "contested maritime rights and responsibilities in other maritime areas of the world" [7].

Yet the analysis is shadowed by a procedural discomfort that goes to the award's legitimacy rather than its logic. China refused to appear, maintaining that disputes should be resolved "only through negotiation and consultation, not through arbitration," and consistently denied the tribunal's jurisdiction while declining to argue the point [7]. Under Annex VII, non-appearance does not defeat jurisdiction and the tribunal may proceed — but, Schoenbaum notes, there could be no default judgment either: unlike a domestic court entering judgment for failure to appear without deciding the law, this tribunal "had the duty of deciding the legal issues of the case without the benefit of China's counterarguments" [7]. That duty created a structural asymmetry he does not soften: the tribunal had itself to "'make up' what arguments China could be presumed to make," and then evaluate those self-formulated arguments against the Philippines' case, argued by an outstanding legal team — "a lawyer's dream—litigating a case before a court that will hear only one side of the case," in which "[t]he outcome in such a case is not hard to guess" [7]. A second advantage compounded the first: China's non-participation also shaped the very make-up of the tribunal under the Annex VII constitution rules [7]. None of this implies conscious bias — Schoenbaum is explicit that there was none — but "the perception of unfairness surrounding the case may linger, obscuring the landmark rulings of the judgment itself" [7].

The graver criticism is forward-looking. International law should ideally ground peaceful settlement; the Philippines' former foreign minister Albert del Rosario hoped the award would provide "a basis for further talks and cooperation by all parties" — but Schoenbaum warns "quite the opposite may happen": the judgment "may have the effect of deepening and exacerbating the dispute," even increasing the danger of conflict [7]. The mechanism of that failure, in his reading, is the tribunal's treatment of the parties' future conduct (Merits Award paras. 1191–1201): confronted with "this crucial question, the tribunal—to put it bluntly—decided to 'punt,' stating only, 'the Tribunal is not persuaded that it is necessary or appropriate for it to make any further declaration'" (para. 1201) [7].

For this report, the episode completes the cartographic argument from the other end. The Convention's machinery — baselines, feature classification, zone formulae, compulsory arbitration — performed exactly as designed: it took an undrawable line and an unclassifiable claim and produced 479 pages of determinate law. What it could not produce was the absent party's assent, and a judgment perceived as one-sided, however rigorous, enters the world as another contested document — a map of the law that one of the two states refuses to read. The phantom island was cured by sending a ship; there is no equivalent voyage that can ground-truth a verdict.

A plan for peaceful resolution of the South China Sea disputes

A judgment is not a settlement, and the essay on the 2016 Award pivots from critique to construction on exactly that point. The tribunal's judgment "contains most of the elements necessary to resolve the disputes, but the tribunal left open the all-important issue of the 'future conduct of the parties'" (Merits Award paras. 1191–1201) — and, as things stand, "China will never expressly accept any aspect of the tribunal's judgment because, in China's view, the Award

is one-sided and ignores China's legitimate rights" [7]. The proposed three-point plan therefore accepts most of the tribunal's rulings as the basis for settlement while expressly not requiring China to accept the judgment: (1) recognition of China's non-exclusive traditional fishing rights in the South China Sea on the basis of the nine-dash line; (2) determinations of sovereignty over the key features; and (3) a multilateral agreement among all concerned States for managing the Sea's living and non-living resources and protecting its environment [7].

The first step operationalizes the previous chapter's legal argument. The riparian States impacted by the nine-dash line — the Philippines, Malaysia, Brunei, Indonesia and Vietnam — should formally offer China good-faith negotiations "to rehabilitate the nine-dash line as a right of China, under international law recognized by UNCLOS Article 62(3), to living resources in their respective exclusive economic zones" [7]. The gesture is designed as magnanimity with a legal core: it recognizes rights the tribunal expressly denied, and it "would also save China's face by giving international status to the nine-dash line, although as non-exclusive traditional fishing rights" [7]. The concession is bounded by Article 62 itself — the coastal States "would still retain full regulatory control of the living resources within their EEZs" — and it is anchored in China's own scholarly voice, Gao and Jia having stated that the line "is not intended to assert historic sovereignty over... sea areas... beyond what is allowed under international law" [7]. The second step sorts sovereignty, which the tribunal deliberately never touched. The groups at issue are the Spratly Islands, the Paracels, Scarborough Shoal and the Pratas Islands; the all-"rocks" holding makes the task manageable because the features generate only territorial seas, and because international law "does not allow these features to be treated as collectivities for purposes of establishing sovereignty" — each claimant must present individual evidence for individual high-tide features [7]. Sovereignty is not a maritime question and UNCLOS compulsory settlement does not reach it, so the parties must negotiate or agree to arbitrate, under the classical jurisprudence of the Island of Palmas (1928), Eastern Greenland (1933) and Clipperton Island (1931) cases, where discovery, historic use and occupation, historic conquest, treaty transfer, recognition and prescription all play their parts; the author adds that China should support Taiwan's claims alongside its own, so that resolution might even enhance cross-strait cooperation [7].

The third step gives the settlement an institution. The South China Sea "qualifies as a 'semi-enclosed sea' under UNCLOS Article 122 since it is connected to other segments of the ocean by narrow outlets," and Article 123 mandates that bordering States cooperate to create "an appropriate regional organization" for four purposes: managing the living resources, implementing environmental protection rights and duties, coordinating scientific research, and cooperating with interested States [7]. On the essay's reading, China and its neighbours accordingly "have a duty under international law to create a multinational organization for the management of the resources and to protect the environment of the South China Sea" — the long-term solution being "an ongoing process to manage the South China Sea on a comprehensive basis," as commentators had long urged (Joyner 1998) [7]. The conclusion is candid about the Award it builds on: a landmark clarification that enhances freedom of navigation and environmental enforcement, yet "one-sided in favor of the Philippines, neglecting or minimizing the rights of China and Taiwan" — through no fault of the tribunal, since China refused to participate — and therefore "unlikely to have much effect with regard to settlement of the South China Sea disputes" [7]. A great power's loss of face "can be dangerous for the world"; China is technically bound but will never expressly affirm the judgments; the three-point plan exists "to allow China to save face in this controversy," so that the rulings may be accepted for what they are — "the basis for a settlement" [7].

As a specimen of this paper's subject, the plan is the drawing put to its final use. The nine-dash line, void as a title, is recycled as the cartographic basis of a negotiated fishing right; the tribunal's feature-by-feature rocks jurisprudence converts an unsortable archipelago-wide claim into a finite list of provable titles; and the Convention's own semi-enclosed-seas cooperation clause supplies the institution the judgment could not. Whether any of it happens is politics. But the structure is the Convention's deepest habit, visible from baselines to Aqaba: take the map as the parties have drawn it, price every claim by the rulebook, and build the settlement out of the priced pieces.

China – Japan – Philippines triangle

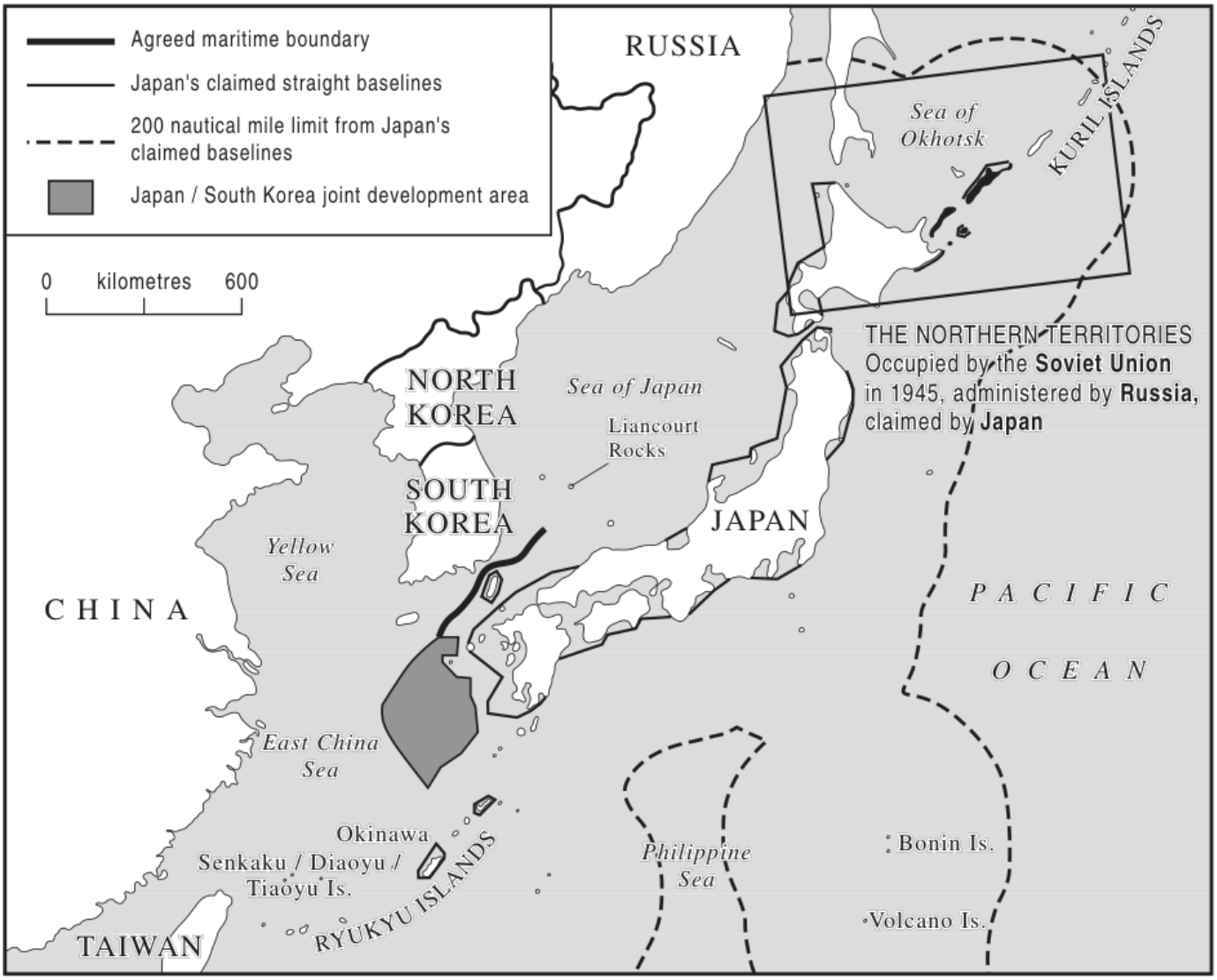
The Asian Rim briefing's largest single construction is the undelimited East China Sea, where the equidistance instrument is run twice — once for each possible owner of the Diaoyu Dao/Senkaku Islands — and the space between the two lines is the dispute made visible. The features themselves are worthless as territory: the islands and rocks "have little value as territory that might be settled, cultivated or mined," and their real value is that "they permit claims to the surrounding seas and seabed which respectively might contain valuable fish stocks and reserves of natural gas and oil" [6]. From islands a 200-mile EEZ may be claimed; from rocks only if Article 121(3)'s habitation or economic-life conditions are met — and here the briefing takes its generous reading to its conclusion: "in our estim-

ation all these features may be used as basepoints in accordance with Article 121," because some are islands and the rest are rocks close to them, and Article 121(3), badly drafted as it is, was designed (per Charney 1999) to prevent extended claims from "fragments of rock standing above high tide in the open sea," a category into which none of the Diaoyu/Senkaku rocks falls [6].

If China owned all the islands, the central section of the equidistance line would trend southwards for about 145 nautical miles from the northern section's terminus, then eastwards for about 170 — generated on the Japanese side by Tori Shima, Nishime Saki, Yaebi Se, Minna Shima, Hirakubo Saki, Iriomote Shima and Yonakuni Shima, and on the Chinese side by Yushan Liedao off the mainland, Chiwei Yu, Nan Xiaodao and Diaoyu Dao in the group itself, and Mien-hua Yu off Taiwan's northeast coast, reaching a southwestern terminus near 25°18′N, 122°52′E equidistant from Mien-hua Yu, Yonakuni Shima and Diaoyu Dao [6]. If Japan owned them all, the line would instead trend southwest about 150 miles and then south about 105 to the same terminus, the Senkaku features now serving as Japanese basepoints against mainland points (Yushan Liedao, Taizhou Leidao, P'i Shan, Beijishan) and Taiwan's P'eng-chia Yu and Mien-hua Yu [6]. The two lines enclose 19,800 square nautical miles (67,800 km²) of almost pure sea and seabed, straddling the junction where the 200-metre isobath divides geological shelf from slope; the western shelf zone is 74 per cent of the island-attached area, both zones prospective for oil and gas and valuable as fishing grounds [6]. The asymmetry of stakes is sharp: the area would be 4.7 per cent of China's EEZ (excluding the Spratlys) but under 2 per cent of Japan's — yet only about 39 per cent of Japan's EEZ is continental margin, against some 80 per cent of China's, so prospective shelf is far scarcer for Tokyo than for Beijing [6]. South of the conjunction, the line runs a further 210 miles to the China–Japan–Philippines tri-junction off the eastern entrance to the Bashi Strait (about 22°10′N, 123°40′E) — the triangle of the section's title — generated almost entirely by Taiwan's east-coast features and offshore islands (Lu Dao, Lan Yu and a string of mainland-Taiwan points) against just two Japanese basepoints, Yonakuni Shima and Haderama Shima [6].

Japan's optimum is the equidistance line itself; China's almost certainly is not — at least for the seabed. "Geologically and geomorphologically the continental margin bounded by the Okinawa Trough is Chinese. It stretches seawards from the mainland coast of China and it has been formed mainly by the filling of marginal basins with sediment provided by Chinese rivers," so that "the imperfect concept of natural prolongation fashioned in the North Sea by the International Court of Justice in 1969 is perfectly illustrated by the continental margin of the East China Sea" [6]. The briefing marshals the Timor Sea precedent: Australia, in China's structural position, won a seabed boundary from Indonesia in 1972 based on geomorphology rather than equidistance, and again in 1989 and 1997 secured control or shared control beyond the equidistance line — notwithstanding Libya/Malta (1985) seeming to reserve natural prolongation for areas beyond 200 miles [6]. China's preference ladder follows: first, a split regime — the water column divided by equidistance, the seabed by the axis of the Okinawa Trough (rare, but Australia–Indonesia and Australia–Papua New Guinea show split boundaries can operate given careful treaties and cordial relations); then seabed lines at the 200-metre isobath, or between the isobath and equidistance; and fourth, a joint seabed zone bounded by equidistance and the Trough axis or the isobath [6].

But the central section cannot even reach the equity argument until sovereignty is settled, and "there is no public evidence to suggest that a solution to this dispute is near" — so the probable future is stalemate: "Japan will continue to occupy the islands but will be unable to make significant use of the waters and seabed that lie within lines of equidistance," a policy of masterly inactivity on both sides [6]. The one live alternative the briefing endorses is a joint development zone for the islands' marine domain, squarely within the provisional-arrangements spirit of Articles 74 and 83, whose merit would be to let exploration establish the hydrocarbon potential — a matter "probably a more urgent matter for Japan than for China" [6]. A working prototype already exists in miniature: the bilateral fisheries agreement of 11 November 1997 circumvented the sovereignty issue entirely, shelving EEZ questions, establishing a jointly controlled provisional sea zone between 30.4° and 27°N (excluding areas within 52 miles of either coast), and simply continuing current fishing activities south of 27°N, around the disputed islands themselves [6]. The triangle's lesson generalizes the Asian Rim chapters: where sovereignty over a handful of rocks can swing 67,800 square kilometres between two lines, the map cannot settle the question — it can only state, with unusual precision, exactly what the question is worth.



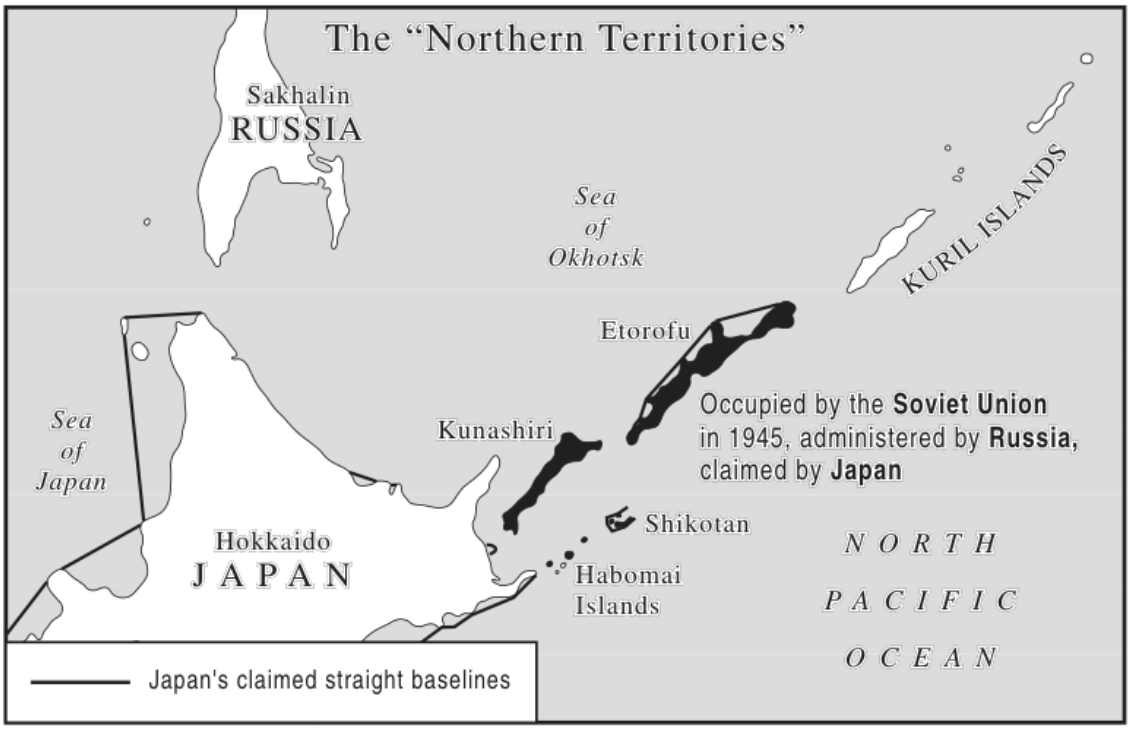


Exhibit. Japan's maritime disputes, including the Northern Territories claimed from Russia and Japan's claimed straight baselines. Source: S0008.

Indonesia's maritime boundaries: Malaysia, Vietnam, Palau, and the Philippines

Indonesia is the archipelagic State in its purest form, and the briefing's longest national chapter follows its unfinished boundaries around the compass. To the northwest, the Indonesia–Malaysia problem is a layered one: a 1969 agreement divided the seabed north of Tanjong Datu, but the water column above it was never delimited. The equidistance line for the water column would follow the 1969 seabed boundary for the first 40 nautical miles, then drift progressively east of it for about 200 miles to a tri-junction with the Spratly features Amboyna Cay and Ladd Reef — Point B, in the vicinity of 5°48′N, 110°52′E — generated by Indonesian basepoints (Tanjong Datu, Pulau Kepala, Pulau Subi Kechil, Pulau Bunguran Besar, Pulau Laut) against Malaysia's Tanjong Datu and Tanjong Sirik; and no factors appear that would let either side call that water-column line inequitable [6]. The seabed, however, strands Malaysia with a geometric puzzle. Its 1969 boundary terminates at Point A; the water-column line terminates at B; and Malaysia's own 1979 claim chart continued the seabed boundary along a segment A–D — yet the triangle B–C–D so created, some 3,200 square nautical miles with depths to 1,750 metres, "lies closer to Ladd Reef, which Malaysia does not claim, than to Pulau Kecil Amboyna, which Malaysia does claim but does not occupy" [6]. The briefing offers two cleaner answers: join A and B by a geodesic (Indonesia has already relinquished the seabed east of A), or run east along the parallel through A to the line B–C and then follow B–C southeastwards to B — on the logic that near the 1969 terminus Malaysia should claim no seabed north of it, and none that belongs to countries other than Indonesia [6]. Split-level delimitation — one line for the seabed, another for the water — turns out to generate its own second-order disputes at the seams.

Across the Celebes Sea, Indonesia faces the Philippines, and the geometry is hostage to two contingencies. Both are archipelagic States, no part of the sea is more than 200 miles from a coast, and both claim 200-mile EEZs; but Indonesia claims a 12-mile territorial sea while the Philippines "appears to claim all waters within its treaty limits as territorial waters which means that they are up to 285nm wide at one point" [6]. Whether a territorial sea boundary is needed at all therefore depends, first, on the International Court of Justice deciding whether Pulau Ligitan belongs to Indonesia or Malaysia, and second, on whether Manila abandons the treaty-limits claim for a 12-mile sea measured from its archipelagic baselines — if Ligitan is Malaysian and the treaty limits lapse, no territorial sea boundary is required [6]. The equidistance line itself, drawn graphically on Admiralty charts BA 4507 and BA 943, runs from the northwest Celebes Sea (origin at Point S or Q, again depending on Ligitan) to the equidistant tri-junction of Indonesia, Palau and the Philippines near 6°40′N, 129°30′E — the Palau boundary being, at this stage, a point where three equidistance systems meet — and measures about 760 nautical miles if Ligitan is Indonesian against 670 if it is Malaysian [6]. It threads two narrow southern passages of Mindanao: only 38 miles separate the Philippine island of Tinina Balut from Indonesia's Pulau Marore, and 50 miles separate Pulau Miangas from Cape San Agustin [6].

Miangas concentrates the chapter's jurisprudence. The island — the Island of Palmas of the famous 1928 arbitration, in which it was confirmed as Dutch, hence Indonesian, against the United States acting for the Philippines — lies inside the Philippines' claimed treaty limits, which is one reason negotiations "will be very difficult… so long as the Philippines insists on claiming all the waters within its treaty limits" [6]. Philippine academics and officials have floated curtailing Indonesia's entitlements from Miangas as Britain's were restricted from the Channel Islands; the briefing dismantles the analogy point by point — the Channel Islands sat in a narrow sea where they would have wrenched the equidistance line severely against France, while Miangas "is simply the northernmost feature in a chain of Indonesian archipelagos called Sangihe, Talaud, Kawio and Nanusa trending northwards" from Sulawesi and Halmahera, claimed by an archipelagic State entitled to archipelagic baselines [6]. Since both parties are archipelagic States legitimately claiming such baselines, "there seems to be little reason why these baselines (rather than the low-water line) wouldn't be used as the basis for generating an equidistance line" — it is in both States' interest to see archipelagic baselines given full weight elsewhere [6]. And the prize is small: without Miangas the Philippines would gain 3,300 square nautical miles, 0.6 per cent of its claimable EEZ; Miangas delivers Indonesia 0.2 per cent of its own [6]. No strong inequity argument attaches to the line itself [6]. Finally, to the west, Indonesia and Vietnam face each other across the Natuna Sea as opposite States — both claiming 12-mile territorial seas and 200-mile EEZs, Vietnam adding a 24-mile contiguous zone — but their nearest coasts are over 240 miles apart, so any boundary between them will separate EEZs only: the simplest configuration in the chapter, and the one where equidistance operates with the fewest encumbrances [6]. Indonesia's boundaries thus display the full grammar of the region: a seabed line outliving its water column, a boundary whose length changes by 90 miles with a court's answer about one small island, an archipelagic neighbour whose colonial treaty polygon swallows the other's island whole, and — in the tri-junction with Palau — three equidistance systems converging on a point none of the three has yet agreed to visit.

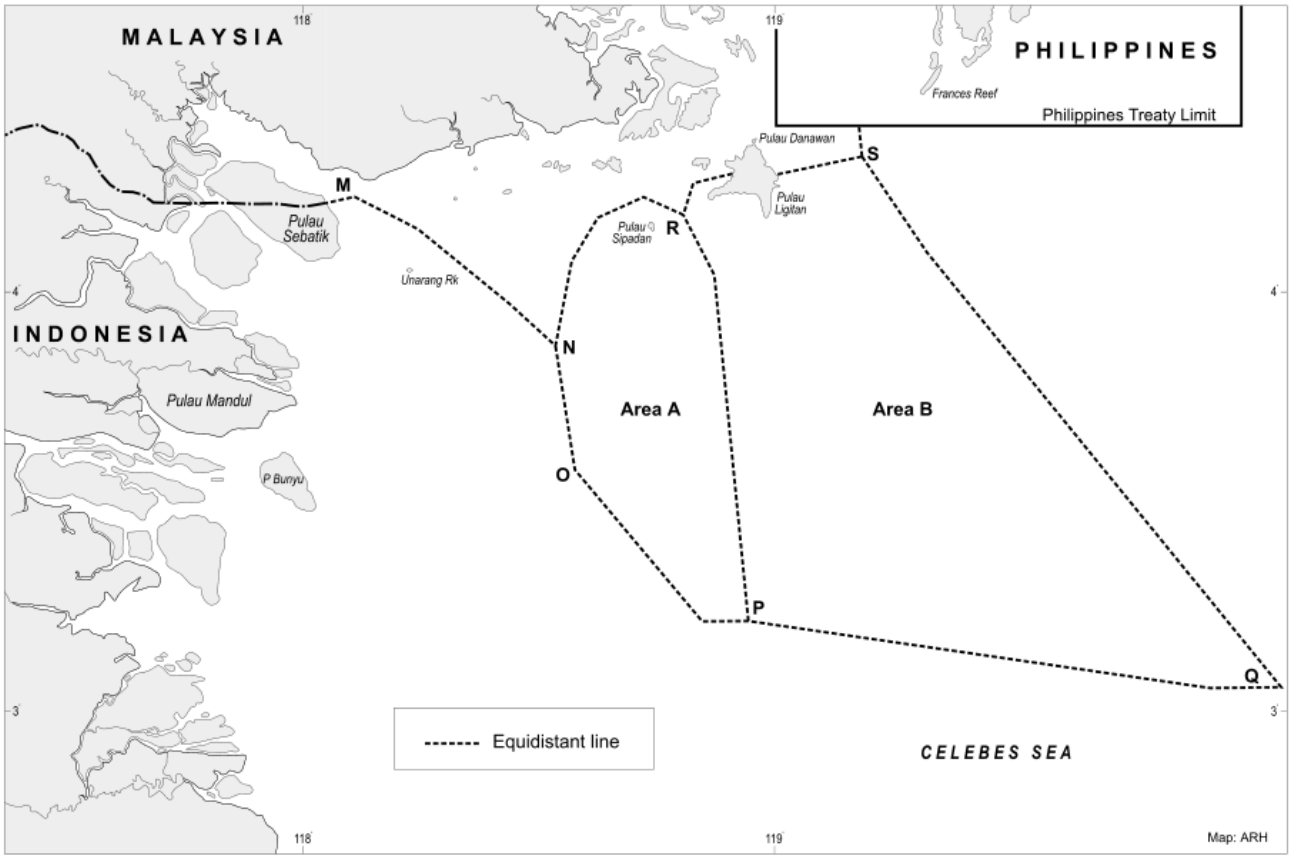


Exhibit. The Indonesia-Malaysia equidistance line in the Celebes Sea, with the two areas marked A and B on the chart. Source: S0011.

Cambodia – Vietnam historic waters area

In the Gulf of Thailand, the equidistance line between Cambodia and Vietnam is a biography of specks. Running seaward from the land-boundary terminus, it is controlled in turn by a low-tide elevation near the entrance to the Kompong Trach River, by Hon Dung, Hon Keo-Ngu'a and its offshore rock, by Rocher Déchiré and Rocher Rosita on the Cambodian side, and by the Mui Nai peninsula and points among the Pirate Islands on the Vietnamese; farther out, Cambodia's Îlots du Sud Est divert the line southwards, Depond Reef pushes it further south still, Vietnam's Tho Chu group turns it southwest, and the furthest offshore section is governed jointly by Cambodia's Poulo Wai group and Vietnam's Tho Chu [6]. Around Phu Quoc the controlling points multiply into dozens of rocks, reefs, and cape tips on both sides [6]. No stretch of this boundary is drawn from anything a layperson would call a coast.

One feature deserves its own paragraph, because it is this report's thesis in miniature. Depond Reef "is a low-tide elevation whose charted position is almost exactly 12nm from the nearest feature in the Poulo Wai island group (British Admiralty chart 3879)" — but "[h]owever, its position is approximate and it would only be a valid basepoint for territorial sea delimitation purposes if it lies within 12nm of the nearest island or mainland coast" [6]. Everything turns on the word if. Recall the rule of Article 13: a low-tide elevation anchors a baseline only within a territorial-sea breadth of land. Depond Reef sits, on the chart, almost exactly on that legal knife-edge — inside, it drags the equidistance line south and enlarges Cambodia's side; outside, it is nothing, and the line retreats to Poulo Wai control. The briefing accordingly plots both alignments (Figure 6): two different maritime boundaries, separated by the survey error of a single reef whose position nobody has fixed precisely [6]. Sandy Island asked whether a charted feature exists; Depond Reef asks whether an existing feature is close enough to matter, and the chart cannot answer either question to the precision the law requires.

The political history layered over this geometry is equally double. When (South) Vietnam and Cambodia advanced continental shelf claims in 1971 and 1972 respectively, both claimed sovereignty over Phu Quoc and its islets, the Poulo Wai group, and the Tho Chu group, so that "their claims overlapped to a considerable degree" from the same basepoints; neither claim has ever been officially retracted [6]. The island sovereignty dispute was nonetheless resolved in 1982 — Poulo Wai confirmed as Cambodian, the other islands as Vietnamese (Schofield 1999) — and the

same settlement manufactured the anomaly in this section's title: "a roughly oblong-shaped area of joint (and still disputed) 'historic waters' projecting into the Gulf of Thailand," whose seaward southwestern limit is "the straight baseline linking the Tho Chu archipelago and Poulo Wei [Wai] Island" [6]. Both states trimmed their straight-baseline claims in 1982 to terminate at Poulo Wai and Tho Chu respectively, joining at "a floating 'Point O'" on the line between them — "floating" because "[i]t was agreed that the precise position of Point O would be determined by mutual agreement," which is to say it remains, decades later, a point defined by a promise [6].

Beneath even that lies a colonial stratum: when both states were ruled by France, a dispute over islands between Ha-tien Province (the future Vietnam) and Kampot Province (Cambodia) was settled by Governor-General Jules Brevié's decision of 31 January 1939 — the Brevié Line, which the briefing still plots as one of the candidate constructions (an equidistance line run along it, with 3 km around Phu Quoc) [6]. A 1939 administrative ruling inside one empire thus survives as a live geometric option between two sovereign states. The historic waters area is, in the end, three maps laminated together — a colonial file decision, a 1982 joint-sovereignty compromise with an unlocated point, and an equidistance construction hostage to the approximate position of one reef — and the boundary exists in the overlap of all three.

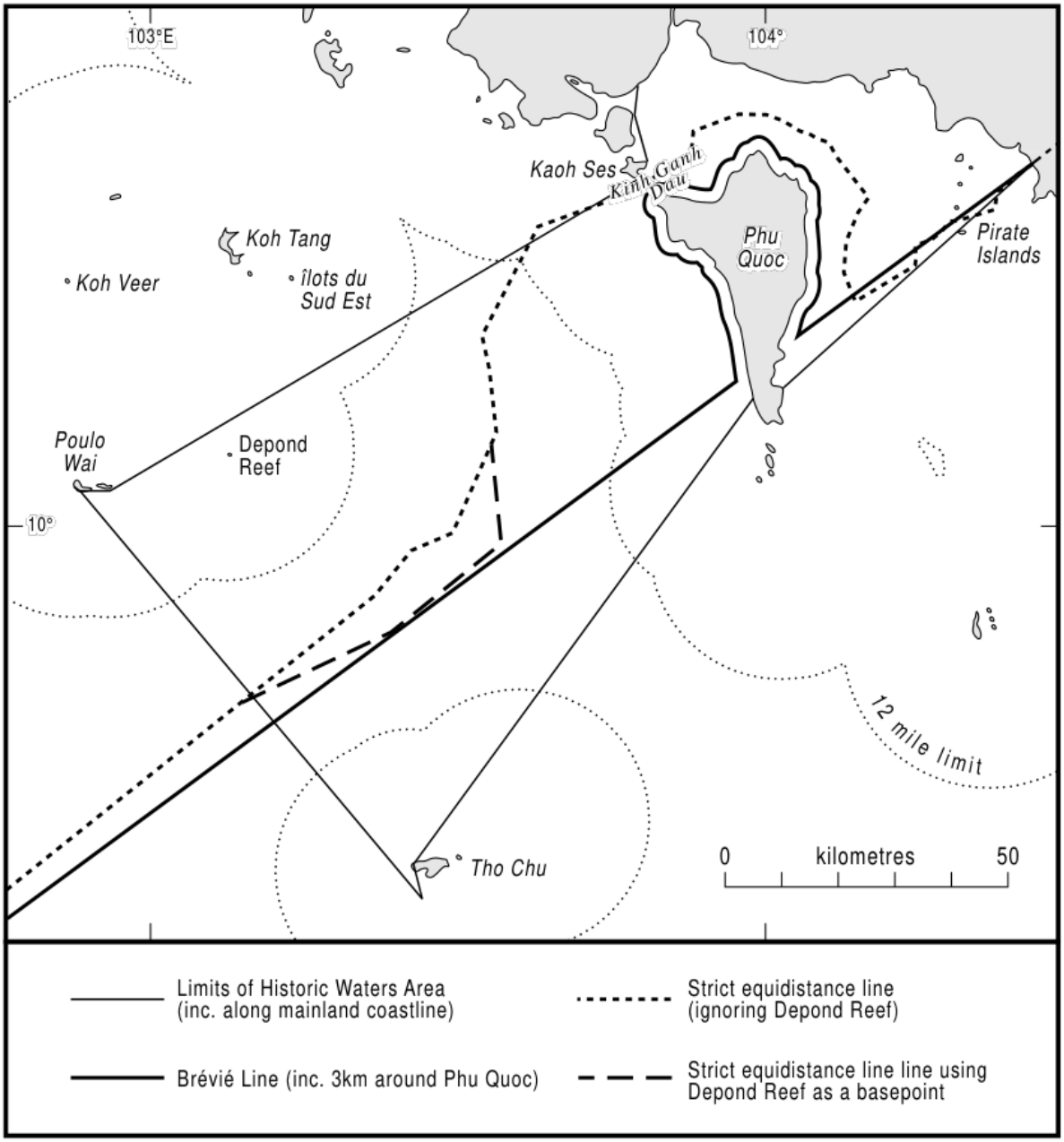


Exhibit. The Cambodia-Vietnam historic waters area around Phu Quoc, showing the Brevie Line and strict equidistance alternatives. Source: S0011.

Japan – North Korea – South Korea and China – North Korea

In the Sea of Japan, a single pair of islets reassigns two countries' neighbours. The Liancourt Rocks — Takeshima to Japan, Tok Do to South Korea, "a pair of islets and scattered rocks" contested between them — sit at the hinge of four potential boundaries, and the briefing's method of dual construction, drawing the equidistance lines for each sovereignty in turn, shows the rocks deciding not merely where a boundary runs but who borders whom [6]. The dispute already truncates one concluded boundary: Japan and South Korea delimited a seabed boundary in the Korea Strait and a joint development zone in the East China Sea in 1974, but the seabed line terminated at the Strait's eastern entrance, Point 35 (36°10′N, 131°15.0′E), precisely "because the two countries dispute sovereignty over the Liancourt Rocks" [6]. It also contaminates a third party's concluded boundary: the 1986 North Korea–Russia EEZ delimitation terminates at 39°39.3′N, 133°45′E, which lies 12 nautical miles southwest of the strict equidistance tripoint (about 39°49′N, 133°54′E) among Russia's Mys Povorotnyy, the Liancourt Rocks and Japan's Hegura Shima — the Rocks being 18 miles nearer that terminus than Mys Povorotnyy, and Hegura Shima 5 miles nearer — so the briefing assumes Japan and South Korea "will not accept the selected terminus of the North Korea-Russia EEZ delimitation as a starting point for any boundary with Russia," sitting as it does some 12 miles to Russia's advantage [6] [6].

The two constructions diverge immediately. Assuming the Rocks are South Korean, the equidistance line — Korean basepoints Changgi Gap and the Liancourt Rocks against Japanese Mi Shima, Hino Misaki, Mitabe Saki, Okino Shima near Dogo and Hegura Shima — runs about 300 nautical miles to the tri-junction of Japanese, South Korean and Russian claims at about 39°49′N, 133°54′E, equidistant from the Rocks, Hegura Shima and Mys Povorotnyy; in this world South Korea holds a short boundary with Russia (Points C–D of the briefing's Figure 16) [6]. Assuming the Rocks are Japanese, the line — Japanese basepoints Mi Shima and the Rocks against Korean Changgi Gap and Ullung Do — runs about 230 miles to a different tri-junction entirely, that of Japan, North Korea and South Korea at about 39°41′N, 132°33′E, equidistant from Ullung Do, Tok Do and North Korea's Musu Dan [6]. The elegant consequence deserves emphasis: "if Japan owns Liancourt Rocks it has a potential maritime boundary with North Korea" — the short B–C segment up to a Japan–North Korea–Russia tri-junction near 39°54′N, 133°11′E — "while if South Korea owns Liancourt Rocks it has a potential maritime boundary with Russia"; the quadrilateral enclosed by the two constructions and their tri-junctions measures about 15,000 square nautical miles [6]. A sovereignty answer over two islets thus does not shift a line; it swaps the region's adjacency matrix, giving Tokyo a boundary with Pyongyang or Seoul a boundary with Moscow.

The inter-Korean line in the Sea of Japan inherits the same dependency. Drawn from its origin near Suwon Dan, where the land boundary reaches the coast, its seaward terminus is the Russia tri-junction (39°54′N, 133°11′E, equidistant from Musu Dan, the Rocks and Mys Povorotnyy) if the Rocks are South Korean — some 174 miles from Tok Do — but the Japan tri-junction (39°54′N, 132°33′E, equidistant from Ullung Do, the Rocks and Musu Dan) if they are Japanese [6]. North Korea's basepoints throughout are merely Suwon Dan, Nan Do and Musu Dan, and since "no North Korean feature has a pronounced influence on the alignment of the line of equidistance," South Korea is unlikely to plead inequity; Pyongyang's available argument runs the other way — that the Rocks and Ullung Do, lying respectively 70 and 115 miles from the mainland, should be discounted when a boundary is delimited [6]. On the Yellow Sea side the two Koreas are adjacent States, each claiming 12-mile territorial waters and 200-mile EEZs, with the negotiation complicated from the first mile because "both countries claim islands that have been under United Nations control since the end of the war in Korea" — and it is there, in the Yellow Sea's island-dotted shallows, that North Korea's other neighbour, China, faces the same undelimited geometry across the Yalu approaches [6].

The briefing's equity discussion for the whole corner is brief because the obstacle is not equity but sequence: "it seems unlikely that a maritime boundary could be delimited before the sovereignty of Liancourt Rocks is settled by agreement" [6]. Its proposed solvent is by now familiar — leave sovereignty unsettled and delimit a joint development zone occupying the area within the various equidistance lines, which "would give symmetry to their 1974 boundary arrangements that has a joint development zone in the East China Sea" [6]. The fallback possibilities are equally characteristic: a withdrawal of claim might be purchased by agreeing the Rocks ground no claims beyond territorial waters, and, absent the sovereignty dispute, the non-owner would predictably argue the Rocks' effect is disproportionate and they should be discounted or ignored [6]. Few chapters state the paper's thesis as cleanly. Equidistance is presented throughout the Asian Rim study as an impartial instrument; here the instrument works perfectly and answers everything except the one question put to it — because the question was never geometric. It was, and remains, whose two islets these are.

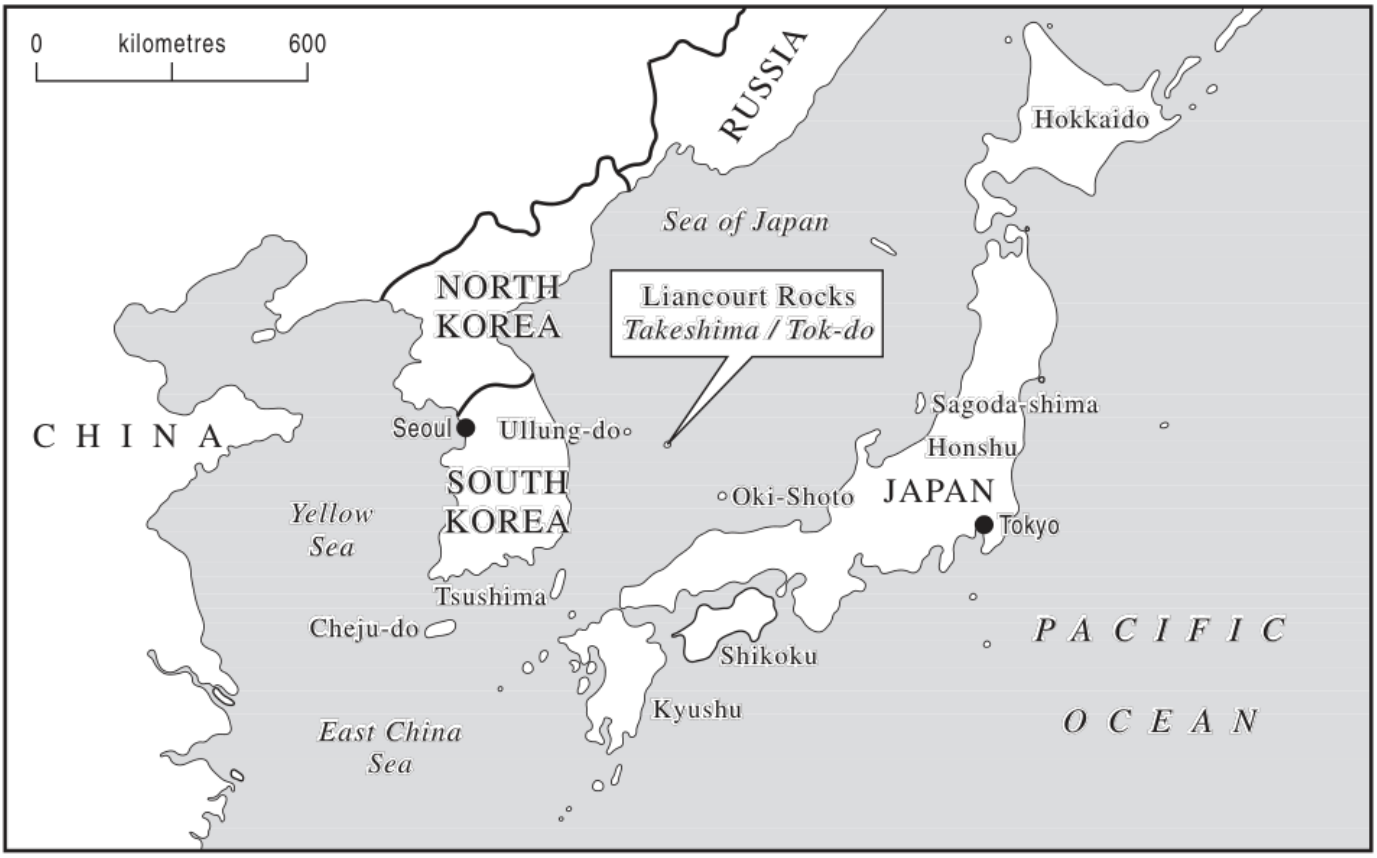


Exhibit. The Liancourt Rocks (Takeshima/Tok-do) in the Sea of Japan, between the Korean Peninsula and Japan. Source: S0008.

The Red Sea and the Gulf

Maritime claims of Red Sea countries: claimed versus delimited

The Red Sea briefing's balance sheet is a study in asymmetry: a full suite of claims, asserted early and expansively, against a delimitation record that remains mostly blank. On the claims side, "most of the Red Sea countries claim the full suite of maritime rights and jurisdictional zones" [5]. All the littoral states except Eritrea and Israel are parties to UNCLOS, though Egypt, Saudi Arabia, Sudan and the two Yemens all made statements on signature or ratification [5]. All claim a 12-nautical-mile territorial sea except Jordan, which maintains a 3-mile claim; Eritrea, which has made no explicit territorial sea claim, "can be assumed to have inherited Ethiopia's 12nm claim" as Ethiopia's successor state (independence 27 April 1993), coastline and claims alike [5] [5]. Djibouti, Egypt and Yemen claim contiguous zones to 24 miles, while Saudi Arabia and Sudan claim contiguous jurisdiction only to 18 miles; and Djibouti, Egypt and Yemen (PDRY) have made explicit continental shelf claims notwithstanding Article 77(3)'s rule that shelf rights need no proclamation, the same three also claiming EEZ rights [5]. Practice outruns paper in both directions: Eritrea has never formally claimed EEZ jurisdiction, yet the Eritrea–Yemen tribunal delimited a boundary between the two States "with reference to their territorial seas, continental shelves and EEZs" [5]. Even the declarations carry disputes inside them — the Yemen Arab Republic's signature declaration asserting sovereignty over all Red Sea and Indian Ocean islands that had been its dependencies under Turkish administration drew a formal Ethiopian protest in November 1984, an early paper skirmish in the island fight the tribunal would later settle [5]. Yemen itself is a merged claimant: the Republic of Yemen was established on 22 May 1990 from the YAR and the PDRY, with both predecessors' treaties remaining in force [5].

Against that full-claims landscape, the briefing's status table records thirteen potential boundaries, of which three are delimited and one is substituted by a resource regime. The concluded entries are Eritrea–Yemen (opposite coasts, continental shelf, 380 km, settled by arbitration in 1999), Israel–Jordan (adjacent, territorial sea, 12 km, 1996) and Saudi Arabia–Yemen (adjacent, territorial sea/continental shelf, 2000), with Saudi Arabia–Sudan (opposite, continental shelf) governed since 1974 not by a line but by a joint development zone [5]. Everything else in the table reads "Unresolved": Djibouti–Eritrea (adjacent, territorial sea, 22 km), Djibouti–Yemen (opposite, territorial sea, 41 km), Egypt–Israel (adjacent, territorial sea, 5 km), Egypt–Jordan (opposite, territorial sea, 10 km), Eritrea–Saudi Arabia (opposite, continental shelf), Eritrea–Sudan (adjacent, territorial sea) and Jordan–Saudi Arabia (adjacent, territorial

sea, 8 km) [5]. Egypt–Saudi Arabia — opposite coasts, the region's heavyweight pairing, comprising 159 km of territorial sea boundary in the Gulf of Aqaba plus territorial sea and shelf reaches in the Red Sea proper — is unresolved, and Egypt–Sudan is listed with the table's most honest status: "In disputed," hostage to the Halaib Triangle [5].

Read as a whole, the table inverts the technician's expectations, and the inversion is this paper's Red Sea thesis. The shortest, geometrically trivial boundaries — five kilometres between Egypt and Israel, eight between Jordan and Saudi Arabia, ten between Egypt and Jordan — remain undrawn, because their endpoints are tripoints entangled with neighbours and with politics, while the boundaries that were concluded arrived by the heavy machinery of statecraft rather than by cartography: a peace treaty (Israel–Jordan 1996), a comprehensive settlement after decades of desert-frontier dispute (Saudi Arabia–Yemen 2000), an arbitration preceded by armed conflict over the Hanish islands (Eritrea–Yemen 1999), and, where even that was impossible, a 1974 joint development zone in place of a line (Saudi Arabia–Sudan) [5]. Claimed-versus-delimited is thus the region's diagnostic ratio: claims cost a decree; boundaries cost a relationship. The chapters that follow take up the two straits through which the Red Sea's undelimited waters must be navigated — Bab al-Mandeb and Tiran — where the operative question is not who owns the water but who may pass through it, and on what terms.

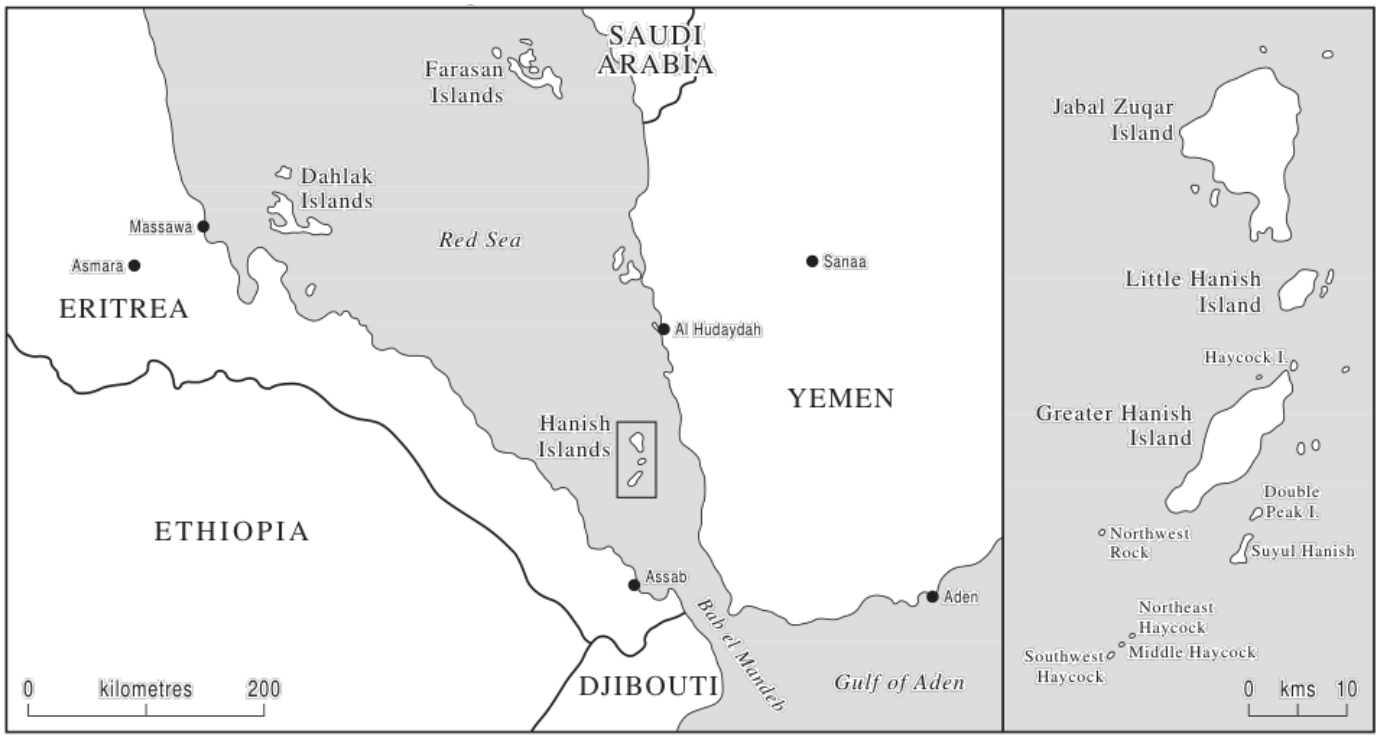


Exhibit. The Hanish Islands in the southern Red Sea, between Yemen and Eritrea, with an inset of Greater and Little Hanish. Source: S0008.

The Saudi Arabia – Sudan common zone

The Red Sea's most original boundary instrument contains no boundary at all. In 1964 and 1966, surveyors found three depressions deeper than 2,000 metres in the central Red Sea, between 21°10′N and 21°30′N — the Atlantis II Deep, Discovery Deep and Chain Deep — whose metalliferous sediments "hold high concentrations of zinc, copper, silver, gold, and some trace elements" [5]. More promising locations followed, and the geology posed a legal problem no equidistance line could solve: the richest seabed in the sea lay in its deep central trough, roughly equidistant from the Saudi and Sudanese coasts, in water far beyond either state's unilateral reach — a prize that any median-line delimitation would have to award, arbitrarily, to one side or split through the middle of the deposits themselves.

The opening moves were competitive. In 1972 "Saudi Arabia passed legislation claiming trusteeship for all non-living resources in the Red Sea, until its coastal states could agree on ownership and development of these resources" — a unilateral claim dressed, notably, as custodianship pending agreement rather than as sovereignty [5]. Sudan answered commercially, concluding an agreement with the exploration company Preussag AG in 1973 for development of nearshore deposits [5]. Then, instead of litigating the overlap, the two states negotiated: they "concluded an Agreement for Joint Exploration in 1974, and formed the Saudi-Sudanese Red Sea Commission in 1975" [5].

The architecture they built deserves attention, because its geometry is bathymetric rather than cartographic. “The agreement provided for national zones from shore out to the 1,000m isobath and a common zone in the central Red Sea between the two countries” — jurisdiction allocated not by coordinates or equidistance but by a depth contour, each state sovereign over its shallows and the deep trough held in common [5]. The Commission was charged with the regime’s working parts: “delimiting the boundaries of the common zone, reviewing exploration and exploitation licenses, and supervising the common zone” [5]. As institutional design this is the joint development zone in its purest form — the resource unit defined by geology, the boundary question dissolved into a licensing question, and both claimants converted from rivals into co-owners. It predates, and in conception exceeds, most of the provisional arrangements the law of the sea would later commend to deadlocked neighbours.

Its practice, however, is the paper’s familiar anticlimax. “The Commission has not formally delimited the common zone boundaries along the 1,000m isobath or specified the northern or southern limits of the zone” — a quarter-century after its creation, the zone’s own edges remained undrawn, the one delimitation the regime existed partly to perform [5]. The Commission did undertake exploration activities and a programme to develop the technology for working the sediments, but “more recently, the Commission appears to be moribund, perhaps because the commercial viability of exploiting the sediments, given their great depth, remains in question” [5]. The institution did not fail legally; it starved economically — the metals stayed two kilometres down, the prices never justified the engineering, and a joint regime with nothing to license slowly stopped meeting.

The common zone’s significance for this paper is the contrast it draws with everything around it. In the same sea, Eritrea’s president could confirm in 1996 that the maritime boundary with Sudan would simply wait on better relations, and that no negotiations with Saudi Arabia had even occurred [5]; the Eritrea–Yemen tribunal, when it finally drew its 380-kilometre line of 29 turning points in December 1999, “stopped short of putative tripoints with Djibouti in the Bab al-Mandeb and with Saudi Arabia, north of Jabal al-Tayr” [5]. Lines stop where third parties begin; zones need no tripoints. The Saudi–Sudanese answer of 1974 — share the trough, split only the shallows, and let a commission hold the question open indefinitely — remains the region’s standing proof that delimitation is a means rather than an end, and sometimes not even the best one. That it worked by never finishing its own map is a fitting emblem for a sea whose charts, as this paper keeps finding, are most honest where they admit what they have not drawn.

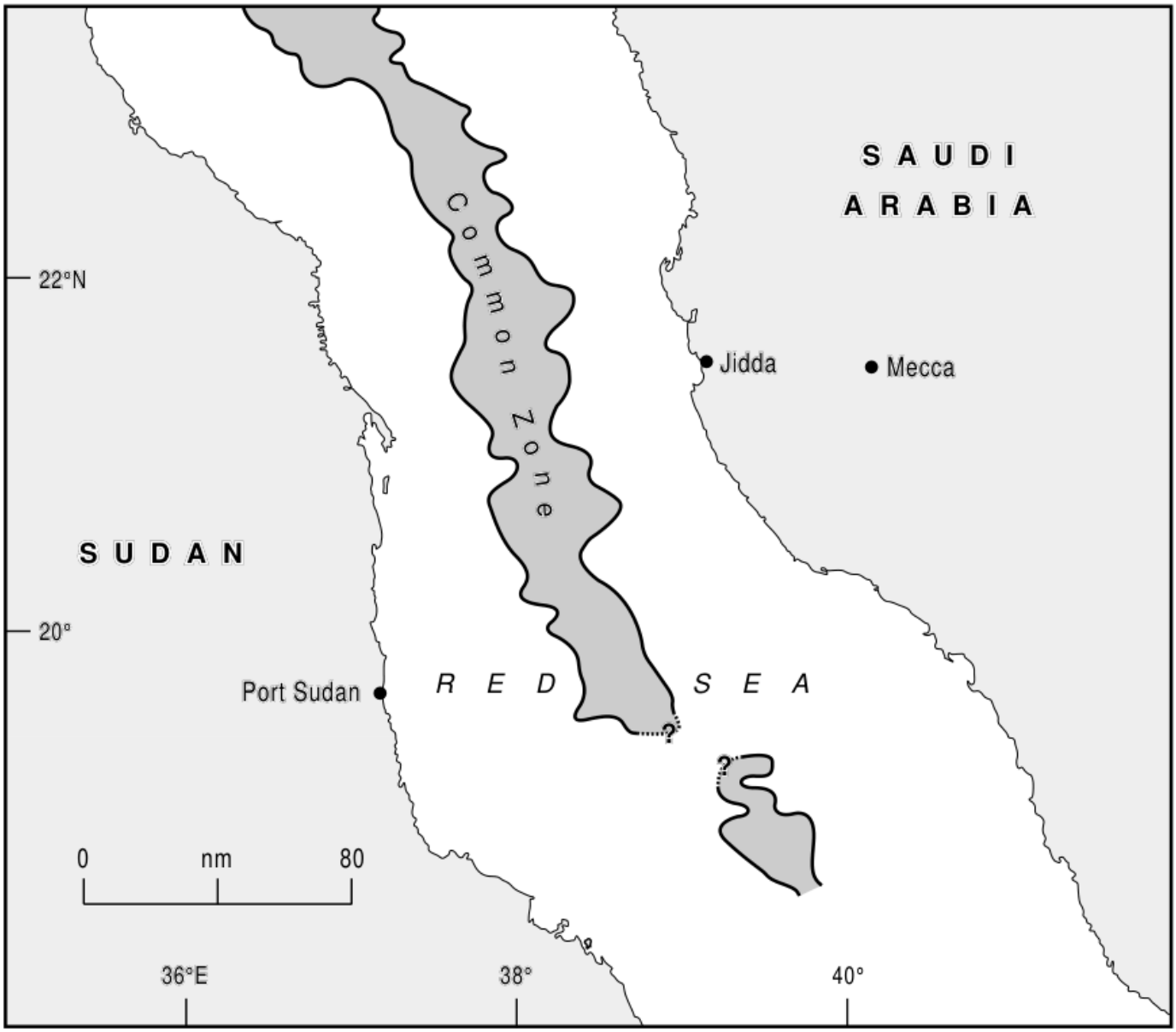


Exhibit. The Saudi Arabia-Sudan Common Zone in the Red Sea. Source: S0009.

Saudi Arabia – Yemen and Egypt – Saudi Arabia

The Saudi Arabia–Yemen boundary is one of the Red Sea's rare finished objects, and its finish line is political, not geometric. Of thirteen potential boundaries in the Red Sea and Gulf of Aqaba, "[o]nly three… have been delimited," and the briefing lists the Saudi–Yemeni case among the region's two "notable… resolved" sovereignty disputes: "Saudi Arabia and Yemen have also resolved their longstanding boundary dispute, including its maritime sector" [5] [5]. The resolution (the 2000 treaty) closed a frontier argument that had run for most of the twentieth century, on land first and at sea as its final component — the maritime line was the tail of a land settlement, which is the Red Sea's recurring pattern: the sea gets divided when the land argument ends, and not before. Its neighbourly contrast is immediate. Saudi Arabia's other Red Sea relationships remain works in progress: with Sudan it substituted a joint development agreement for a line, and with Egypt — the subject of this section's second half — nothing is delimited at all.

Egypt–Saudi Arabia is the region's heavyweight non-boundary: opposite coasts across both the Gulf of Aqaba and the Red Sea proper. The Aqaba reach is geometrically the simplest water in this report. "Because the Gulf of Aqaba is less than 26nm wide, most boundaries within it will separate the territorial seas of the littoral states"; the coastline "is generally smooth, with no significant islands," and "[g]iven its simple geometry, median line boundaries would appear to divide the Gulf equitably" [5]. Only at the gulf's wider southern end would "small strips of continental shelf/EEZ for Egypt and Saudi Arabia" arise [5]. The proportionality arithmetic is equally tame — in the northern gulf, coastline shares run Egypt 23%, Israel 4%, Jordan 5%, Saudi Arabia 19% [5]. And yet the boundary does not exist, for three stacked reasons, none of them geometric.

First, the entrance is occupied by a sovereignty dispute: “at its southern entrance, where there is a latent island sovereignty dispute between Egypt and Saudi Arabia” — Tiran and Sanafir, Egyptian-held, Saudi-claimed, sitting exactly where any Aqaba delimitation must begin [5]. Second, the two states’ baselines are asymmetrically unknowable. Egypt defined its system in 1990 and filed the coordinates; Saudi Arabia decreed straight baselines on 21 February 1958 in terms that treat “any feature not below lowest low-water as an island” — a definition “at variance with UNCLOS language” — and “[n]o charts or coordinates of the Saudi baselines have been published to date. Without further data, it is not possible to determine where these baseline segments are located” [5]. A median line is a function of two baselines; when one party’s baseline exists only as an unpublished decree, the function cannot even be evaluated, let alone negotiated. The briefing’s verdict on priorities is blunt: “[w]hat effect, if any, to attribute to Egypt’s straight baseline system and to the undelimited Saudi straight baseline system may be the most significant issues in talks” — ahead of any question of equidistance geometry [5]. Third, and decisive: “[t]he greatest impediment to dividing the Gulf of Aqaba lies in Israeli-Arab relations” [5]. Every Aqaba line terminates at a tripoint with Israel or Jordan, so the Egypt–Saudi file cannot move faster than the region’s politics.

Set side by side, the two relationships state the Red Sea’s rule. Saudi Arabia–Yemen shows a boundary completed the moment a comprehensive political settlement made it a clause rather than a cause; Egypt–Saudi Arabia shows a boundary that is cartographically trivial — smooth coasts, a sub-26-mile gulf, an equitable median ready-made — and politically impossible, blocked by an island dispute at its mouth, an unpublished baseline on one shore, and a four-state tripoint problem at its head. The map is not wrong here; it is unfinished, and the unfinished parts are drawn in a different ink entirely.

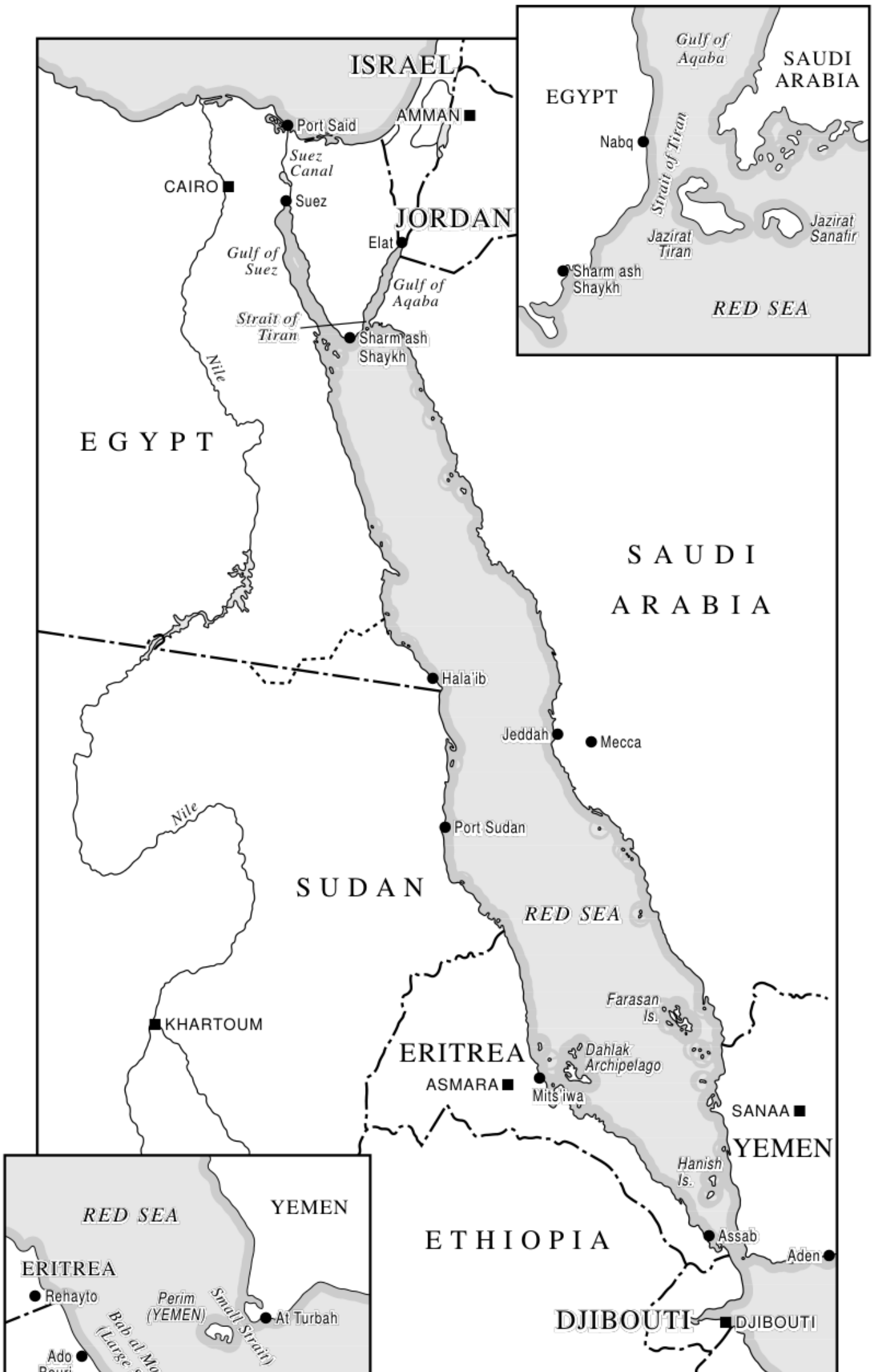

ISRAEL
Port Said
AMMAN
Suez Canal
CAIRO
Suez
JORDAN
Elat
Gulf of Suez
Gulf of Aqaba
Strait of Tiran
Sharm ash Shaykh
Nile
EGYPT
SAUDI ARABIA
Hala'ib
Jeddah
Mecca
Nile
Port Sudan
SUDAN
RED SEA
KHARTOUM
Farasan Is.
ERITREA
Dahlak Archipelago
ASMARA
Mits'iwa
SANAA
YEMEN
Hanish Is.
ETHIOPIA
Assab
Aden
DJIBOUTI
DJIBOUTI
Gulf of Aqaba
SAUDI ARABIA
EGYPT
Nabq
Strait of Tiran
Jazirat Tiran
Jazirat Sanafir
Sharm ash Shaykh
RED SEA
RED SEA
YEMEN
ERITREA
Rehayto
Perim (YEMEN)
Small Strait
At Turbah
Bab al Mandab (Large Strait)
Ado

Exhibit. The Red Sea region, from the Gulf of Aqaba and the Suez Canal to the Bab el-Mandeb. Source: S0009.

The Arctic Frontier: The Northern Sea Route

Defining the Northern Sea Route and its history

The Northern Sea Route defeats the premise of every other chapter in this paper: it is a route whose location is a variable. The IBRU briefing that anchors these Arctic chapters begins with the estimates — "the sailing distance of the entire route can vary between 2,100 and 3,400 nautical miles, depending on the severity of ice conditions" — and treats that variability not as a defect of the data but as "an important element of any definition" [8]. Ivanov and Ushakov define the route as "a series of shipping routes along the Russian coast passing through the seas of the Arctic Ocean… which can be located, depending on ice conditions, in the Russian economic zone, territorial and inland waters, and include near-polar voyages," noting that its geographic definition "has changed significantly with time" [8]. Østreng goes furthest: the route is a "series of individual seas… linked by straits [with] no single set channel to be followed," its shifting course through the ice the defining fact about it [8]. Against these scholarly definitions stands the official one, from the 1991 Regulations: "the essential transportational line of the USSR that is situated within its inland seas, territorial seas (territorial waters), or exclusive economic zone adjacent to the USSR Northern Coast and includes seaways suitable for leading ships in ice," bounded in the west by the western entrances to the Novaya Zemlya Straits and the meridian through Mys Zhelaniya, and in the east, in the Bering Strait, by the parallel 66°N and the meridian 168°58′37″W [8]. Note what the legal definition does: it fixes the route's endpoints and its jurisdictional wrapper, while leaving the sailed line itself to the ice.

The route's history, as the briefing frames it, is a Soviet epic entering a commercial afterlife. The Northern Sea Route "has long played a significant role in Soviet and Russian economic history," the subject of Armstrong's classic The Northern Sea Route: Soviet Exploitation of the North East Passage (1952) and Krypton's studies of its place in Russian economic history before 1917 and in the economy of the Soviet North [8] [8]. By the 1990s the story had changed genre: "the Russian Federation, continuing an initiative begun by the Soviet Union, is attempting to open the Northern Sea Route, the shipping route along the Arctic coast of Siberia from the Norwegian frontier through the Bering Strait, to international commerce," intending eventually to operate it year-round as "an alternative, substantially shorter route from northern Europe to the Pacific Ocean" and to earn hard currency "in exchange for pilotage, icebreaking, refuelling, and other services" [8]. The internationalisation project — pursued through the International Northern Sea Route Programme (INSROP) and a 1992 special issue of International Challenges — had, the briefing notes, "been on the Soviet and Russian agendas, in one form or another, for a quarter of a century" [8] [8].

That project is what makes the route a legal problem rather than merely a logistical one, and it is why Dunlap's 1996 study — the source of these chapters — inventories 43 significant straits of the Northeast Passage one by one [8]. The hinge is Article 37 of the Law of the Sea Convention, in force only since 16 November 1994, which grants transit passage through straits "used for international navigation" [8] [8]. The phrase descends from the Corfu Channel case, where the International Court "abolished the distinction" between indispensability and usage: that most traffic was local, and the strait merely an alternative route, "were irrelevant to the channel's legal status as an international strait" [8]. The trap for Moscow is therefore structural, and the briefing states it exactly: the study concerns the Arctic straits "to the extent that they form a part of the Northern Sea Route, for it is these straits whose legal status may be irrevocably transformed if the Russian government succeeds in attracting foreign shipping to its Arctic shores" [8]. Success is self-defeating under the use criterion — every foreign transit Russia sells makes the straits more "used for international navigation" and Russia's control over them weaker, the same trap Canada faces in the Northwest Passage, where the identical scenario "has already ignited a dispute between Canada and the United States" [8]. The chapters that follow take the inventory strait by strait; this one need only fix the paradox they share: the Northern Sea Route is a line that exists only when sailed, drawn on water that is mostly ice, whose legal character its owner can change — irrevocably, and against its own interest — simply by succeeding in selling it.

Transit passage in the Russian Arctic straits and its conditions

The Arctic chapters of this report begin with a commercial proposition colliding with a legal one. The commercial proposition: carrying forward a Soviet-era initiative, Russia is working to open the Northern Sea Route — the Arctic passage running along Siberia's coast from Norway's frontier through the Bering Strait — to international shipping, with the long-term aim of operating it year-round as a substantially shorter alternative link between northern Europe and the Pacific, earning hard currency by selling pilotage, icebreaking, refuelling and other support services. [8]. The

legal proposition arrived almost simultaneously: the law of the sea "creating, among other things, a right of transit passage that allows, subject to specified conditions, the relatively unrestricted passage of all foreign vessels - commercial and military - through straits used for international navigation," extending even to "submerged transit by submarines and overflight by aircraft, practices with implications for the national security of states bordering straits" [8]. William Dunlap's IBRU study (1996) sits exactly on the collision: it "describes 43 significant straits of the Northeast Arctic Passage, identifying the characteristics of each that are relevant to a determination of whether the strait will be subject to the transit-passage regime" [8]. Russia wants customers; the Convention may entitle the customers to pass without asking.

The conditions turn on definitions, and the definitions have a genealogy. The Corfu Channel court had protected innocent passage through straits "connecting two parts of the high seas"; the International Law Commission codified that there "must be no suspension of the innocent passage of foreign ships through straits normally used for international navigation between two parts of the high seas"; and the 1958 Territorial Sea Convention widened the net to straits ending in "the territorial sea of a foreign State" (Article 16(4)) — a clause written, in effect, for Tiran, so that states in the position of "Egypt, Jordan, or Saudi Arabia" cannot "clos[e] off the Israeli port of Eilat from the high seas" [8]. UNCLOS then rebuilt the regime in two tiers: the Convention's straits section "applies to straits which are used for international navigation between one part of the high seas or an exclusive economic zone and another part of the high seas or an exclusive economic zone" (Article 37), attracting full transit passage; but straits connecting the high seas or an EEZ to a foreign state's territorial sea get only non-suspendable innocent passage under Article 45 — "this two-tier system has come under criticism (Reisman, 1980)" [8]. Whether an Arctic strait earns tier one or tier two — or no regime at all — depends on what lies at its two ends, which is a question about baselines as much as about water.

That dependence creates the study's central loophole: enclosure. "If a strait is enclosed at each end by newly drawn straight baselines, then under article 37 transit passage may apply if the other conditions are satisfied" — though the ship must still cross "a 12-mile belt of territorial sea" at each end, leaving it genuinely arguable whether such a strait is "used for navigation between two parts of the high seas or EEZ" [8]. Nine of the studied straits sit in that configuration [8]. The harder case is "when a strait lies well within straight baselines": the ship "must pass through internal waters at either end," and the question becomes "whether a state should be permitted to insulate a strait otherwise available to international navigation from transit passage by enclosing it and the surrounding waters with straight baselines" [8]. Twenty-four straits fall into that category [8]. A third configuration — a strait lying behind an island enclosed by baselines — splits the analysis again [8]. In other words, the coastal state's own cartographic act, the drawing of straight baselines examined throughout this report, can determine which legal regime the world's shipping enjoys: the map does not merely record the right of passage; drawn aggressively enough, it can abolish it.

Dunlap is candid about the exercise's limits: the transit-passage regime "is quite recent (in fact, the convention that created it did not enter into force until 16 November 1994) and not yet the subject of any international litigation or incident," so the study "does not provide any definitive answers. It aims, instead, at raising and investigating pertinent questions before they become contentious" [8]. Even its raw materials are provisional — coordinates "are meant only to facilitate location on maps and charts, few of which identify all the straits and islands mentioned here, and the choice of coordinates is not methodical," with names transliterated from Russian charts because those are "most of the useful maps and charts of the area" [8]. The Northern Sea Route is thus a passage regime suspended between a seller and a rulebook: Russia offering escorted navigation as a service, the Convention offering unescorted passage as a right, and 43 straits whose status depends on where baselines nobody has yet tested are drawn. The next sections take the straits one by one.

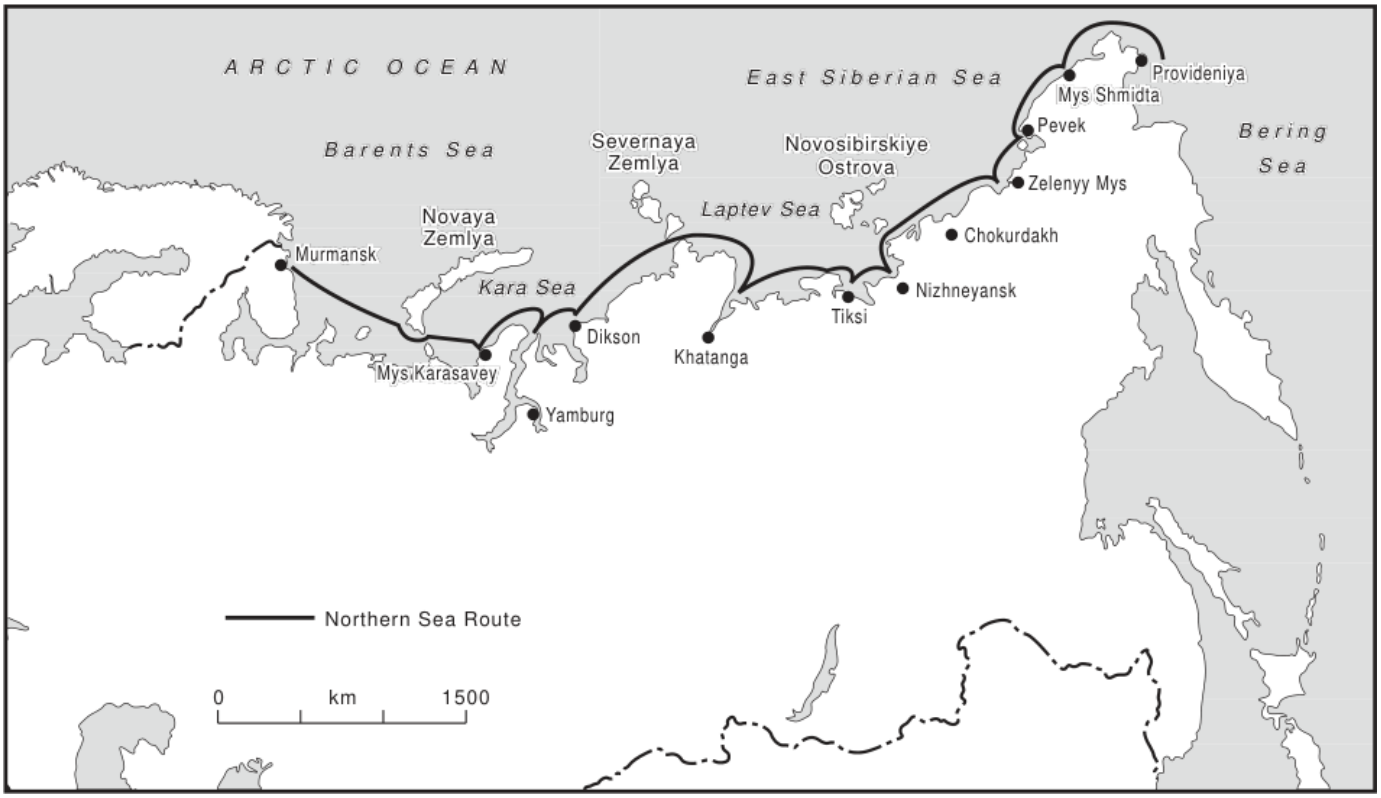


Exhibit. The Northern Sea Route along Russia's Arctic coast, from Murmansk through the Siberian straits to Provideniya. Source: S0010.

Russian jurisdiction over the Arctic straits

Moscow's jurisdictional case over the Northern Sea Route is a layered construction, and Dunlap's briefing dismantles it layer by layer. The foundation is cartographic: Russia's 12-mile territorial sea is measured from "an extensive set of straight baselines, the international validity of which is not settled," which "enclose, among other waters, the major straits connecting the Barents, Kara, Laptev, and East Siberian Seas, relegating the straits to internal waters" [8]. Most of those baselines, the briefing concedes, arguably satisfy the Anglo-Norwegian Fisheries criteria; the exceptions are the enclosures of the three great archipelagos — Novaya Zemlya, Severnaya Zemlya and the New Siberian Islands — which "appear to face the same objections as the straight baselines that enclose much of the Canadian Arctic archipelago," neither resembling the conventions' "fringe of islands along the coast" nor obviously following the coast's general direction [8].

Above the baselines sits the grander claim: the route itself as a national transport route. The argument is old — it "dates back at least to 1582, when Russia granted to English merchants exclusive trading privileges around the Arctic mouths of some Siberian rivers," followed between 1616 and 1620 by four tsarist edicts "forbidding commercial navigation in the Kara Sea," a regime foreign governments took more than three centuries to protest [8]. Its modern form is Kolodkin and Volosov's: "the integral nature of the Northern Sea Route as a transport route is not affected by the fact that individual portions of it, at one time or another, may pass outside of the aforesaid boundaries… (ie it may pass into the high seas)," and the historical record of Russian exploration and outfitting supplies "an aggregate of legal and other material circumstances which enable the Northern Sea Route to be relegated to the category of national transport routes" — regulation of its navigation being therefore "the prerogative of the USSR as the coastal state" [8]. Their precedent is Norway's Indreleia, held by the International Court in 1951 to be "not a strait at all, but rather a navigational route prepared as such by means of artificial aids to navigation provided by Norway," its enclosure sanctioned where other states knew of the claim and did not react against it; the Court's positive reply, they insist, "is fully applicable to the USSR" [8] [8]. Butler's cooler version — in 1971 he called the coastal nature of the route the strongest Soviet argument, on Vyshnepol'skii's criteria of usage, cargo destination and ports of origin and terminus — stops well short of sovereignty over the whole line [8].

The briefing's verdict is brisk. It is "not entirely clear" whether Kolodkin and Volosov argue for enclosing the entire route as internal waters — "a rather expansive claim given the constantly shifting position of the route" — or merely for regulatory jurisdiction beyond the territorial sea [8]. Their companion argument, that no innocent passage survives enclosure by straight baselines, has "little legal foundation": the Soviet Union ratified the 1958 Territorial Sea Con-

vention, whose Article 5(2) expressly retains innocent passage through newly enclosed waters, a route Canada escapes only because it never ratified and argues from custom [8]. The historic-waters door is closed too: despite "the virtual unanimity of Soviet commentators and a few official statements that the straits were internal waters, the Soviet government never claimed them as historic waters, and Soviet practice in the region was not consistent with their having been internal waters before their enclosure" [8]. Hence the summary judgment: "there is, in short, no strong argument that the straits were regarded as internal waters before their enclosure by straight baselines in 1985," and unless they were, "innocent passage and perhaps transit passage will apply" [8]. The national-transport-route doctrine, for all its scholarly unanimity, "has not been Soviet or Russian governmental policy" — surfacing officially only in episodes such as 1985, when a US Coast Guard icebreaker's Northwest Passage transit drew a Canadian internal-waters objection that "Soviet public comment supported," contending the Northern Sea Route too was an internal waterway, after which "little has been heard officially of this somewhat broader claim" [8] [8].

What Russia does instead is regulate, and the regulation is itself a jurisdictional argument in waiting. The 1991 Regulations require advance requests for permission to transit, impose special vessel requirements, assign state pilots, demand proof of financial security against pollution liability, reserve inspection rights, make icebreaker-assisted pilotage compulsory in the Vil'kitskogo, Shokal'skogo, Dmitriya Lapteva and Sannikova straits, and oblige vessels "to follow assigned routes and route corrections issued in response to changes in ice conditions" [8]. Butler suggested the 1967 opening to foreign shipping was itself an attempt "to accentuate the sovereign authority of the USSR over the northern sea route"; Zhudro had already argued that foreign vessels accepting Soviet navigation instructions thereby accepted Soviet authority, building "a generally recognised international custom" [8]. The briefing closes the loop with a warning that cuts against the previous chapter's trap: "if foreign vessels routinely adhere to these regulations, it is likely that Zhudro's assertions will resurface in support of the national-transport-route argument" [8]. Internationalisation thus arms both sides — every foreign transit builds the "use" that triggers transit passage, and every foreign compliance builds the acquiescence that sustains coastal control. Which custom the traffic creates depends on the terms on which it is accepted — and Russia drafts the terms.

The straits themselves: Minin Skerries and Dikson Island in the Kara Sea

After the doctrines, the water. Dunlap's gazetteer works through the Northern Sea Route strait by strait, and in the Kara Sea it reaches the Shkhery Minina — the Minin Skerries, "an archipelago comprising three main groups of islands: Ostrova Plavnikovyy (Plavnikov Islands), a central group, and a northeastern group. They are all enclosed by straight baselines" [8]. The channels among them are measured in the units of pilotage, not of oceans: Proliv Dubravina, "2 miles wide and 20 to 26 metres deep in the fairway"; Proliv Glubokiy, "narrow and 10 to 13 metres deep"; Proliv Iney, "about 1 mile wide and 7 miles long, but about midway the fairway narrows to 2 cables" [8]. Proliv Stalintsa, separating the northeastern group from the mainland, is "7 cables wide and at least 11 metres deep" — seven-tenths of a nautical mile of water carrying the legal weight of an international strait [8]. Farther along the archipelago chain, Proliv Palander offers the skerries' one generous channel, "2¼ miles wide and 38 to 49 metres deep in the fairway," while neighbours remain frankly unexamined: Proliv Sverdrup "not surveyed but known to be deep," Proliv Matisena unsurveyed at "12 to 18 metres," Proliv Toros "unexamined but known to be about 16 miles long and 1 to 2 miles wide with depths of about 20 metres" [8]. These are the "international straits" of the Arctic: leads between ice-foot islands, several of them never properly sounded.

Off Dikson Island, near the Yenisey mouth, the pattern repeats in miniature: "Proliv Lena and Proliv Preven, both narrow and lying north of the island, and Proliv Vega on the south side" separate Ostrov Dikson from the mainland [8]. Twenty significant Russian Arctic straits in all "lie between the mainland and an island," and for every one of them the same legal question arises, because Article 38 withdraws transit passage from such a strait where a seaward alternative exists: "if the strait is formed by an island of a State bordering the strait and its mainland, transit passage shall not apply if there exists seaward of the island a route through the high seas or through an exclusive economic zone of similar convenience with respect to navigational and hydrographical characteristics" [8]. The clause "was designed to apply to such straits as the Strait of Messina... and Pemba Channel" — Mediterranean and tropical water, where the seaward route is simply more sea [8].

In the Kara Sea the exemption collapses twice over. First, cartographically: Stalintsa "is so far within the straight baselines that any similarly convenient seaward route will be within internal waters"; Frama is likewise "[w]ell within the straight baselines that enclose Arkhipelag Nordenshel'da to the north, so the seaward route is in internal waters," and Vostochnyy repeats the formula verbatim [8]. An alternative route that lies in internal waters is no alternative at all in the Convention's terms, since Article 38 requires the substitute to run "through the high seas or through an exclusive economic zone" [8]. The baselines thus do double duty: they enclose the strait, and they enclose the es-

cape route from enclosure. Second, physically: “it is far less likely in the Arctic than in other parts of the world that seaward routes of similar convenience will exist, as the thickness, extent, and strength of sea ice are unpredictable” [8]. The neighbouring straits make the point concretely — in Proliv Malygina, “a light draft vessel can save 50 miles by not having to pass around Ostrov Belyy, suggesting that the seaward route is not of similar convenience,” while Proliv Kil’dinskiy is “deep in the fairway and never freezes” yet “frequently blocked by ice carried in by tidal currents” [8]. Convenience in the Arctic is a seasonal variable, and the Convention supplies “little guidance as to standards, such as minimum width or depth, for determining similar convenience” [8].

The width census completes the picture of a route legally constructed rather than naturally given. “Only five straits, measured between baselines, are wider than 24 miles at the narrowest point” — Vil’kitskogo, Dmitriya Lapteva, and Sannikova at 30 miles, Blagoveshchenskiy at 25, Proliv Longa at 75 — and “[o]f these, only Proliv Longa and Proliv Blagoveshchenskiy appear to contain any belt of EEZ, as the others are entirely enclosed by straight baselines”; strike the Severnaya Zemlya and New Siberian baselines as invalid, and “all five straits will contain belts of EEZ” [8]. Every other strait on the route, the Minin Skerries’ cables-wide channels included, sits inside territorial sea or internal waters on the Russian drawing. The straits themselves — shallow, shifting, half-surveyed — are almost incidental to their legal status. What governs them is the line drawn around them, and the ice that decides, month by month, whether any alternative to them exists at all.

Undiscovered petroleum resources of the Barents Sea shelf

Every boundary dispute in this paper has a prize, and the Barents prize has been measured. The US Geological Survey’s assessment of the Barents Sea Shelf — completed in 2008 as part of its Circum-Arctic Oil and Gas Resource Appraisal and published as Fact Sheet 2009–3037 — estimates the mean undiscovered, conventional, technically recoverable resources at “more than 76 billion barrels of oil equivalent, which includes approximately 11 billion barrels of crude oil, 380 trillion cubic feet of natural gas, and 2 billion barrels of natural gas liquids” [9] [9]. The figures are means of probability distributions, fully risked, and exclude discovered reserves; the resources are “assumed to be recoverable regardless of the presence of sea ice or depth of water” — a modelling convenience that the Arctic chapters of this paper have already taught the reader to discount, but the geology is the geology [9]. The shelf itself is vast and, by Arctic standards, kind: some 1,760,000 square kilometres between 0° and 80°E, lying entirely north of the Arctic Circle offshore northern Norway and Russia, “most of this area… under less than 500 meters of water” [9]. It is also, the Survey concedes in the same breath, a place of “sparse data, high geologic uncertainty, substantial petroleum-resource potential, and technical barriers that impede exploration and development” [9].

The distribution matters more than the total. The shelf is divided into four geologic provinces — East Barents Basins, Novaya Zemlya Basins and Admiralty Arch, Barents Platform, and Norwegian Margin — subdivided into eight assessment units, and “most undiscovered petroleum is estimated to be in the East Barents Basins Province” [9] [9]. That province alone carries mean estimates of roughly 7.5 billion barrels of oil, 321 trillion cubic feet of gas and 1.45 billion barrels of natural gas liquids, dominated by two units: the North Barents Basin (mean oil about 5.3 billion barrels, with a 5-percent fractile above 21 billion — a measure of the uncertainty’s width) and the South Barents Basin and Ludlov Saddle, whose gas endowment (mean about 184 trillion cubic feet) is the shelf’s single largest prize [9]. The petroleum system is a Paleozoic–Mesozoic composite sourced chiefly from Middle Triassic and Upper Jurassic mudstones, reservoired in sandstones from Carboniferous to Paleogene plus Upper Permian spiculite, and trapped in everything from structural highs and salt structures to stratigraphic pinchouts and submarine fans [9]. The western provinces are an order smaller: the Barents Platform totals about 2 billion barrels of oil and 26 trillion cubic feet of gas at the mean, the Norwegian Margin about 1.4 billion barrels and 32 trillion cubic feet — and one unit, Novaya Zemlya Basins and Admiralty Arch, “was not quantitatively assessed because of a less than 10 percent probability of the existence of one undiscovered accumulation exceeding the defined minimum size of 50 million barrels of oil equivalent” [9] [9].

Read against the boundary history, three features of the assessment explain the politics of the preceding chapters. First, the prize is overwhelmingly gas — 380 trillion cubic feet against 11 billion barrels of oil — and gas at Arctic distances from market is the hardest hydrocarbon to monetise; the Barents dispute could be left to ripen for four decades partly because neither side was leaving pumpable oil revenue on the table. Second, the endowment concentrates in the east, in the East Barents Basins offshore Russia — the long-disputed central zone mattered intensely as sovereignty and strategically, but the USGS means suggest the shelf’s centre of gravity never sat inside it; both capitals were negotiating hardest over the smaller part of the prize. Third, the uncertainty is the widest in this paper: a province whose oil estimate runs from 389 million barrels at the 95-percent fractile to 24 billion at the 5-percent is not an asset but a lottery ticket, and it was priced as one — the assessment resting on sparse data, with Norwegian

Petroleum Directorate data for the western provinces and commercial databases current only to December 2007 [9] [9]. The Barents chapters that follow return to the line itself; this one fixes what the line was about: 76 billion barrels of oil equivalent on paper, mostly gas, mostly eastern, mostly hypothetical — a phantom endowment, one might say, awaiting its 2012 moment of verification, with the difference that here both governments agreed to split the ocean before checking whether the treasure was in it.

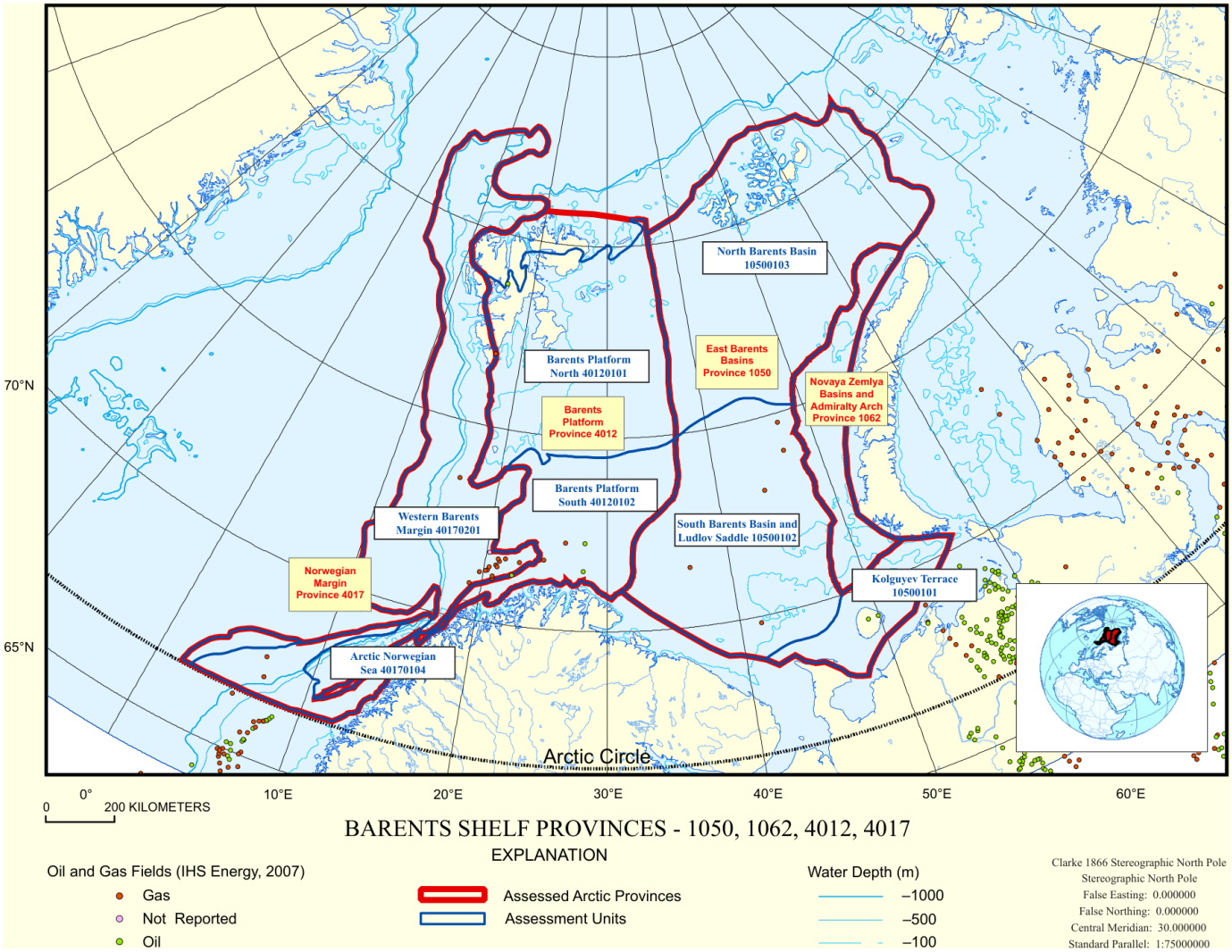


Exhibit. USGS assessment units and petroleum provinces of the Barents Shelf. Source: S0024.

Straits and the Rights of Passage

Transit passage versus innocent passage

The two passage regimes are answers to the same question — what may a foreign ship do in waters another state owns? — calibrated four decades apart, and the distance between them is the distance between 1949 and 1982. Innocent passage is the older settlement, crystallised in the Corfu Channel case. After Albanian shore batteries fired on two British cruisers on 15 May 1946 and a returning British squadron struck mines on 22 October with the loss of 44 lives, the International Court stated the customary rule: “States in time of peace have a right to send their warships through straits used for international navigation between two parts of the high seas without the previous authorisation of a coastal State, provided that the passage is innocent,” and a coastal state has “no right… to prohibit such passage through straits in time of peace” [8]. Even in the exceptional tension of the Albania–Greece frontier, Albania “would have been justified in issuing regulations in respect of the passage of warships through the strait, but not in prohibiting such passage or in subjecting it to the requirement of special authorisation” [8]. Innocence was judged by manner, not motive: that the British passage was mounted to challenge Albania’s attempted closure “did not render it not innocent,” while the subsequent minesweeping fell outside innocent passage altogether, the Court rejecting the urgency defence in words that still sting — such intervention would “in the nature of things… be reserved for the most powerful States, and might easily lead to perverting the administration of international justice it-

self” [8]. Within the general territorial sea, innocent passage remains conditional and fragile: it is defined by the absence of prejudice to the coastal state, it can be policed act by act, and the coastal state retains the wider jurisdiction that makes it, in Moore’s calibration quoted by the briefing, the regime under which the coastal state holds more power over the passing ship than under transit passage [8].

Transit passage, the LOSC’s invention for straits between two parts of the high seas or EEZs, is defined in Article 38(2) as “the exercise… of the freedom of navigation and overflight solely for the purpose of continuous and expeditious transit of the strait” [8]. The briefing lists the three great differences. First, dimension: it “applies to overflights by aircraft as well as to navigation” — innocent passage has no aerial component at all [8]. Second, security against the coastal state: transit passage “may not be hampered or suspended (article 44),” where innocent passage in the territorial sea is suspendable [8]. Third, and most fundamental, the disappearance of innocence as a test: “there are no criteria of innocence to be met,” replaced by three behavioural duties under Article 39(1) — proceed “without delay,” “refrain from the threat or use of force against the coastal state,” and “refrain from any activity other than those incident to their normal modes of continuous and expeditious transit” [8]. The regimes interlock rather than merely coexist: under Article 38(3), any other activity “remains subject to the other applicable provisions” of the Convention, “which appears to mean that the rules of innocent passage take effect once a vessel engages in any unauthorised activity, permitting the coastal state to bar the vessel or to assume jurisdiction over the offending behaviour” [8]. Transit passage is thus a lane, not a liberty: step outside the transit purpose and the older, stricter regime reattaches.

The “normal modes” phrase carries the regime’s most consequential passenger. Because Article 39(1)(c) protects activities incident to a vessel’s normal mode of transit, and a submarine’s normal mode is submerged, the briefing reports the consensus — Moore and others find the text “undeniably establishes the right of submerged transit through straits” — reinforced by the failure of Greece’s attempt at UNCLOS III “to require submarines to transit on the surface and to show the flag” [8]. Friedheim adds the seaman’s rationale: a surfaced submarine sits low, shows a small radar profile and manoeuvres badly, so submerged transit is also the safer mode [8]. Against these ship’s rights the coastal state keeps a defined residue: it “may designate sea lanes and prescribe traffic separation schemes” (Article 41), and legislate on safety and traffic, on pollution only by giving effect to existing international regulations, on fishing, and on customs, fiscal, immigration and sanitary offences — more jurisdiction “than do the rules governing freedom of the high seas or EEZs but significantly less than over vessels in innocent passage” [8].

Why the distinction matters for this paper is that the two regimes price the same strait differently, and the Arctic chapters are a fight over which price applies. A strait under innocent passage can be regulated, its submarines surfaced, its passage in principle suspendable and its users judged act by act; the same strait under transit passage is a corridor the coastal state may channel but never close, open to aircraft above and submarines below. That is why Russia’s whole jurisdictional strategy, examined in the preceding chapter, aims to keep its straits out of Article 37’s category entirely — as internal waters behind baselines, where no passage regime applies — and why the “used for international navigation” test from Corfu Channel, where the Court held that predominantly local traffic and mere alternative-route status sufficed, is the tripwire: the straits’ legal character turns on their use, and use is the one variable the traffic itself controls [8] [8].

The Corfu Channel case

Every straits argument in this report descends from an ambush. “The dispute leading to it began on 15 May 1946, when Albanian shore batteries opened fire on two British cruisers passing through the Corfu Channel between Albania and the Greek island of Corfu, and in so doing, through Albania’s territorial sea” [8]. Britain protested; then, on 22 October, it sent a squadron back through the north Corfu Channel, “cleared for action but with the guns in the normal stowage position” — a deliberate test of the right Albania had denied. “Two of the ships struck mines and were seriously damaged, with the loss of 44 lives” [8]. Britain then announced it would sweep the channel and did so over Albanian protest, “detecting 22 mines and destroying 20,” the inspection of the remaining two producing “evidence strongly suggesting Albanian responsibility” [8]. The United Kingdom sued for compensation, alleging Albania had either laid the mines or known of them in a channel it knew other states’ shipping used; Albania denied the mining and denied the premise, arguing the channel “was not an international strait but a means of lateral traffic of secondary and limited importance,” and that its position on the frontier with a Greece that considered itself at war with Albania made national security a legitimate ground to refuse passage [8].

By special agreement the Court answered two questions — Albania’s responsibility for the explosions, and whether Britain had violated Albanian sovereignty on 22 October and in the sweeping — and its answer to the second be-

came the foundation of the modern straits regime. The holding, in the customary law the Court declared, is quoted wherever straits are argued: “States in time of peace have a right to send their warships through straits used for international navigation between two parts of the high seas without the previous authorisation of a coastal State, provided that the passage is innocent. Unless otherwise prescribed in an international convention there is no right for a coastal State to prohibit such passage through straits in time of peace” (Corfu Channel, 1949: 28) [8]. The concession to the coastal state was carefully bounded: given the exceptional Albania–Greece tensions, Albania “would have been justified in issuing regulations in respect of the passage of warships through the strait, but not in prohibiting such passage or in subjecting it to the requirement of special authorisation” (1949: 29) [8]. Regulation yes; permission no. The distinction is the whole law of straits in embryo, and Russia’s advance-permission regime for the Northern Sea Route is intelligible only as an attempt to live on the other side of it.

The case’s most consequential move, for the Arctic chapters, is its threshold for what counts as an international strait at all. Albania’s description was factually plausible — an alternative route of secondary importance, used almost exclusively by local traffic — and the Court held it sufficient: “the decisive criteria being (1) its geographical situation as a strait connecting two parts of the high seas, and (2) the fact that it was actually used for international navigation. The actual volume of international traffic through the strait was not relevant to its legal status” [8]. Geography plus any genuine international use; volume irrelevant. Applied to the Kara Sea, that is a tripwire set at ankle height: a strait does not become international when the shipping arrives in fleets, but when it arrives at all.

On innocence the Court split Britain’s two voyages with a distinction that still structures the law. The 22 October transit was innocent: the ships “were not in combat formation,” and “the fact that the purpose of the passage was to challenge the Albanian government’s attempt to close the channel to British warships did not render it not innocent. What governed innocence was not the purpose of the voyage but the manner in which it was carried out” [8]. The minesweeping was not innocent passage at all, and the self-help defence failed in language aimed well beyond the case: such intervention would “in the nature of things… be reserved for the most powerful States, and might easily lead to perverting the administration of international justice itself” (1949: 33) [8]. Passage to assert a right: protected. Passage to enforce one: a violation. Corfu Channel thus hands both sides of every later dispute their best sentence — the mariner’s, that warships may pass without asking and that local traffic suffices to internationalise a strait; and the coastal state’s, that passage is a regulated privilege of manner, forfeited the moment the transiting state reaches for the tools of enforcement. The Russian Arctic argument of the preceding chapters is, at bottom, a seventy-year-old Albanian argument with better ice.

Sea lanes, traffic separation, and the laws of states bordering straits

Between the mariner’s right to pass and the coastal state’s sovereignty sits a negotiated middle layer: the bordering state’s power to organise the passage it may not prevent. Under the transit passage regime the LOSC grants states bordering straits a defined regulatory residue. They “may designate sea lanes and prescribe traffic separation schemes for safety reasons (article 41(1)),” and “may adopt laws and regulations regarding maritime safety and traffic (article 42(1)(a)); the prevention of fishing (article 42(1)(c)); loading or unloading in violation of customs, fiscal, immigration, or sanitary laws (article 42(1)(d)); and the prevention, reduction and control of pollution but only by giving effect to existing international regulations (article 42(1)(b))” [8]. The list is closed and deliberately asymmetrical: traffic management is the coastal state’s to run; pollution standards are not its to invent, only to enforce from the international book. And the whole residue is bounded by Article 44’s prohibition — transit passage “may not be hampered or suspended” — so the sea lane is an instrument of channeling, never of closure [8]. The calibration, in Moore’s phrase carried by the briefing, leaves the bordering state with “significantly more jurisdiction over foreign vessels than do the rules governing freedom of the high seas or EEZs but significantly less than over vessels in innocent passage” [8].

The territorial sea’s neighbouring regime shows both the parallel and the gap. There, too, the coastal state legislates from a list — Article 21’s categories of “navigational safety; protection of navigational aids, cables, and pipelines; conservation of living resources; environmental protection; scientific research; and preventing infringement of fisheries, customs, immigration, and sanitary laws” — and there too the list under-describes raw sovereignty: “article 21 does not limit the prescriptive jurisdiction of the coastal state, which may, for example, still declare homicide on a foreign ship in the territorial sea to be murder,” though enforcement against a ship that remains in innocent passage is tightly restricted, and breach of an authorised regulation “may render the passage non-innocent,” exposing the vessel to the full criminal law [8]. The straits regime withdraws precisely that lever: in transit passage there is no innocence to forfeit, only the transit duties of Article 39, and the Article 42 list is the whole of the bordering state’s legislative reach over the transiting ship as such [8].

The sea-lane power itself carries a diplomatic history the briefing uses to mark how far the 1989 settlement reached. When the United States and the Soviet Union exchanged notes in 1989 confirming that both recognised warships' right of innocent passage "without prior authorisation or notification" — apparently "the first time in history that they had been on the same side of the question at the same time" — the same exchange "confirmed that coastal states might not establish laws or regulations impairing innocent passage through the territorial sea, but that a coastal state might designate sea lanes and enforce traffic separation schemes when necessary to protect the safety of navigation" [8]. Lane designation, in other words, was the agreed consolation prize of the freedom-of-navigation wars: the superpowers traded away permission regimes and kept traffic management, and the United States simultaneously "confirmed that it would refrain from exercising the right of innocent passage in Soviet territorial waters of the Black Sea," as its Freedom of Navigation Program had been doing — rights asserted globally, exercised with discretion locally [8]. The settlement was never universal: China, ratifying the LOSC in 1996, declared that the innocent passage provisions "shall not prejudice the right of a coastal state to request… a foreign state to obtain advance approval from or give prior notification to the coastal state for the passage of its warships" — the permission model, formally preserved by a rising maritime power [8].

For the Arctic chapters the details that matter most are two. First, designation and traffic schemes are the legal form Russia's Northern Sea Route regime most nearly fits: assigned routes, route corrections, prescribed pilotage in named straits — the 1991 Regulations read, in their least objectionable aspect, as an Article 41–42 system, and read, in their permission requirement and inspection regime, as the thing Corfu Channel and the 1989 notes forbid [8]. The argument over the Russian Arctic is thus not whether Moscow may organise the traffic — it may — but whether organisation has swollen into authorisation. Second, the briefing flags a classification with outsized Arctic consequences: coast guard vessels "appear to qualify as warships under the terms of the LOSC (1982: article 29)," broader than the 1958 High Seas Convention's wording [8]. In waters where the most frequent foreign state vessels are icebreakers and coast guard cutters — the Polar Star of the 1985 Northwest Passage dispute among them — the warship question is not a doctrinal corner; it is the main case, and the sea-lane rules are the ground on which it will be fought.

Submarines, nuclear-powered ships, and dangerous cargoes

The passage regimes are usually debated in the abstract; they bite on hulls. Three classes of vessel — submarines, warships generally, and the nuclear-powered and dangerous-cargo ships that concentrate every coastal anxiety — are where the difference between innocent and transit passage stops being doctrinal and becomes operational, because each regime prescribes not just whether these vessels may pass, but how visibly.

Start with the demotion that transit passage was invented to prevent. When territorial seas spread to 12 miles, "116 international straits would be enclosed by a worldwide adoption of the 12-mile territorial sea," and foreign ships in them would fall back on innocent passage — "significantly more restrictive of ships' movements and activities" [8]. For aircraft the downgrade was total: it "would have eliminated the right of overflight, as there is no right of innocent passage for aircraft above the territorial sea" [8]. For submarines it was a demand for exposure: enclosed straits "would have required transiting submarines to surface and to show the flag (LOSC, 1982: article 20; TSC, 1958: article 14(6))" [8]. Article 20 states the rule for the territorial sea generally: "a submarine in the territorial sea must navigate on the surface and show its flag" [8]. For a ballistic-missile submarine, whose entire military value is concealment, a surfaced, flagged strait transit is not a formality; it is the mission negated on schedule, in named waters. Straight baselines threatened worse: straits enclosed as internal waters "would have been altogether inaccessible to foreign vessels without the authorisation of the coastal state," tempered only by the 1958 Convention's Article 5(2) preservation of innocent passage through newly enclosed waters [8].

Transit passage answers hull by hull. Overflight exists. Suspension is barred. And the submarine question is resolved by three words — Article 39(1)(c)'s "normal modes" of transit: "by reference to 'normal modes' of transit, article 39(1)(c) permits submarines to transit submerged" [8]. The reading was contested early — "Reisman (1980) argued… that this was not unambiguous" — but "Moore and others have found that the language of the convention's text undeniably establishes the right of submerged transit through straits," a reading reinforced by "the failure of Greece's attempt at the fourth session of UNCLOS III to require submarines to transit on the surface and to show the flag" [8]. Friedheim adds the mariner's case beneath the lawyer's: surfaced submarines "are low in the water, provide a small radar profile, and are not easily manoeuvrable, and thus pose a hazard to other surface vessels. For these reasons… submarines are permitted to transit in their normal mode, submerged" [8]. The nuclear-powered submarine, the Arctic's most consequential traffic, is thus the vessel the transit regime most deliberately protects — which is

precisely why straits states that fear it fight the classification instead, reaching for internal waters where no regime applies.

Warships occupy the regime's deliberate ambiguity. Beyond the submarine rule, "the innocent passage of warships has been a matter of contention," the customary law "both unclear and controverted" (Jennings and Watts), and neither convention resolves it: the 1958 text "has been read as granting a right of innocent passage to warships… but has also been interpreted as not resolving the question at all," while the LOSC, despite separate rule-sets for merchant and government-commercial ships (Part II, section 3(B)) and for "Warships and Other Government Ships Operated for Non-Commercial Purposes" (section 3(C)), "does not directly address the question of whether warships enjoy the right of innocent passage" [8]. States filled the silence from both directions: the United States "has recently, but not always, argued that the right of innocent passage extends to all ships, including warships," while "[a] number of states have contended that innocent passage does not extend to warships and that they require authorisation before entering the territorial sea" — the Soviet Union filing "a reservation to that effect upon its ratification of the Territorial Sea Convention" [8]. The briefing's bibliography maps the running fight in its titles alone: Franckx's successive studies of "the clarification of the Soviet position on the innocent passage of foreign warships through its territorial waters" and of "new developments in the North-east Passage" [8]. A nuclear-powered warship carrying dangerous cargoes is where all three anxieties stack — the hull that must not surface, the flag that demands permission, the cargo that makes the coastal state's fear rational — and the Convention's answer is the Arctic's unresolved question in miniature: in a true international strait it passes submerged, unannounced, and unboardable under transit passage; behind Russian baselines, on Moscow's theory, it does not pass at all without leave. Everything else is commentary.

Enclaves, Exclaves, and the Lines Between

The Land Boundary Agreement of 2015: a post-implementation analysis

The India–Bangladesh enclaves were the world's most extreme cartographic residue: pieces of each state sealed inside the other, some of them nested — enclaves within enclaves, and one third-order enclave, a Bangladeshi fragment inside an Indian fragment inside Bangladesh. Their origin lies in the unfinished business of the Kingdom of Cooch Behar and the Mughal Empire: territories conquered back and forth until, by one account, "a confused outcome of a 1713 treaty" ended a war "without determining a single boundary for what territories had been gained or lost" — popular legend prefers the version in which the slivers were stakes in card and chess games between the Raja of Coochbehar and the Maharaja of Rangpur [10]. The 1947 Radcliffe partition froze the pattern into international law: Cooch Behar acceded to India in 1949, Rangpur to Pakistan in 1952, and the enclaves — 111 Indian inside Bangladesh, 51 Bangladeshi inside India, 53,384 people in all — became islands whose residents could not reach their own state without crossing a foreign one [10] [10]. The post-implementation literature is blunt about what that meant: people living "in 'no-man's land' as 'nowhere people'" for 67 years after independence, "sustaining their existence under a hardnosed situation of statelessness without support of the state aids and administrations from either country" — no constitutional rights, no services, no amenities, because the surrounding state owed them nothing and their own state could not reach them [10].

The road to 2015 is a case study in ratification asymmetry. A first exchange was envisaged in the 1958 Nehru–Noon Agreement and stalled on an Indian Supreme Court case; the 1974 Indira–Mujib Land Boundary Agreement settled most of the 4,096.7-kilometre border but expressly left three issues outstanding — the enclave exchange, adverse possessions, and some 6.1 kilometres of undemarcated boundary in three sectors [10]. Its one executed exchange demonstrated the blockage precisely: Bangladesh handed over South Berubari (7.39 sq. km) instantly in 1974, while India could not transfer the Tin Bigha Corridor — a strip 178 by 85 metres giving Bangladesh access to its largest enclave, Dahagram–Angarpota — because cession required a constitutional amendment that politics blocked; the corridor was finally leased to Bangladesh in perpetuity only in 1992 [10]. The 2011 Manmohan–Hasina protocol revived the full package — swap the 162 enclaves, regularise adverse possessions, let residents choose their nationality — and India's 119th Amendment Bill failed again in 2013 before passing both houses in May 2015 [10].

The exchange itself, executed at midnight on 31 July 2015 under the agreement of 6 June 2015, was arithmetically lopsided and deliberately so: India transferred 111 enclaves (17,160.63 acres) and 35 parcels under Bangladeshi adverse possession (2,267.682 acres), receiving 51 enclaves (7,110.02 acres) and 42 parcels under Indian adverse possession (2,777.038 acres) [10]. The analysis stresses what kind of act this was: "basically a switch from a de-facto situation to a de-jure one, since these enclaves and adverse possessions were already occupied by both the coun-

tries" [10]. One exception proves the rule's negotiability — Dahagram–Angarpota, 18.68 sq. km and some 20,000 people, "has not yet been exchanged and it still remains as an integrated part of Bangladesh, although completely surrounded by Indian Territory," its position governed by the 1974 settlement and the Tin Bigha lease [10]. For the residents, the agreement's humane core was elective identity: each person could stay on their birth soil under the incoming citizenship with full benefits, or migrate to the mother state, which "would offer adequate rehabilitation packages" — India establishing Enclave Settlement Camps at Haldibari, Mekhligunj and Dinhata for those who chose to move [10].

Post-implementation scholarship, of which Bhattacharya's study is the anchor here, evaluates the agreement on exactly those human terms — how residents of the erstwhile Bangladeshi enclaves accepted the exchange, how far India's rehabilitation packages were implemented, and how satisfied the migrants in the settlement camps are — surveying 23 erstwhile enclaves across the Mekhligunj, Sitalkuchi, Mathabhanga and Dinhata blocks of Cooch Behar and all three camps, with a companion comparative study interviewing 328 inhabitants of 20 erstwhile Bangladeshi enclaves [10] [11]. For this paper's argument, the case is the enclave chapters' resolving chord. Every other boundary instrument in these pages allocates jurisdiction over space; the 2015 agreement is the rare one that allocates people out of a cartographic accident, converting statelessness into citizenship by redrawing the map to match the ground. Where the Northern Sea Route showed a line that moves with the ice and Liancourt showed a line waiting on sovereignty, Cooch Behar showed the opposite extreme: a boundary so excessively, absurdly fixed — in a 1713 war's unfinished arithmetic — that it imprisoned three generations, until two states simply agreed to exchange the fiction for the fact.

Comparative impact of the agreement on the people of the erstwhile enclaves

What did the swap feel like from inside? The comparative field studies — surveys across the erstwhile Bangladeshi enclaves in Cooch Behar and the three Enclave Settlement Camps, conducted in June 2016 and May 2017 — give the agreement an almost unheard-of verdict from the population that stayed, and a more complicated one from the population that moved. Start with the stayers. "The entire population of erstwhile Bangladeshi enclaves in Indian territory covered under the field survey cordially accepted exchange of enclaves through LBA, 2015, as they got rid of their exiled life of long 67 years in enclaves" — and the acceptance was not merely attitudinal: "all 14,863 residents of the Bangladesh enclaves in India opted for Indian nationality and opted to stay in India as Indian citizen," with the study emphasising that "not a single person from erstwhile Bangladeshi enclaves in India opted to go back to Bangladesh" [11]. In a report full of populations gaming, evading or outlasting the lines drawn around them, this is the unique case of a population unanimously ratifying the line's removal — "cent percent highly positive feedback" [11].

The material transformation was real but shallow — citizenship arrived faster than the state. Residents "duly got their Voter ID Card and Aadhar Card in support of their Identity as Indian Citizen after execution and implementation of LBA, 2015"; most received ration cards, some job cards, and some were engaged in the government's 100-days work programme [11]. Beyond the documents, however, the survey found "no infrastructural developmental works within these enclaves have been noticed" — the enclaves had acquired a sovereign but not yet roads, and two structural problems dominated. The first was land: "a settlement of their land records within these erstwhile enclaves" was the foremost anxiety, because during 67 years of stateless limbo, land — "the only source of generating economic resources of most of the enclaves' people" — had changed hands informally: "many original land deed holders already sold out their lands to other enclaves' dwellers on the basis of mutual written agreement just on a white paper," leaving occupants without original title deeds from the Bangladesh Land Settlement Authority and facing the open question of whether white-paper transfers would be recognised in India's settlement process [11]. The District Magistrate of Cooch Behar himself "admitted… that the major challenge to resolve erstwhile Bangladeshi enclaves' problem was the disposal of land records and settlements" [11]. The second problem was opportunity: "no job reservation has since been made for the erstwhile enclaves' dwellers, nor has any alternative job opportunity yet been initiated for them," and even school and college admission for their children ran into trouble for want of any reservation [11]. De jure belonging, in short, was delivered on schedule; de facto incorporation lagged at exactly the points — title, work, schooling — where a population's paper past had never existed.

The movers' story is the mirror image: a small, self-selected minority exercising the agreement's other option. Of the 38,521 residents of the erstwhile Indian enclaves in Bangladesh identified by the joint survey, just 989 persons "opted to retain their original nationality of Indian citizenship and migrated to Indian mainland from Bangladesh" [11]. They were received in the three Enclave Settlement Camps at Haldibari, Mekhligunj and Dinhata, where the researchers interviewed 134 of them across the 2016 and 2017 survey rounds [11]. The asymmetry between the two populations is itself the finding. Where the Bangladeshi enclaves' residents faced a choice between staying home under a new flag

or abandoning home for an ancestral state most had never seen, 100 per cent chose home; where Indian enclaves' residents faced the same geometry in reverse, 97.4 per cent also chose home, accepting Bangladeshi citizenship — and the 989 who chose the flag over the soil traded settled lives for camp accommodation and a rehabilitation process. Read comparatively, the agreement's deepest success is demographic: given a free choice with full information, enclave people overwhelmingly preferred place to passport — which is precisely why the de facto-to-de jure reading of the previous section holds. The states did not, in the end, move borders over people; they recognised that the people had never really lived under the borders at all, and the residual work of the settlement — deeds, jobs, classrooms — is the state now catching up, on the ground, to a fact its own map denied for 67 years.

The human dimension of boundary settlements

A boundary settlement is usually audited in kilometres and acres. The enclave settlement invites a different audit, because its subjects kept books of their own — in anxiety. Bhattacharya's fieldwork returns to the camps and erstwhile enclaves in June 2016 and May 2017, a year and two years after the midnight exchange, and records what the state delivered and what it merely promised. The documents arrived: both stayers and migrants "duly got their Voter ID Card and Aadhar Card in support of their Identity as Indian Citizen" [10]. Citizenship, the thing 67 years had withheld, was the easy part — a card, printed. Everything that makes citizenship usable lagged behind it, and the lag distributed itself with a cruel asymmetry between the two populations the agreement created.

The migrants — the 989 who chose the flag over the soil — received the state's most visible care and its most visible failure. The Government of India "arranged temporary accommodations for them by setting up three Enclaves Settlement Camps in the District of Coochbehar... with free electricity facility and also provided essential food commodities for them through rationing system," but "[a]s per opinion of camps' dwellers, those food commodities provided by the Govt. of India were really insufficient for maintenance of their family" [10]. They had left a known poverty for an unknown one: in the erstwhile Indian enclaves "they had their own lands for cultivation and they could move to Bangladeshi lands in the hunt for their bread and butter," whereas on the mainland they are "completely 'landless' as well as 'jobless'" [10]. No permanent shelter had yet been provided; no job reservation created; their children faced the same admission obstacles as the stayers'. The regret is on the record: "A year after enclave dwellers got Indian citizenship, several of these people want to go back to Bangladesh as they are miffed over the lack of job and other opportunities in India" — "Those who have come from Indian enclaves... are feeling alienated here. They don't have either any job opportunities or any proper source of income... So they feel that going back to Bangladesh is the best option to secure their future" [10]. The population that ratified the settlement most dramatically — by walking away from home for it — became the population most failed by its implementation: "migrated people... were really scared about their permanent settlement in Indian mainland," "feeling confused about their decision for migrating" [10].

The stayers' deficit is quieter: deeds and development. Their land — "the only source of generating economic resources of most of enclaves' people" — sits in the white-paper limbo described in the previous section, and the then District Magistrate, P. Ulganathan, "admitted that the major challenge to resolve erstwhile Bangladeshi enclaves' problem was disposal of land records and settlements," while assuring researchers "that adequate funds would be released by the Govt. of India for resettlement of migrated and non-migrated erstwhile enclaves' people and for infrastructural development" [10]. The comparative study adds the cross-border asymmetry: "Govt. of Bangladesh has initiated lots of infrastructural developmental works within erstwhile Indian enclaves; whereas Govt. of India could do little infrastructural developmental works within erstwhile Bangladeshi enclaves so far," and neither state created job reservations for its new citizens [11]. The same treaty, signed once, is being implemented at two speeds on its two sides of the line it erased.

The comparative study's conclusion refuses both triumphalism and despair, and its phrasing deserves to stand as this section's verdict. The enclave people "have started to forget those days of the nightmare that they passed on," yet they remain "anxious whether 'delay in justice' would ultimately be ended with 'denial of justice' for them or not. A ray of hope is still blinking on their hazy eyes dreaming for their luminous future in the long run" [11]. That is the human dimension of boundary settlements generally, stated by its most extreme case: the line can be settled in a night — 31 July 2015, midnight, flags exchanged — but a settlement is not finished when the map is corrected. It is finished when the people the map imprisoned have deeds the state will honour, work the economy will offer, and schools that will take their children; and until then, the agreement exists in the same de facto/de jure gap it was designed to close, one level down. The cartographers' part took sixty-seven years. The administrators' part is the part still being timed.

Ferghana Valley: enclaves and border conflicts in Central Asia

If Cooch Behar is the enclave problem solved, the Ferghana Valley is the same problem with the solvent withheld. The valley — "an intermontane bowl of fertile land, roughly 300 km long and up to 70 km wide," ringed by the Tien Shan and Pamir ranges, home to "over 10–15 million people" where Uzbekistan, Kyrgyzstan, and Tajikistan meet — is "the most densely populated area in post-Soviet Central Asia," holding nearly a third of the region's people [12]. Its borders have the same imperial provenance as Bengal's, with intent added: "the modern boundaries in the Fergana Valley were drawn in the 1920s by Soviet mapmakers with little regard for local demographics or geography," and the delimitation of 1924–1927 "sought to divide ethnic groups among different Union republics as a means of divide-and-rule, while also ensuring each republic had a mix of resources" [12]. Lines drawn as internal administrative conveniences under "a unitary Soviet authority" became, in 1991, "international borders guarded by militaries and border police" [12]. Radcliffe improvised in five weeks; Moscow engineered over a decade. The results are comparable.

The enclave inventory is smaller than Bengal's and stranger. "[T]here were eight enclaves as of the 2010s. The largest is Sokh, a 352 km² pocket of Uzbek territory inside Kyrgyzstan – populated overwhelmingly (~99%) by ethnic Tajiks" [12]. Sokh is a triple mismatch — Uzbek sovereignty, Tajik population, Kyrgyz surroundings — a jurisdiction answering to a capital its residents do not ethnically belong to, inside a state they do not legally belong to. Nearby lie Shohimardon, another Uzbek exclave in Kyrgyzstan, and "Tajikistan's own enclave Vorukh nestled within Kyrgyzstan's Batken province," while "Kyrgyzstan even had a tiny exclave of its own (Barak village) trapped inside Uzbekistan" until recently [12]. "[T]he enclaves pose extreme governance challenges: they are connected to their home country only by thin corridors or by goodwill of the surrounding state, and their presence has been a constant source of misunderstanding and conflict" [12]. Overlaid on the jurisdictions is the resource geometry: "Control of water in upstream areas (mostly in Kyrgyzstan and Tajikistan) versus downstream irrigation (largely in Uzbekistan) has been a perennial strategic concern," and "infrastructure built in the Soviet period often crosses today's borders: an irrigation canal or road might originate in one country but serve communities in another," an interdependence that "without strong coordinating mechanisms, can breed zero-sum thinking" [12]. Hence the valley's escalation physics: "seemingly minor incidents – a dispute over a patch of land, a scuffle at a border crossing, a shortage of irrigation water – carry an outsized risk of escalation" [12].

The casualty record shows the physics at work. In June 2010, Osh and Jalal-Abad "were convulsed by clashes between Kyrgyz and Uzbeks in the wake of a Kyrgyz political revolution. The conflict left at least 470 people dead… over 400,000 civilians (mostly ethnic Uzbeks) fled, including around 100,000 who took refuge across the border in Uzbekistan" [12] — the valley's titular minorities (Uzbeks dominant in parts of Osh province yet "historically… underrepresented in government and security structures") bearing the cost of a capital-city revolution [12]. The Kyrgyz–Tajik frontier then militarised in steps: "A turning point came in April 2021 when a dispute over a surveillance camera at a water intake escalated into sustained firefights between the two countries' security forces. The confrontation resulted in over 50 deaths and hundreds of injuries. In September 2022, an even more intense outbreak occurred, with tanks, mortars, and armed drones used. Nearly 100 people were killed, and over 140,000 Kyrgyz residents were displaced" [12]. A camera at a water intake to artillery in seventeen months: the boundary itself was not even the object of either war — the water infrastructure straddling it was. Uzbekistan, which "min[ed] parts of its borders" in its hardline decades, has since 2016 shifted "toward reconciliation, resolving long-standing disputes through diplomacy," though "incidents involving Uzbek enclaves (such as Sokh) continued to occasionally provoke tensions" [12].

Set beside the previous chapters, Ferghana is the control case that proves the enclave rule. Bengal's enclaves generated misery but little interstate violence, and were ultimately exchangeable because both states could treat the swap as bookkeeping — de facto for de jure. Ferghana's enclaves sit inside a live three-way competition for water and pasture, among populations whose ethnic maps cross-cut the jurisdictional ones, along borders two of the three states have repeatedly fought over. The map is wrong in the same way in both valleys; what differs is everything around the map. The valley's own analysts conclude that repair "requires not only formal border agreements and resource-sharing mechanisms but also sustained political will, inclusive governance, and multilateral engagement" [12] — which is to say, the cartography is the easy part, and it has not been done either.

The thalweg: where a river draws the border

Most boundaries in this report are drawn by surveyors, diplomats, or courts. The thalweg is drawn by the river. The TALOS Manual defines it exactly: "The Thalweg is defined as the line of maximum depth along a river channel or lake but may be considered in any coastal channel. The principle has been employed for centuries where such water bodies form the boundary" [3]. Its rationale is access, not symmetry: "Its most obvious use as a boundary lies in

areas of shallow water in the territorial sea where it is desirable that the navigational channels giving access to both States should not be under the sole control of one State" [3]. A median line between banks could strand the deep channel wholly inside one state and leave the other's ports landlocked by shallows; the thalweg instead splits the channel itself, giving each riparian a navigable road to the sea. Where the water deepens and access is no longer scarce, the logic evaporates: "[i]n the deeper waters seaward of rivers or estuaries, justification of using the Thalweg method is uncertain" [3]. It is a boundary with a physical argument built in — the only line in this report whose fairness claim is hydrographic. Its line "may be determined from charts. In some cases a special survey may be necessary" [3].

The price of letting the river draw the border is that the river keeps drawing. "If the Thalweg is following a navigable channel in unstable areas, it will change as the channel changes. It can only remain as a stationary line if it is used in waters too deep for changes in bathymetry to matter" [3]. An ambulatory boundary is a legal commitment to a moving object, and where the thalweg forms the inshore end of a maritime boundary, the movement must somehow be coupled to the fixed offshore line. TALOS prescribes the hinge: terminate the thalweg "at a defined line (a line of bearing from a fixed point, or a meridian or parallel) which passes through the position determined as the landward end of the fixed section of boundary," so that "[t]he inshore section of the boundary, at any particular time, will be the Thalweg as it is at that time, and will terminate at whatever point it intersects the defined line" [3]. The border thus has a living segment and a dead segment, joined at a line that absorbs the river's wandering without transmitting it seaward.

The Cameroon–Nigeria judgment of 10 October 2002 is the doctrine's modern showpiece, and it uses the hinge exactly as the manual describes. Having held, by thirteen votes to three, that sovereignty over the Bakassi Peninsula lies with Cameroon and that the boundary there is delimited by Articles XVIII–XX of the Anglo-German Agreement of 11 March 1913, the Court decided — again by thirteen to three — that "the boundary between the Republic of Cameroon and the Federal Republic of Nigeria in Bakassi follows the thalweg of the Akpakorum (Akwayafe) River, dividing the Mangrove Islands near Ikang in the way shown on map TSGS 2240, as far as the straight line joining Bakassi Point and King Point" [13]. The maritime boundary then begins "from the point of intersection of the centre of the navigable channel of the Akwayafe River with the straight line joining Bakassi Point and King Point" and follows the "compromise line" of the 1971 Yaoundé II Declaration through twelve numbered points [13]. Thalweg to the fixed line, fixed line to the sea: the TALOS hinge, executed by the World Court. The same river logic runs up the whole land boundary the Court confirmed — delimited by the Thomson–Marchand Declaration of 1929–1930, the British Order in Council of 2 August 1946, and the Anglo-German Agreements of 1913 — including a segment running "to the point on the thalweg of the Akpa Yafe which is opposite the mid-point of the mouth of Archibong Creek," and a Lake Chad sector drawn in straight lines from a tripoint at 14°04′59.9999″E, 13°05′N to the mouth of the River Ebeji and on to "the point where the River Ebeji bifurcates" [13] [13].

The thalweg is this report's limiting case of a self-drawing map. A phantom island is a feature that does not exist; a shifting channel is a feature that will not hold still, and the law's answer is not to freeze it but to draft around its motion — boundary as a function of bathymetry, re-evaluated every time the river re-surveys itself. Where the channel is stable and deep, the thalweg hardens into an ordinary line and the fiction of permanence goes unchallenged. Where it wanders — through mangrove islands, across tidal flats, down the Akwayafe — the border is less a place than a rule for finding the place, and sovereignty attaches not to coordinates but to the deepest water, wherever the river has most recently put it.

Kashmir: Learning From the Past

Internal dynamics of the Kashmir dispute

Every other dispute in this paper is, at bottom, an argument between states about a line. Kashmir is an argument inside the line — about who Kashmiris are, who speaks for them, and what the former princely state is for — and the ICG's 2003 anatomy, this chapter's principal source, shows the external stalemate being continuously regenerated by internal ones. Its first finding is an identity in dissolution. Kashmiriyat, "the basis of a secular Kashmiri identity, was waning by the middle of the 1980s": Islamic revivalism in the Iranian revolution's wake, Muslim educational institutions expanding into the vacuum of state failure, declining youth employment, and a secular mainstream leadership discredited by "corruption and nepotism" left "a new generation of Kashmiris… open to new forms of political mobilisation, premised on mullah and mosque" [14]. The religious current was older than the crisis — the Mirwaiz's return from Pakistan in the late 1950s, and the 1963 theft of the Hazratbal relic, a hair of the Prophet, which convulsed

the Valley and showed "how close to the surface of Kashmiri culture religious sentiments were" [14]. Nor is the state internally singular: Jammu and Ladakh nursed their own resentment of Valley domination and sought autonomy from Srinagar through hill councils — a reminder that "Kashmiri opinion" is a plural noun [14].

The detonator, on the ICG's account, was the systematic closure of institutional politics by New Delhi. Indira Gandhi dismissed Farooq Abdullah's elected government in 1984 "in flagrant violation of Article 370," installing the compliant G. M. Shah, whose incompetent administration deepened alienation while Islamic social movements grew in the space party politics vacated [14]. When Farooq then allied with Rajiv Gandhi's Congress for the 1987 election against the Muslim United Front's challenge, "New Delhi decided to manipulate the election": the result "was widely perceived in Kashmir to be the result of massive electoral fraud," with "significant MUF electoral gains… annulled" — and the consequence is the report's pivotal sentence, that a generation "unable to express dissent in an institutional context" took dissent elsewhere [14]. The uprising followed from 1989.

India's answer manufactured the grievance it policed. By 1994 "in excess of 500,000 security personnel" were deployed; the early counter-insurgency brought cordons separating men from women, "arbitrary arrest of Kashmiri Muslim males and a high incidence of custodial deaths," in measures that "resulted in massive human rights violations… producing a groundswell of anti-Indian sentiment" [14]. The legal instrument matched: TADA saw "over 76,000 people… arrested… with less than 2 per cent of detainees being convicted" before its 1995 repeal, and parliament's March 2002 replacement was "denounced by its own Human Rights Commission" [14]. Simultaneously the conflict mutated from insurrection to "full insurgency, with numerous foreign militant groups fighting both each other and the Indian security forces," more than 50 groups active since the mid-1990s — including side-switchers running counter-insurgency for the state — while the Valley's Hindus, over 200,000 of them, remained refugees, tourism collapsed, and "an entire Kashmiri generation has been exposed to the immediacy and apparent glamour of violence" [14].

The political vehicles all failed in sequence. The APHC, formed in 1993 as an alliance of some 38 parties behind a three-option referendum (independence, Pakistan, India), "left crucial issues undecided such as whether the new state it aspired to create would be secular or theocratic" and boycotted the polls [14]. The National Conference returned in 1996 on a low turnout amid intimidation from both sides, platformed on Article 370 with "virtually no conviction that Article 370 could solve the Kashmir crisis on its own," compromised itself by supporting the BJP-led coalition whose elements "wanted to abolish Article 370 altogether," and was defeated in 2002 [14]. The PDP–Congress government of Mufti Mohammed Sayeed then attempted the political track — overtures to the APHC, released detainees, an assembly committee on "revitalising Article 370, 'full autonomy', and greater regional autonomy" — while New Delhi sent mixed signals plausibly "deliberate" in design, denouncing APHC leaders as "'Pakistani stooges' one week, while inviting them for talks the next," pressure that fed the APHC's split into pro-independence and pro-Pakistan leaderships [14]. The ICG's verdict frames the whole chapter: bilateral settlements failed "in important part because Kashmiri representatives were excluded from the negotiating process," and "unilateral attempts to resolve the Kashmir dispute provide at best interludes of deceptive calm, but they do not affect the roots of the dispute" [14] [14]. For this paper's argument, Kashmir is the limiting case of the delimitation problem: a boundary whose legal geometry was settled, after a fashion, in 1949 and 1972, but whose interior — identity, representation, consent — was never drawn at all. The LoC holds not because it was agreed, but because everything inside it remains undecided.



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Exhibit. United Nations map of the Jammu and Kashmir area, showing the Line of Control promulgated in the 1972 Simla Agreement. Source: S0014.

Pakistani intervention and Islamic extremism

If the previous section showed the dispute's internal fuel, this one traces its external accelerant — and the ICG is careful about the direction of causation: Pakistan did not invent Kashmiri alienation, but from 1989 it organised,

armed and progressively rebranded it. The rebranding began under General Zia-ul-Haq (1977–1988), whose regime "aligned itself with new radical Islamic movements" and "prioritised and promoted the issue of Indian 'occupied' Kashmir as primarily a religious issue of Muslims under Hindu rule," working through the Jamaat-i-Islami: "official Pakistani patronage of the Hizbul Mujahidin in Jammu and Kashmir, the Jamaat's militant wing, was seen as a way of undercutting the secular forces initially at the forefront of the Kashmir agitation, such as the JKLF" [14]. The Afghan jihad against the Soviets supplied the machinery and the manpower pipeline — "Pakistan's experience of training and controlling insurgents during the Afghan insurrection of the 1980s… provided ready experience for intervention in Indian-administered Kashmir when the opportunity arose after 1989" — run through the ISI, revamped under Zia to command "a considerable budget," headed by "a senior army officer accountable to Zia alone" and remaining "unaccountable to parliament" even after democracy's 1988 restoration [14]. The newer groups the pipeline produced — Lashkar-e-Tayaba and Jaish-e-Mohammad — "emerged out of a tradition… premised on the Sunni and Wahhabi radical theology of some Pakistani religious seminaries," preaching "a jihad or a purified religion stripped of non-Islamic culture – an intolerant, militant theme alien to wider Pakistani Sunni traditions" [14].

The system's political insulation is as important as its tradecraft. Civilian governments of 1988–1999, "deprived of a policy-making role on sensitive issues and concerned about an adverse military response… acquiesced in this strategy," while Islamic parties converted coalition indispensability into influence "out of all proportion to their actual power" [14]. The military-backed MMA's 2002 electoral success — government in the NWFP, coalition partner in Baluchistan — meant "the religious right's political resurgence ensures the continuation of the Kashmir jihad policy, with the Pakistan military's blessings" [14]. Scale, as the report notes, is genuinely contested: Indian official estimates put foreign militants at 1,000–2,500 with a foreign-to-indigenous ratio of 2:3, while "another calculation reverses this ratio to 3:2" [14] — the intervention is decisive in capability and direction even where it is a minority in headcount.

Kargil in 1999 exposed the arrangement's logic at full stretch: Pakistan-backed insurgents and regular troops intruded "several miles inside the Indian side of the LOC" near the town midway between Srinagar and Leh; India threatened to carry the war across the line; Clinton's mediation extracted an unconditional withdrawal; Sharif's attempt to blame Musharraf triggered religious-ally riots over a "Kashmir sell-out" and, in October 1999, the coup that installed Musharraf himself [14]. The ICG draws the historical rhyme explicitly — Kargil "had close parallels to the tribal invasions of 1947 and Pakistan's infiltration of the ceasefire line in 1965. The policy decision on both occasions was taken by the military," and in 1947 as in 1999 "the civilian leadership was ill-informed of the military implications of intervention" [14].

After 11 September 2001, the policy survived its own public renunciation. Musharraf distanced himself from the Taliban and pledged a clampdown, but "many of the steps taken by the military-led government have… been cosmetic": over 2,000 radicals arrested in January 2002, "most… released"; LeT and JeM banned, yet "they re-emerged under different names and remained active across the LOC," requiring fresh bans in November 2003; the ISI chief was removed in an "ostensible" restructuring (and given a military industrial conglomerate to run) while the ISI "still continues… to play a proactive role in implementing interventionist policies in Kashmir," with "independent evidence to confirm that infiltration continues across the LOC" [14]. Only escalation worked, briefly: after the December 2001 Parliament attack attributed to Jaish and Lashkar, India's massing of troops brought war close enough that the military imposed "some, most likely temporary, curbs on militant crossings" [14]. By late 2003 the structure was stable and legible — a military running Kashmir policy in line with its institutional preferences, religious allies supplying ideological cover and street power, banned groups persisting through renames, and India's privately preferred LoC-as-border outcome blocked not least because, as the report's Indian-preferences analysis concedes, New Delhi's own dialogue offers stumbled over definitions and over militant wings "not under their direct or immediate control" on the Kashmiri side [14]. For this paper, the Kashmir chapters complete the typology of undrawn lines: a maritime boundary can await a datum, an enclave can await a parliament, but a ceasefire line manned by proxies has acquired a constituency for its own non-resolution — the map is wrong, and powerful institutions on both sides of it are paid, in legitimacy and budget, to keep it that way.

The Indian crackdown

The crackdown is best understood as delimitation by other means: unable to settle the line, New Delhi undertook to hold the territory behind it, at a density that itself became a fact of Kashmiri life. "India's immediate response to the upsurge in Kashmiri militancy after 1989 was to deploy and use force. By 1994 there were in excess of 500,000 security personnel in Jammu and Kashmir, drawn from the army and paramilitary units such as the Border Security Forces" [14]. The early methods were, in the ICG's words, "crude and ineffective counter-insurgency measures… that resulted in massive human rights violations, dislocating life in and around Srinagar and producing a groundswell of

anti-Indian sentiment. Such measures included security cordons that separated men from women for long periods, the arbitrary arrest of Kashmiri Muslim males and a high incidence of custodial deaths associated with detentions without trial or judicial review" [14]. The legal architecture matched the street practice: the Terrorism and Disruptive Activities Prevention Act, under which "between 1987 and 1995, over 76,000 people had been arrested… with less than 2 per cent of detainees being convicted," was scrapped in 1995 under pressure and in recognition "that it had not worked" — yet in March 2002 parliament passed a replacement anti-terrorism bill at a special joint sitting that "was denounced by its own Human Rights Commission" [14].

The occupation of the interior was literal as well as legal. A special report the Indian government itself submitted to the National Commission on Human Rights in 1998 "pointed out that 758 educational institutions within the state had been destroyed" — omitting, as the ICG notes, "that in part this reflected the conversion of schools and colleges to army barracks"; most schools around Srinagar, Poonch and Anantnag eventually reopened, "but enrolment is poor," and a 2001 government report "disclosed that nearly all the sports and recreational facilities of the state were closed or damaged" [14]. Layer onto this the social ledger already recorded: over 200,000 Kashmiri Hindus still refugees in 2000, the tourist industry devastated, and "an entire Kashmiri generation… exposed to the immediacy and apparent glamour of violence" [14]. A state that converts its schools into barracks has announced, in bricks, which of its functions it expects the territory to need.

Militarily, the crackdown presided over the conflict's mutation rather than its suppression. From 1995–96 India "sought to regain Kashmiri trust by holding elections," but "the crisis in Kashmir shifted from insurrection to full insurgency, with numerous foreign militant groups fighting both each other and the Indian security forces"; the shoot-to-kill policy "encouraged, in the opinion of independent observers, faked 'encounter' killings," and "since the mid-1990s, more than 50 militant groups have been active in Kashmir, including militants who have switched sides and are involved in counter-insurgency operations on behalf of the Indian state" [14] — the state franchising coercion to renegades, a further blur in a battlespace already defined by blurred lines. Elections, the intended normalisation ritual, were held inside the violence rather than after it: the 1996 poll ran on low turnout amid "widespread violence and intimidation by both Indian security forces and the militant groups," and in 2002, as in 1997, "militant Islamic groups carried out disruptive attacks on polling stations and against candidates. Hundreds were killed, including political activists and even several candidates, including the much-respected Law Minister Lone" [14] [14]. Turnout in 2002 nonetheless reached around 44 per cent, "higher than expected" though varying widely by district [14] — participation that coexisted with, and did not dissolve, the underlying refusal: standing for election entailed accepting the state's constitutional status, which is precisely what the APHC's boycott declined to concede [14].

Accountability arrived late and partial: national and state Human Rights Commissions brought "greater accountability and oversight of the activities of Indian security forces," yet "continued abuses have held back the normalisation of state politics" [14]. The crackdown's balance sheet, on the ICG's evidence, is therefore double-entry. It preserved India's possession of the Valley — the line held — while systematically devaluing the possession: every cordon, custodial death and converted school converted another cohort to the proposition that the line was a prison wall rather than a border. In the vocabulary of this paper, the crackdown treated Kashmir as a delimitation problem — acreage to be retained — when the preceding sections show it to be a legitimacy problem, and legitimacy is the one quantity that half a million troops cannot survey into existence.

UN initiatives and stalemate

Kashmir is the dispute in this paper that was formally handed to the world's referee — and the ICG's history is, in large part, an account of why the referee's whistle stopped mattering. India itself made the referral: in late December 1947 it took the dispute to the UN under Article 35 of the Charter, inviting investigation and "advisory, non-mandatory findings," and itself recommended a referendum once peace was restored and tribal and Pakistani forces withdrawn, confident it could win one [14]. The report's judgement is blunt: "Nehru's decision to submit the Kashmir dispute to the UN, thus inviting its internationalisation, was a mistake from the Indian viewpoint" — though it immediately cautions that "there is a tendency to exaggerate the UN's role in the Kashmir dispute… The dynamics of the crisis have always remained within India and Pakistan, and the UN has little power to influence them directly" [14]. What the UN built was real but conditional: Security Council Resolutions 38, 39, 47 and 51, the UNCIP commission, an observer group that "remains in existence today," and the 1949 Karachi ceasefire — followed by the substantive project of setting conditions, beginning with Pakistani withdrawal, for a referendum "to 'ascertain the wishes of the people' as to whether they wished to belong to India or to Pakistan" [14].

The referendum died of its own preconditions, and of a suppressed premise: neither state "entertained the idea of allowing it to become independent," so the question on offer excluded what would become the Valley's loudest answer [14]. India would not accept a referendum under UN auspices; Pakistan, holding roughly a third of the state, "refused to believe that a referendum held under Indian administration would be free and fair"; and the compromise device — a referendum under a "neutral administration" staffed by the UN — India rejected as equating its administration, grounded in the Dogra accession's legality, with Pakistan's "unlawful occupation" of Azad Kashmir and the Northern Areas [14]. The one document that saw past the impasse was Sir Owen Dixon's 1950 report, "one of the first documents to grasp the necessity of dealing with Kashmir as a series of overlapping territories that needed to be disaggregated down to the district and tehsil levels" to honour its pluralism — with "a candid recognition that the solution to the Kashmir problem does not lie in a referendum at all," and an implicit partition along the ceasefire line, whose flaw Dixon also saw: it "gives Pakistan no stake in the Valley, which remains the heart of the former kingdom and by far its most populous and prosperous area" [14]. Thereafter the parties sorted themselves by advantage: India, holding the Valley and disappointed that the UN never condemned Pakistani aggression or secured its withdrawal, "abandoned its multilateral approach for one premised on bilateralism and physical occupation"; Pakistan, marginal in the Valley, "remained attached to the UN approach" it had gained from; the Council debated "without any sense of direction or purpose, and without even reaffirming earlier resolutions," its logjam thickened by vetoes — India's position increasingly buttressed by the Soviet Union — until by the 1966 Tashkent Declaration the debate "had become a sideshow" [14].

Bilateralism's founding document then repeated the exclusion that doomed multilateralism. The 1972 Simla Accord — after 1965's Operation Gibraltar foundered on the Valley's refusal to rise, and 1971's war transformed the balance — converted the ceasefire line into a modified LoC intended as a "'soft border,'" with the two Kashmiri governments to develop relations, tacit cross-line movement, the referendum principle apparently abandoned, and "this de facto partition… meant to conclude the matter" [14]. But the soft border "was not actually spelled out" anywhere in the text; India's attempt to bury the UN route was fudged by Article 1's own reference to the UN Charter, which Delhi read as a no-war commitment and Pakistan as "a continuing commitment to resolve the issue in line with the various UN resolutions"; and "the problem of the Simla conference was the lack of any explicit Kashmiri representation" — an omission repaired by neither side since [14]. The stalemate's symmetry completes the picture on Pakistan's side of the line, where Azad Kashmir's "quasi-autonomous" status was hedged from the start: the Northern Areas (Gilgit, Hunza, Baltistan) were hived off as a centrally administered region whose people could not vote in AJK elections, and Islamabad, like Delhi, "has not shied from dismissing Azad Kashmiri leaders or imprisoning them when they threatened the status quo" [14]. Hence the report's bleakest convergence: "since the mid-1970s, the only common ground between New Delhi and Islamabad has been their determination to marginalise and indeed to repress calls for Kashmiri independence" — a consensus that "emphasises the hollowness of those resolutions as a way forward," since the resolutions never contemplated the option both capitals unite to suppress [14]. In this paper's terms, the UN phase bequeathed Kashmir a line recognised by everyone and consented to by no one it enclosed — the precise inverse of Cooch Behar, where the people were asked, and of the Saudi–Sudan Common Zone, where the map was deliberately never finished.

What view should be taken of colonial treaties?

The Cameroon–Nigeria judgment of 10 October 2002 is this paper's courtroom centrepiece, and its hardest question is not where the boundary runs but what the law should make of the instruments that drew it — treaties signed in the partition of Africa, by powers that regarded the continent as disposable, with chiefs who were never asked about lines at all. The documentary chain the Court worked through is itself a century of empire: the Franco-British Convention of 1906 and its 1910 Protocol, the Franco-German instruments of 1894 and 1908, the Anglo-German Agreements of 1893 and 1906, and the two 1913 Agreements — London, 11 March 1913 (the frontier "from Yola to the Sea," some 1,100 km, plus Cross River navigation) and Obokum, 12 April 1913 (demarcation, with eight maps) — followed, after Versailles transferred Germany's territories to French and British mandates, by the Milner-Simon Declaration of 1919 and the Thomson-Marchand Declaration of 1929–30, approved in the Henderson-Fleuriau Exchange of Notes of 9 January 1931 [13].

The Court's operative answer was austere: it tested the 1913 Anglo-German Agreement against the parties' own later conduct and found it "valid and applicable in its entirety" — pointing, among other acknowledgments, to the formal requests submitted into the 1980s by the Nigerian Embassy in Yaoundé and Nigerian consular authorities before visiting their nationals in Bakassi, conduct acknowledging Cameroon sovereignty over the peninsula [13]. Nigeria's three alternative titles failed in sequence. The claimed ancient title of the Kings and Chiefs of Old Calabar could

not vest in Nigeria at independence in 1960, because "on the date of its independence Cameroon succeeded to title over Bakassi as established by the Anglo-German Agreement of 11 March 1913"; historical consolidation "cannot in any event vest title to Bakassi in Nigeria, where its 'occupation' of the peninsula is adverse to Cameroon's prior treaty title and where, moreover, the possession has been for a limited period" [13]. The Court accordingly found "no need to address the arguments… as to the severability of treaty provisions" and "not… necessary to pronounce upon the arguments of uti possidetis advanced by the Parties in relation to Bakassi" [13] — treaty title, once found, simply outranked everything else.

A separate opinion appended to the judgment supplies the philosophy the majority left implicit, and it is bracing in both directions. On the protection treaties with African rulers, it is dismissive: the agreements signed "with the numerous 'kinglets'… were simply intended to warn the other colonial Powers that specific parts of the black continent were now a possession falling within the zone of influence of a given European State," under the Berlin Act's notification regime; "They had no validity in international law. To accord them such validity now would be to open a Pandora's box. The chiefs themselves had no precise idea of the territorial boundaries of the areas which they governed" [13]. It adds the anthropological root: "the notion of a linear frontier is not an African one. It was imported into the continent by the colonial Powers" — pre-colonial boundaries were natural features and chiefly power ran over people, not surveyed land — and the signing practice was correspondingly casual, the instruments "rarely signed by the duly authorized representatives of the colonial State" but by "explorers, merchants, navigators, sometimes junior military officers" [13]. Gonidec's doctrine completes the picture: annexed territories became integral parts of the colonial state, freely disposable (the 1911 Franco-German swap is the example), and "in international terms, the annexed countries lost all personality… considered never to have been legal persons" — even though the very act of concluding "treaties" with their authorities implied the personality the doctrine denied [13].

Yet the same opinion insists on the rehabilitation of the delimitation treaties through intertemporal discipline: "the problem here is not to make a value judgment today concerning those rules and practices but rather, in the context of intertemporal law, to take note of them. The Court does not have the authority to revise international law" [13]. The political settlement of 1964 does the rest: the OAU's Cairo resolution AGH/Res.16(1), by which member states "pledge themselves to respect the frontiers existing on their achievement of national independence" — uti possidetis juris, whose purpose the Frontier Dispute Chamber stated as preventing "the independence and stability of new States being endangered by fratricidal struggles provoked by the challenging of frontiers following the withdrawal of the administering power" [13]. Frontiers so inherited "can certainly be modified, not by invoking their technical defects but only in accordance with the rules of international law — in other words, by mutual agreement or by judicial decisions," and a court's office is bounded: it "may remedy material defects but not alleged legal errors… A court cannot change a clear provision" [13]. The opinion's conclusion endorses the result — the boundary derives from Thomson-Marchand, the 1913 Agreements and the 1946 Order in Council: "Bakassi is Cameroonian" [13]. That is the view this paper takes away: colonial treaties are to be read as law of their time and as title for ours, not because their origins were just — the opinion's own account makes that untenable — but because the alternative, reopening every inherited line to equitable revision, prices stability at zero. The map is wrong in a hundred documented ways; the Court's answer is that wrongness, without more, is not a cause of action.

Frozen Conflicts and Administered Territories

Western Sahara: the UN referendum mission and its military and police strength

MINURSO is the frozen conflict rendered as an organisation chart: a mission whose very name — the United Nations Mission for the Referendum in Western Sahara — memorialises a vote that has never been held, deployed along a sand berm that was supposed to be a ceasefire line and has become, in practice, the most heavily observed undrawn border on earth. The Secretary-General's report S/2024/707, covering September 2023 to August 2024, prices the observation. "As at 31 August 2024, the military component of MINURSO consisted of a total of 229 military personnel, of which 57 were women," with women observers at 25 per cent of deployed strength, "exceeding the target of 20 per cent for 2024 set in the uniformed gender parity strategy 2018–2028" [15]. The annexed strength table distributes those 229 across some forty contributing countries — almost all as unarmed experts on mission, with token staff and troop contingents (Bangladesh's twenty troops, Ghana's seven staff officers) — while the police component's columns, formed units and individual officers alike, record zeros against country after country [15]: the referendum's civilian-policing arm exists, three decades on, essentially on paper.

What the 229 actually do is patrol a line in two unequal halves. West of the berm, in the Moroccan-held territory, access is comparatively deep: 9,049 visits in the year to Royal Moroccan Army headquarters, units, strong points and observation posts, ground patrols across a mission total of some 386,675 kilometres, and 140 aerial reconnaissance flights — though observers "continued to be advised by Morocco not to visit forward units without an Army escort," and the reconnaissance flights shrank from a 15-kilometre to a 10-kilometre standoff from the berm for safety [15]. East of the berm, in the POLISARIO-held strip, the regime is rationed: ground patrols, suspended in August 2022, resumed only in December 2023 at once — later twice — weekly frequency, escorted by a POLISARIO liaison officer, totalling 1,392 patrols over 46,597 kilometres; POLISARIO "continued to restrict MINURSO ground patrol movements to a 20-kilometre distance from each team site," and "helicopter reconnaissance flights have not been permitted east of the berm since November 2020" [15]. The liaison asymmetry is starker still: while coordination with the Moroccan Army ran to headquarters visits and seven working-level meetings, "the MINURSO Force Commander remained unable to establish direct contact with the military leadership of the Frente POLISARIO and all communications continued to be conducted through written correspondence" [15].

The mission's workload now measures a war resumed at low intensity. The Moroccan Army reported 164 firing incidents at or near the berm in the year — "approximately 75 per cent of which were concentrated in Mahbas" — and MINURSO mounted 127 investigation patrols to 357 reported sites; it also received information on "14 alleged aerial strikes conducted by uncrewed aerial vehicles of the Royal Moroccan Army east of the berm," visiting twelve sites after security assurances from both parties and being refused assurances twice [15]. Firing incidents near the Smara, Aousserd and Mahbas team sites drove the mission to install safety bunkers with sandbags and overhead protection [15]. The one unambiguously functional channel is mine action: full demining east of the berm resumed in January 2024 for the first time since September 2019 — 14,032 kilometres of routes assessed, 3,466,915 m² cleared, 359 missions, against a remaining ledger of 23 of 62 known minefields and 40 of 539 cluster-strike areas still uncleared [15]. A Department of Peace Operations capability study (August–November 2023) recommended professionalising the headquarters — staff officers in place of observers for dedicated functions — and enlarging the Medical Unit to make aeromedical evacuation real for the isolated team sites [15]: the adaptations of a mission settling in.

The Secretary-General's own observations supply the verdict this section exists to record: "the lack of a full ceasefire in effect between Morocco and Frente POLISARIO remains a major setback," the "continued entrenchment of a deteriorated state of affairs… is alarming and unsustainable," and, with "the fiftieth anniversary of the conflict approaching," the goal remains a solution "that will provide for the self-determination of the people of Western Sahara" under the resolution chain running from 2440 (2018) to 2703 (2023) [15]. For this paper's typology, Western Sahara completes the set that Kashmir began. Kashmir's line has constituencies for its non-resolution; the berm has something quieter — a 229-person institution whose mandate, patrol patterns, capability studies and bunker-building all presuppose the line's permanence, while its founding purpose, the referendum, recedes into its own acronym. The map here is not wrong. It is unfinished, and the unfinishedness has acquired staff, a budget line, and a gender-parity target.

Humanitarian activities and human rights in Western Sahara

Behind the berm's military geometry sits the population the unfinished map governs, and the Secretary-General's 2024 report prices their condition with the same flat precision it applies to patrol kilometres. On the Algerian side of the line, UNHCR, UNICEF, WFP and WHO sustained assistance to Sahrawi refugees in the five camps near Tindouf "despite critical funding challenges": the first consolidated Sahrawi Refugee Response Plan (2024–2025) "was funded at about 37 per cent of the $110.5 million required for the first year" [15]. Food security is the sharpest edge: the 2024 Food Security Assessment found worsening conditions driven by "a 30 per cent reduction in the dried food basket owing to the rise in food prices," with "over 80 per cent of the camp population relied on food assistance" [15]. The system's longer investments persist alongside the emergency — a five-year education strategy (2021–2025) benefiting 40,000 children and teachers, WFP mid-morning snacks for 33,000 schoolchildren, UNICEF emergency-preparedness training for 200 Sahrawi health workers, and UNHCR water provision through boreholes and distribution networks [15]. What does not persist is contact: the confidence-building measures mandated since resolution 1282 (1999) "to allow family links between Sahrawi refugees in the Tindouf camps and their communities of origin in Western Sahara, remained on hold," with neither party expressing "immediate interest in further work on these issues" [15] — the population divided by the line may not even visit across it.

The human-rights record is defined, first, by an absence. "The Office of the United Nations High Commissioner for Human Rights (OHCHR) was not able to visit Western Sahara for the ninth consecutive year despite multiple official requests and despite the Security Council, in its resolution 2703 (2023), urging enhanced cooperation, including

through facilitating such visits" — and that lack of access to "independent, impartial, comprehensive and sustained monitoring… remained detrimental to a comprehensive assessment" [15]. What reaches OHCHR from outside is a consistent pattern: "reports of hindrance, intimidation and harassment against Sahrawi activists advocating for the right to self-determination, pointing to an increasingly shrinking civic space," corroborated in specifics — the Working Group on Arbitrary Detention's finding (November 2023) that a Sahrawi journalist's fair-trial rights were violated, and a Moroccan court's June 2024 decision upholding the denial of a law licence to a Sahrawi defender "due to his political opinions and affiliation," with its "possible deterring effect on lawyers to freely represent clients" [15]. The Committee on the Elimination of Racial Discrimination added its concern at the "targeting, intimidation and surveillance faced by Sahrawi activists, human rights defenders, student movements and Sahrawi organizations," recommending non-interference and investigations with redress [15]. Detention conditions are the gravest file: overcrowding, inadequate healthcare and training, denied transfers nearer home, and the Gdeim Izik group "dispersed and held in prisons outside of Western Sahara while serving lengthy prison sentences," their isolation and discriminatory treatment reportedly damaging their health — imprisonment the Working Group considered arbitrary for 18 members, calling for "immediate release of the detainees and appropriate redress and reparations" [15]. Reports also described natural-resource exploitation "without meaningful consultations and participation by affected communities," with investments "frequently prioritizing interests that do not address the needs of the local Sahrawi population" [15].

The report records the contest of accounts as faithfully as the allegations. Morocco replied through four notes verbale emphasising its treaty-reporting cooperation and its New Development Model — "sustainable development and generating employment… by leveraging natural resources exploitation and local participation" — refuted the Gdeim Izik detention allegations, and counter-charged violations inside the Tindouf camps themselves: "severe restrictions on fundamental freedoms, including freedom of movement, assembly, opinion and trade, and absence of access to justice" [15]; its National Human Rights Council reported detention visits producing improvements [15]. POLISARIO's letters of July and August 2024 alleged civic-space restrictions, arbitrary detention and harsh sentencing west of the berm, and "expropriation of land and property belonging to Sahrawi civilians, demolition of homes and the denial of the rights to peacefully protest" [15]. Around this contested interior, the diplomatic weather hardened during the same year: France's July 2024 letter endorsing autonomy under Moroccan sovereignty as "the only basis" for settlement drew Algeria's "profound disapproval" and the withdrawal of its ambassador, and POLISARIO's declaration that France stood "in the same place as the occupier" and was unwelcome even in MINURSO [15]; Chad's consulate in Dakhla was condemned as a breach of the territory's non-self-governing status [15]. And the material base of observation itself is thinning: the General Assembly appropriated $69.3 million for the mission's year, while unpaid assessments to MINURSO's account stood at $54.2 million — a shortfall the report says is "increasingly affecting the ability of the Mission to sustain mandate implementation, as well as its presence throughout the Territory" [15]. The frozen line, in other words, is not costless stasis: it is a half-funded humanitarian protraction, an unmonitored rights space entering its ninth year, and a refugee population — fed at 80-per-cent dependence on a shrinking basket — for whom the map's unfinishedness is measured in calories, licences and prison transfers rather than in kilometres of berm.

Movement and access in the West Bank

Every boundary in this paper so far has been a line between jurisdictions. The West Bank presents the inverted case: a boundary system drawn inside a single occupied territory, hundreds of micro-lines whose combined effect is to govern the movement of 3.4 million Palestinians without ever quite constituting a border. OCHA's fact sheet (data as of December 2025) counts the system: "925 movement obstacles that permanently or intermittently restrict the movement of 3.4 million Palestinians across the West Bank, including East Jerusalem. This is 43% more than the annual average of 647 movement obstacles in the preceding 20 years" — with about half the increase since early 2023 recorded by March 2024 [16]. The composition matters more than the count: 89 checkpoints staffed 24/7; 218 partial checkpoints, of which 152 have gates (90 usually open, 62 frequently closed); 232 road gates, 159 of them frequently closed; 114 linear closures; 167 earthmounds; 105 roadblocks [16].

The trend within the trend is hardening. "Checkpoints and road gates comprise nearly 60% of movement obstacles and form the backbone of the current access regime, signaling a further entrenchment of movement restrictions. Two decades ago, this type of elaborate infrastructure accounted for 15% of movement obstacles whereas ad-hoc or temporary obstructions such as earthmounds and roadblocks constituted about 73%" [16]. The gate is the emblematic instrument — in 2025 alone OCHA documented over 120 new road gates installed, their share of all obstacles rising from 34 per cent in February 2025 to over 41 per cent by December [16]. A gate is a border reduced to a hinge: a line that exists only when closed, staffed at another's discretion, and normalised by being usually open. The sys-

tem's geometry is connective, not merely obstructive — 459 obstacles, about half, "block or hinder access between Palestinian towns and villages and main roads," funnelling traffic "onto longer, secondary road networks," with 121 of them blocking Road 60, "the main north-south artery in the West Bank," and about a fifth carrying governorate-wide or multi-governorate impact [16]. The density peaks in Hebron's H2 area, which holds 34 of the 89 checkpoints and 101 of the 925 obstacles for a strictly regulated zone "home to about 7,000 Palestinians and several hundred Israeli settlers," access since October 2023 limited to registered residents through three checkpoints operating 07:00–20:00 [16].

Above the mesh sits the macro-line: "the 712-kilometre-long Barrier (of which 64% is built and 85% runs inside the West Bank), together with its gate and permit regime, remains the single largest obstacle to Palestinian movement" — only 3 of its 13 checkpoints usable by West Bank ID holders with permits "difficult to obtain," and most permits "revoked or suspended since 7 October 2023, including over 800 permits that had been issued for nationally recruited humanitarian workers" [16]. The strip between Barrier and Green Line — the "Seam Zone" — is this paper's enclave logic at territorial scale: over 14,000 Palestinians live inside it and must coordinate every crossing to reach services, markets and family; 62 agricultural gates, "the majority of which are closed year-round," govern farmers' access to their own land (74 per cent opened for limited days during the 2025 olive harvest), while 98 communities must now obtain military "prior coordination" to reach farmland near settlements twice a year [16]. In September 2025 three villages on Jerusalem's side of the Barrier — Beit Iksa, An Nabi Samuel and Al Khalayla, some 2,600 people — were issued "Seam Zone" permits for the first time, formalising their residents' lives as licensed crossings [16].

The friction is continuous and measurable: about 2,000 access incidents in 2025 (38 a week; an average of 15 ad-hoc "flying" checkpoints weekly), WHO's documentation of 233 attacks on health care — 84 per cent involving obstructed access — and a health-permit refusal-or-pending rate that doubled, from 18 per cent of 87,427 applications in 2022's terms to 36 per cent in 2025 [16]. The legal frame is equally explicit in the fact sheet: the occupying power must facilitate free movement, with exceptions "only for imperative reasons of security and only in response to specific security threats"; the Barrier's in-West-Bank sections and their regime were held unlawful in the ICJ's 2004 Wall opinion; and the Court's Advisory Opinion of 19 July 2024 found Israel's continued presence in the occupied territory itself unlawful, its near-complete separation measures in breach of the racial-discrimination convention — a reading OHCHR's January 2026 thematic report extended to the two-tier legal system governing movement, land and livelihoods [16]. Set beside the berm and the LoC, the West Bank's sieve is the third species of frozen line: not a wall between states, nor a scar between armies, but an internal lattice whose purpose is not to keep an enemy out — it is to make a population's own geography conditional, gate by gate, permit by permit, until conditionality itself is the map.

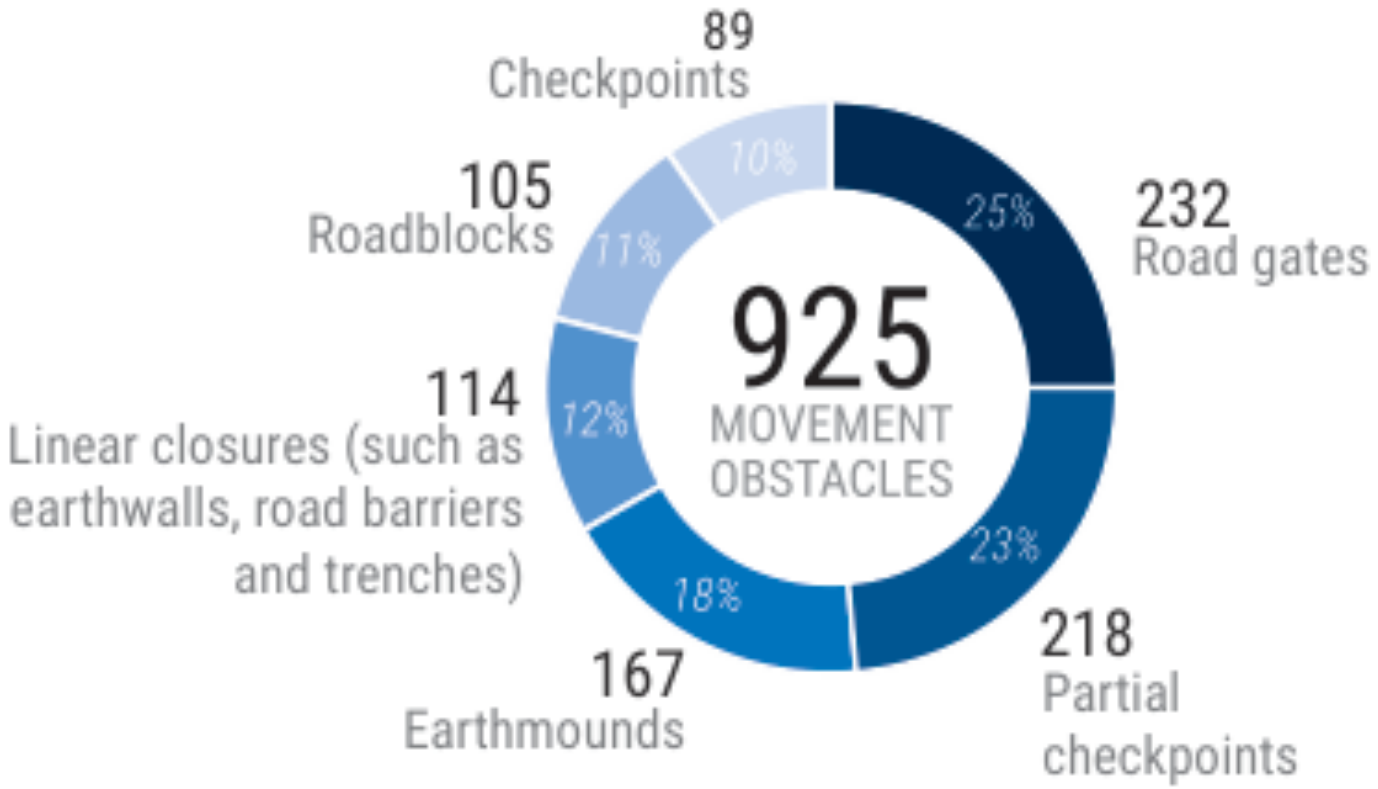


Exhibit. Movement obstacles in the West Bank by type: 925 road gates, partial checkpoints, earthmounds, linear closures, roadblocks, and checkpoints recorded by OCHA. Source: S0020.

Lake Chad Basin: effectivités and historical consolidation of title

At the northern end of the boundary, Nigeria's argument changed shape. Around Bakassi it had asked the Court to let decades of administration displace a treaty; in the Lake Chad sector it argued there was a gap to fill — an absence of delimitation that its effectivités, "which confirmed its historic title," could supply, citing "the occupation of Darak and the surrounding villages by Nigerians, together with a whole series of other facts which, according to Nigeria, clearly demonstrated the exercise of its sovereignty," plus alleged Cameroonian acquiescence [13]. The supporting theory was historical consolidation: Nigeria claimed the villages around the lake on the strength of "legal jurisdiction, taxation, authority of traditional rulers and the fact that the settlements were populated by Nigerian nationals" — long use "coupled with a complex of interests and relations which, in themselves, have the effect of attaching a territory" [13]. One member of the Court recorded "much cause for legal concern" at the refusal even to assess this evidence; the majority's doctrine, endorsed in a separate opinion, was unyielding.

The doctrine has three tiers, and effectivités lose in two of them. Where a legal title exists, "that title must prevail over effectivites": the Chamber in the Frontier Dispute (Burkina Faso/Republic of Mali) had put it beyond ambiguity — "[w]here the act corresponds exactly to law, where effective administration is additional to the uti possidetis juris, the only role of effectivite is to confirm the exercise of the right derived from a legal title. Where the act does not correspond to the law, where the territory which is the subject of the dispute is effectively administered by a State other than the one possessing the legal title, preference should be given to the holder of the title" [13]. Administration therefore counts only as confirmation, or in the residual case "that 'no conventional line of sufficient topographical precision exists or if there are gaps in the frontiers otherwise established, or if a conventional line leaves room for doubt'" [13]. Applied to the lake, the Court held that "any Nigerian effectivites are indeed to be evaluated for their legal consequences as acts contra legem" — acts against the law, not sources of it [13]. Acquiescence, Nigeria's bridge between the two, was construed strictly: it "requires a long period and a clear and unequivocal voluntary acceptance, which is not the situation in the present case," and the very existence of the Lake Chad Basin Commission, "of its work and of the way in which the two Parties continuously collaborated" in it, evidenced the opposite of acceptance [13].

The general theory fared no better. Of historical consolidation the judgment held that, apart from the Fisheries case, the notion "has never been used as a basis of title in other territorial disputes," that "nothing in the Fisheries Judgment suggests that the 'historical consolidation' referred to... allows land occupation to prevail over an established treaty title," and that "the established modes of acquisition of title... take into account many other important variables of fact and law" which the "over-generalized" concept ignores [13]. The evidentiary mass behind the theory was formidable — for Bakassi, Nigeria had catalogued indirect rule through Warrant Chiefs from 1922, native courts, taxes paid to Calabar and Eket, a Methodist school at Abana in 1937, a census under the Eket Division in 1953, electoral wards, oil exploration permits, customs, passports, and the claim that Cameroon had not challenged its administration between 1960 and 1972 [13] — and it changed nothing, because the question was never how much administration there was, but whether administration can outrank an instrument. It cannot.

Beneath the doctrine sits the human fact that makes the lake the hardest sector of all. The "Nigerian" villages were "relatively recent," established "following the retreat of the shores of Lake Chad"; as Nigeria's own counsel conceded, "the villages move with the water." The opinion draws the moral plainly: villagers settling beside or straddling an African boundary do so "because it is a custom in Africa to ignore linear boundaries, which are a foreign importation, especially when men and women of the same ethnic origin live on the other side" [13]. A population that follows a shrinking lake, taxed and judged and schooled by whichever administration is present, generates effectivités the way weather generates erosion — continuously, genuinely, and without legal intent. The Court's answer protects the map from that weather: title drawn in 1913 governs villages founded decades later on land the water abandoned. Whether that is the law's wisdom or its blindness is the question the separate opinions leave open — but it is the premise on which the entire judgment, and much of this report, rests: the line on paper outranks the life on the ground, until a court or a treaty says otherwise.

Islands and Sovereignty

Island disputes and the law of the sea: sovereignty and delimitation examined

The island is the law of the sea's force multiplier, and this paper's cases keep returning to the arithmetic. Under the LOS Convention a small island carries a territorial sea of up to 12 nautical miles and, beyond it, resource jurisdiction to 200: "An exclusive economic zone (EEZ) of 200 miles around a small island can encompass about 125,660 square nautical miles of ocean space," covering "the living and non-living resources [such as fisheries and hydrocarbon deposits], the waters superjacent to the seabed and of the seabed and its subsoil," with a continental-shelf claim that may run beyond 200 miles where Article 76's conditions are met [17]. Hence the Wall Street Journal's charge that the Senkaku/Diaoyu quarrel is "claim-staking mania inspired by the UN Convention on the Law of the Sea" [17]. The IBRU examination on which this section draws accepts the catalyst but rejects the causation: the Convention, in force from 16 November 1994, prompted states to review their offshore situations, yet "almost all of the island disputes that now interfere with certain bilateral relations predate the development of many, if not most, of the LOS concepts" — China dates its South China Sea claims to the second century BC — while what has genuinely changed is that "maritime resource jurisdiction is at least an implicit issue in almost all island disputes," driven by offshore food and energy demand as land resources dwindle [17]. The shelf idea itself is older than the Convention: the Truman Proclamation of 1945, codification in 1958, 52 shelf claims by 1971, 80 EEZ claims by 1990 [17].

The doctrinal core is a sequencing rule this paper has met at every island: the sea follows the land, never the reverse. Article 121(1) defines an island as "a naturally formed area of land, surrounded by water, which is above water at high tide," and Article 121(2) accords it the zones "applicable to other land territory" — so that "in many of the island sovereignty disputes the value of the land is less than the maritime zone that, under the provisions of the LOS Convention, possibly could be claimed from that island" [17]. But the Convention "does not contain any provisions in any of its articles for resolution of disputes over any territory, including islands"; its application "is premised on the assumption that a particular state has undisputed title over the territory from which the maritime zone is claimed" [17]. Title must therefore come from elsewhere — treaty, effectivités, adjudication — and attempts to run the logic backwards are fallacies: Malaysia's claim to Spratly features because they fall within its 1979 proclaimed shelf, or China's implication that the Senkakus are the "natural prolongation of the Chinese mainland," both mistake the consequence for the source, since "there is no rule in international law that prescribes sovereignty over islands on the basis of making a maritime claim" [17]. The Convention's single island-specific discipline is the exception: Article 121(3) provides that "Rocks which cannot sustain human habitation or economic life of their own shall have no exclusive economic zone or continental shelf" — a rule whose operative terms ("rock," "sustain," "economic life of their own") the drafters left undefined, and whose first clear official application came when the United Kingdom, acceding in July 1997, declared Rockall incapable of sustaining human habitation and undertook to redraw the fisheries limits it had claimed from it [17] [17].

Disputes accordingly come in two types — sovereignty over the feature, and the effect a feature should have on a delimitation — and practice keeps merging them: tribunals asked for title are asked for the boundary too, as in the ICJ's 1992 Gulf of Fonseca decision resolving island sovereignty and maritime jurisdiction together, and in the Eritrea–Yemen arbitration over the Hanish Islands, where Yemen demanded that the Red Sea centreline be determined as part of the same award [17] [17]. Where sovereignty is shelved rather than settled, the toolbox is by now standard: joint fishery conservation zones (Argentina and the United Kingdom around the Falklands/Islas Malvinas), joint development zones for hydrocarbons, the Canada–Denmark device of ignoring the feature altogether — proposed for Machias Seal Island as joint management plus a delimitation that disregards it, each side's claim expressly preserved — and the calibrated middle course of limited effect, as in the Italy–Tunisia shelf agreement, where Italian islands close to Tunisia received some jurisdiction but were "otherwise ignored in the boundary delimitation" [17]. Variable breadths do similar work in straits, Turkey claiming 12 miles in the Black Sea and Mediterranean but 6 in the Aegean, matching Greece [17]. Bowett's geography of entitlement — islands as sole units, or conjoined with a mainland, proximate to their own sovereign's coast, straddling a median line, or proximate to a foreign coast — predicts much of the variance, tempered by size, political status and the feature's nature [17].

The examination's verdict, echoing Hodgson's 1973 observation that islands "often have caused problems in the delimitation of boundaries and the extent of national sovereignty and jurisdiction," is that no predictable pattern of island treatment has yet crystallised from state practice: outcomes remain negotiated, arbitrated or endured case by case, and "if the political will exists between the affected parties, then resolution should be achievable for any given dispute" [17]. That is the honest summary of this paper's island chapters. The Convention supplies the multiplier and

the sequencing rule; it deliberately supplies nothing else. Who owns the rock, and what the rock is worth against a neighbour's coast, are questions answered — when they are answered at all — by history, by courts applying other law, or by the parties' willingness to share, shelve or ignore the water the rock commands.

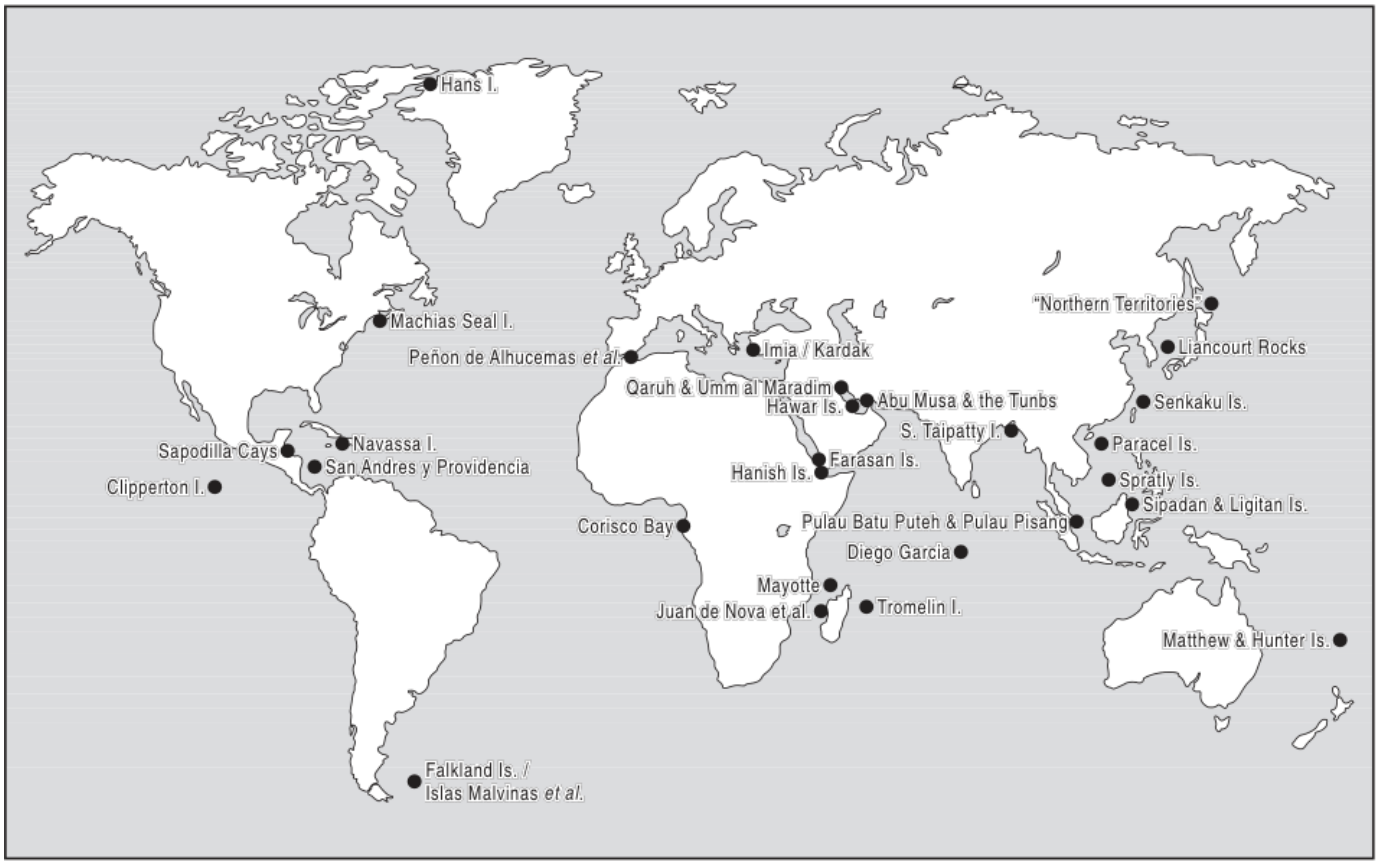


Exhibit. Disputed islands of the world, from Hans Island and the Liancourt Rocks to Abu Musa, the Tunbs, and the Senkaku/Diaoyu Islands. Source: S0008.

Islands and agreed maritime boundaries

If disputed boundaries show the island multiplier at full strength, agreed boundaries show it being negotiated down. Reviewing state practice, the IBRU examination sorts island treatment into three registers: islands "treated as an integral part of states, with no special treatment given to (or taken from) them," islands given "special consideration," and islands given "no consideration at all" [17]. A methodological caution governs the vocabulary: terms like "full weight" and "partial effect" presuppose the equidistance method, yet "international law does not make the use of equidistance mandatory for exclusive economic zone or continental shelf boundaries although there is some preference given to equidistance in Article 15" for the territorial sea — and where agreements use other lines, islands often drop out as factors not because of their own attributes "but to other considerations by the states concerned" [17]. An equidistant line, moreover, "will be only as accurate and precise as the source data" [17].

Full weight is the Pacific norm and the American policy. When the Magnuson Act extended United States fisheries jurisdiction to 200 miles in 1976, "a stroke of a pen" created "approximately 28 maritime boundary situations that formerly did not exist" — from Kiribati and Tonga to the Soviet Union, Canada off four coasts, Mexico, Cuba and Venezuela — and Washington's settled policy was that "in those situations in which the equidistance method provided an equitable solution, all territory would be given equal weight," with only three areas excepted (the Soviet/Russian boundary, the Gulf of Maine, and part of the Bahamas boundary) [17]. The resulting agreements — with Mexico, Venezuela, the Cook Islands, New Zealand for Tokelau, the United Kingdom for the British Virgin Islands and Anguilla, and Niue — use equidistant lines with "the islands of each side treated equally and given full weight," and "virtually all of the negotiated boundaries in the Pacific, which involve island states, are based on the equidistance method" [17]. Bowett's explanation is symmetry: full effect is usually given "where islands of both states have been given full effect in calculating an equidistant line... due to the fact that both parties have offshore islands that balance or offset each other" [17]. Symmetry has its critics — the United States and Netherlands treaties with Venezuela drew complaints from smaller Caribbean states for giving Venezuela's Aves Island full consideration [17].

Special consideration is the discount register, applied by negotiators and courts alike. In the Libya–Malta judgment the ICJ "did not allow Malta, an independent island-state, to receive full consideration" against Libya; in the 1977 Anglo-French Award the Scilly Islands "were not given full weight" [17]. The Italy–Tunisia agreement priced each island separately: the Italian islands near the Tunisian coast "were not considered in the boundary delimitation per se" — uninhabited Lampione received a 12-mile territorial sea, while Lampedusa, Linosa and Pantelleria received territorial seas plus "a one-mile band of continental shelf" [17]. The Netherlands–Venezuela agreement of March 1978 splits the difference geographically: where Aruba, Bonaire and Curaçao lie "only about 30 miles opposite the mainland, the boundary approximates an equidistant line giving full weight to the islands," but in the central Caribbean "the line deviates from the equidistant line, in Venezuela's favour" [17].

The third register is erasure. In the 1988 United Kingdom–Ireland delimitation "the British islet of Rockall was not used" at all [17]. Canada and Denmark drew their equidistance shelf boundary between Canada's eastern Arctic islands and Greenland so that "Points 122 and 123 of the boundary lie on the north and south coasts of Hans Island, respectively" — and because of "problems of sovereignty and the effect this disputed island would have on the delimitation… the parties ignor[ed] the island completely," the line passing through a feature neither would price [17]. Between the registers sits administration without settlement: at the Strait of Hormuz approaches, Iran and Sharjah have jointly administered Abu Musa under a 1971 Memorandum of Understanding and shared an offshore oil concession's revenue, even as sovereignty over Abu Musa and the Tunbs remains disputed between Iran and the Emirates — an arrangement that survived, tensely, Iran's 1992 tightening of control, because the dispute "is essentially a sovereignty issue rather than a maritime resource issue" and the water can be worked while the title sleeps [17].

The lesson for this paper's argument is that "island effect" is not a property of islands. It is a price, quoted case by case: full where islands balance, discounted where they distort, zero where they embarrass, and suspended where the parties prefer revenue to resolution. Equidistance supplies the opening bid; equity — geographic, political, occasionally merely diplomatic — settles the account.

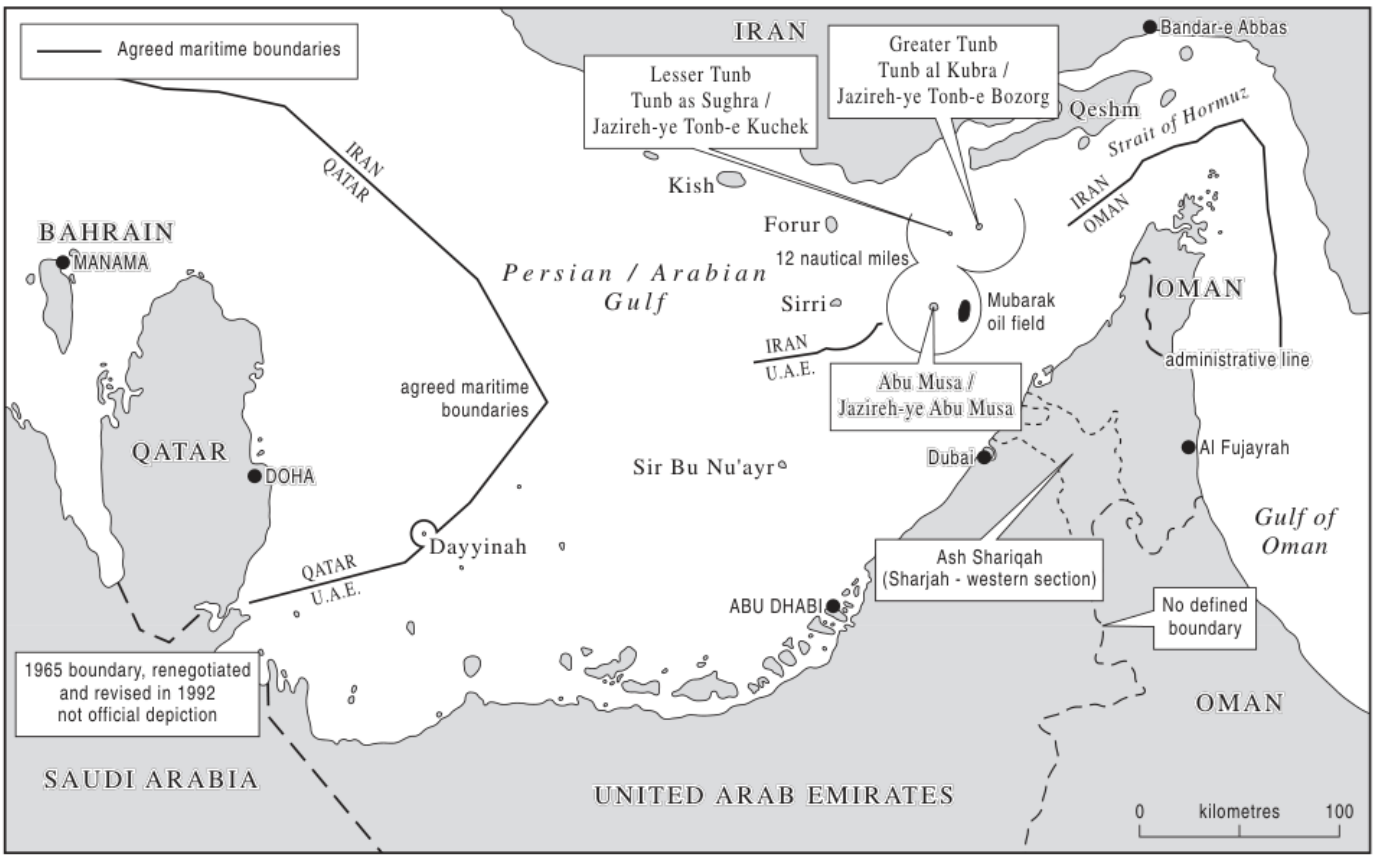


Exhibit. Abu Musa and the Tunb Islands in the Persian/Arabian Gulf, disputed between Iran and the United Arab Emirates. Source: S0008.

Courts, Tribunals, and Peaceful Settlement

The International Court of Justice: institution of proceedings

The Cameroon–Nigeria judgment, whose doctrines fill so much of this report, began as something much smaller than a whole frontier. On 29 March 1994 Cameroon filed an Application concerning a dispute "relating essentially to the question of sovereignty over the Bakassi Peninsula," adding that the maritime delimitation "has remained a partial one" despite many attempts, and asking the Court, "in order to avoid further incidents between the two countries, ... to determine the course of the maritime boundary between the two States beyond the line fixed in 1975" [13]. Jurisdiction was founded on the two parties' declarations under Article 36, paragraph 2, of the Statute — the optional-clause system doing, for once, exactly what it was designed for [13]. Ten weeks later the case changed scale. On 6 June 1994 Cameroon filed an Additional Application extending the dispute to Lake Chad and asking the Court "to specify definitively" the frontier "from Lake Chad to the sea" in a single case; at the presidential meeting of 14 June 1994 Cameroon explained the filing was an amendment rather than a new application, Nigeria declared it "did not object" to that treatment, and by Order of 16 June 1994 the Court set the written timetable [13]. A peninsula dispute had become a 1,600-kilometre adjudication by consent to a procedural characterization — the first lesson of the case being that the scope of a boundary judgment is itself negotiated before it is argued.

What followed was eight years of procedure before the merits were heard, and the procedure was not decoration. Nigeria filed preliminary objections within its Counter-Memorial deadline, suspending the merits phase; Cameroon sought provisional measures on 10 February 1996, which the Court indicated on 15 March 1996 after hearing the parties; and each party, the bench containing no judge of either nationality, chose a judge ad hoc under Article 31 — Keba Mbaye for Cameroon, Bola Ajibola for Nigeria [13]. The objections were heard from 2 to 11 March 1998, and in its Judgment of 11 June 1998 "the Court found that it had jurisdiction to adjudicate upon the merits of the dispute and that Cameroon's requests were admissible. The Court rejected seven of the preliminary objections raised by Nigeria and declared that the eighth did not have an exclusively preliminary character" [13]. Nigeria then asked the Court to interpret that judgment — a request filed on 28 October 1998, constituted as a separate case, and held inadmissible on 25 March 1999 [13]. Even the threshold of the case, in other words, consumed five years and three judgments or orders of consequence.

The merits phase widened the case twice more. Nigeria's Counter-Memorial carried counter-claims, which Cameroon did not oppose and which the Court declared admissible by Order of 30 June 1999, structuring a Reply (4 April 2000), a Rejoinder (4 January 2001), and eventually an additional Cameroonian pleading confined to the counter-claims (4 July 2001) [13]. The same day, Equatorial Guinea applied to intervene under Article 62 "to protect the legal rights of the Republic of Equatorial Guinea in the Gulf of Guinea" and inform the Court of rights "that could be affected by the Court's decision," expressly not seeking to become a party; neither litigant objected in principle, and by Order of 21 October 1999 the Court authorised the intervention, Equatorial Guinea filing its statement on 4 April 2001 [13]. A bilateral boundary case had acquired a third state's shadow, because a maritime line drawn between two states in a concave gulf necessarily leans toward a third. Oral proceedings finally opened on 18 February 2002 [13], and the judgment of 10 October 2002 closed a file that had been open for eight and a half years. For this report's purposes, the procedural record is itself evidence: definitive delimitation by adjudication is available, principled, and slow — a remedy measured in decades of incidents before it and years of pleadings during it, which is one reason so many of the lines in these chapters were never taken to court at all.

Dissenting and separate opinions: Judges Koroma and Al-Khasawneh

The judgment of 10 October 2002 was accompanied by a full bench of appended texts — declarations by Judges Oda, Herczegh, and Rezek; separate opinions by Judges Ranjeva, Parra-Aranguren, Al-Khasawneh, and Mbaye; and dissents by Judge Koroma and by Judge ad hoc Ajibola [13]. Two of them attack the same load-bearing premise from opposite sides of the result: that the 1884 Treaty of Protection between Great Britain and the Kings and Chiefs of Old Calabar had silently extinguished the chiefs' territorial sovereignty, leaving Britain free to cede Bakassi to Germany in 1913. Judge Koroma, dissenting, would have voided the cession; Judge Al-Khasawneh, concurring in the outcome, refused the reasoning that sustained it.

Koroma's charge is institutional before it is doctrinal. The Court exists, he writes, "to declare and apply the law, and... would range outside that task at its peril"; its Bakassi conclusion was "tantamount to a recognition of political reality rather than to an application of the treaty and the relevant legal principles. In my view, it is not the function of the Court to recognize or consecrate political reality but rather to apply the law" [13]. The treaty, read on its terms,

never authorised the cession. In a protectorate, on the account he takes from Oppenheim, sovereignty splits: the protector exercises rights of external sovereignty while "the internal sovereignty would continue to be exercised by the local kings and rulers," and the protected state "always has, and retains, for some purposes, a position of its own as an international person" [13]. The 1884 treaty was written in negative clauses — the Kings and Chiefs undertook not to treat with other Powers, not to maintain relations, not to go to war, "and, most importantly, not to cede territory" — while Britain "was not authorized... to act in their name and on their behalf," and the chiefs retained their treaty-making power subject to informing Britain and obtaining its approval [13]. Since "Bakassi was part of the territorial scope of the 1884 Treaty of Protection and could not have been changed without the consent of the Kings and Chiefs of Old Calabar," and no such consent was evidenced, "there was no basis even under the applicable law at that time for Great Britain to be able to determine its boundaries with Germany in respect of Bakassi" [13]. The 1913 Agreement was therefore invalid as to Bakassi — nemo dat quod non habet — and the maritime boundary "anchored" to it, running west of the peninsula from the Bakassi Point–King Point line, fell with it: ex injuria non oritur jus [13]. Koroma also refused the majority's handling of historical consolidation, whose implication "that conventional title based on the 1913 Anglo-German Agreement is the only valid means of acquiring title or that the mode of territorial acquisition is closed" would leave "no place in the Court's jurisprudence for prescriptive title" [13].

Al-Khasawneh's separate opinion dismantles the jurisprudential shortcut the majority had used to reach the same premise — Max Huber's Island of Palmas generalisation that treaties with local rulers created suzerainty and vassalage. The approach, he writes, is a "sweeping generalization" under which local rulers, "no matter how valid and old their title and how clear the display of their sovereignty," are "deemed to have become virtual colonies or vassal States... even if control over them was nominal and even if in subsequent dealings... they continued to be treated as retaining some sovereignty, for example, for the purposes of sovereign immunity" — citing Mighell v. Sultan of Johore and the Johore litigation, and protesting that "[i]t is difficult to understand how a local ruler would be considered to be entitled to absolute sovereign immunity and to have been divested of his territorial sovereignty at one and the same time" [13]. The generalisation is, further, "clearly rooted in a Eurocentric conception of international law based on notions of otherness": European protected principalities were never supposed to have lost sovereignty to their protectors, and "[i]ntertemporal law is general in its application... and cannot be divided into regional intertemporal law, all the more so when no State in the concerned region... participated in its formation" [13]. Huber's reading rests on "an extreme interpretation of the theory of constitutive recognition," a theory with "as many opponents as it has adherents," and state practice contradicts it: protecting Powers "were not seeking colonial title but merely spheres of influence or dominance," and a treaty of protection was at most "a legal lever for acquiring an inchoate title to territory: a title capable of being perfected more or less at leisure" — "until that happened... they remained a lever and no more" [13].

Against these two jurisprudential challenges, Judge Oda's declaration supplies the technocratic counterpoint: he gave the judgment his "full support" on its three essential questions, reserving only the Court's specification of the Lake Chad tripoint to 14°04′59.9999″ east — a ten-thousandth of a second that "represents less than 3 mm and has no significant effect on any map," where reference to the Lake Chad Basin Commission's tripoint "would be sufficient" — and the manner of Nigeria's withdrawal, which "should be conducted in accordance with arrangements to be agreed upon by the Parties," with "orderly repatriation of those Nigerian nationals wishing to leave" [13]. The range of the opinions is the measure of the judgment: its precision could be faulted to the millimetre, its humanitarian execution left unscripted, and its central act — letting a colonial power's signature outrank the protected rulers it had promised not to dispossess — defended by the majority only at the cost of a doctrine two of its own judges, voting differently, would not sign.

Arbitration, conciliation, and the International Tribunal for the Law of the Sea

Every line drawn in this paper exists in the shadow of a forum. Part XV of the Convention builds that shadow as a menu with a default: under Article 287 a State may declare its choice among "(a) the International Tribunal for the Law of the Sea...; (b) the International Court of Justice; (c) an arbitral tribunal constituted in accordance with Annex VII; (d) a special arbitral tribunal constituted in accordance with Annex VIII," but a party with no declaration in force "shall be deemed to have accepted arbitration in accordance with Annex VII," and where the parties' declarations do not match, the dispute "may be submitted only to arbitration in accordance with Annex VII, unless the parties otherwise agree" [4]. Arbitration is thus the system's gravity well — the forum you get when you chose nothing, or chose differently from your opponent. Declarations persist until three months after a revocation notice, and revocation never disturbs pending proceedings [4]. Whichever forum is seised decides its own jurisdiction, and in scientific or technical disputes may sit non-voting experts alongside the judges [4].

Annex VII's tribunal is engineered against obstruction. Five members: one appointed by each party, three — including the President — by agreement, and if agreement fails within 60 days, the appointments fall to the President of ITLOS, choosing from the Annex's list persons of different nationalities who are not nationals, residents or servants of any party [4]. The tribunal determines its own procedure, the parties must facilitate its work down to site visits, expenses are shared equally, and decisions are by majority with a presidential casting vote [4]. The anti-veto clause is the famous one: "If one of the parties to the dispute does not appear before the arbitral tribunal or fails to defend its case, the other party may request the tribunal to continue the proceedings and to make its award. Absence of a party or failure of a party to defend its case shall not constitute a bar to the proceedings" — tempered by the tribunal's duty, before awarding, to "satisfy itself not only that it has jurisdiction over the dispute but also that the claim is well founded in fact and law" [4]. The award "shall be final and without appeal" and "shall be complied with by the parties," with interpretation disputes returning to the same tribunal [4]. Conciliation under Annex V is the soft track built beside it: a five-member commission (two appointed by each side, a chairman chosen by the four, the UN Secretary-General filling defaults) that hears the parties and "make[s] proposals... with a view to reaching an amicable settlement," reporting within 12 months — its report, "including its conclusions or recommendations, shall not be binding upon the parties" [4]. For the categories Part XV routes to compulsory conciliation, however, the notified party "shall be obliged to submit to such proceedings," its silence is no bar, and the commission decides its own competence [4]. Annex VIII's special arbitration mirrors Annex VII's five-member constitution for expert-heavy categories, its appointing authority the Secretary-General [4]. And the standing court: ITLOS, seated in "the Free and Hanseatic City of Hamburg," composed of "21 independent members, elected from among persons enjoying the highest reputation for fairness and integrity and of recognized competence in the field of the law of the sea," with the principal legal systems and equitable geography represented [4].

The system's stress test arrived in 2013, when the Philippines instituted Annex VII proceedings against China. The five-member tribunal, administered by the Permanent Court of Arbitration, issued a 151-page Award on Jurisdiction and Admissibility on 29 October 2015 and a 479-page Merits Award on 12 July 2016, ruling "in favor of the Philippines on virtually every issue" [7]. China refused to appear, maintaining the tribunal lacked jurisdiction and that disputes were for "negotiation and consultation" alone; as Annex VII permits, non-appearance did not defeat jurisdiction [7]. Schoenbaum's anatomy of the legitimacy cost is unsparing: the tribunal could not enter a default judgment, so it had "the task of 'making up' what arguments China could be presumed to make," then evaluating its own constructions against "the arguments made by the lawyers for the Philippines" — "a lawyer's dream—litigating a case before a court that will hear only one side," leaving the awards with "a whiff of unfairness" despite impeccable reasoning [7]. The jurisdiction award also "broke new ground": objections were overcome "by artful pleading—formulating legal claims that... bear upon sovereignty and the jurisdictional exceptions" without confronting them, and Article 282 — deference to other agreed procedures — was read restrictively, against the Southern Bluefin Tuna precedent, so that the impractical, non-binding alternatives did not oust UNCLOS jurisdiction [7]. On future conduct the tribunal "punted": "the Tribunal is not persuaded that it is necessary or appropriate for it to make any further declaration" [7]. The machinery, in short, works exactly as designed — no veto by absence, no escape by drafting, awards final and reasoned. What it cannot manufacture is the losing party's consent to be bound in fact. That gap, between an award that is legally complete and a sea that remains contested, is the dispute-settlement chapter's honest ending, and this paper's recurring one.

State responsibility and liability

The Convention is generous with jurisdiction and parsimonious with liability. Its responsibility provisions are not a general code but a set of narrow channels, each cut into a specific regime — and Article 304 then expressly steps aside: the Convention's provisions "regarding responsibility and liability for damage are without prejudice to the application of existing rules and the development of further rules regarding responsibility and liability under international law" [4]. The fullest channel is environmental. Under Article 235, "States are responsible for the fulfilment of their international obligations concerning the protection and preservation of the marine environment. They shall be liable in accordance with international law," must "ensure that recourse is available in accordance with their legal systems for prompt and adequate compensation or other relief in respect of damage caused by pollution," and must cooperate toward further liability law, "such as compulsory insurance or compensation funds" [4]. Enforcement itself carries a price: "States shall be liable for damage or loss attributable to them arising from measures taken pursuant to section 6 when such measures are unlawful or exceed those reasonably required in the light of available information. States shall provide for recourse in their courts for actions in respect of such damage or loss" [4]. Around that liability sits a ring of safeguards for the accused vessel: penalty proceedings by a coastal State are suspended if the flag State takes up corresponding charges within six months (unless major damage or habitual flag-State disregard),

lapse after three years, leave civil claims untouched, and may impose "monetary penalties only" on foreign vessels beyond the territorial sea — and within it too, "except in the case of a wilful and serious act of pollution" — with "recognized rights of the accused" observed throughout [4].

The deep seabed channel is the most carefully engineered. Article 139 gives sponsoring States "the responsibility to ensure" that activities in the Area by their nationals and sponsored entities conform to Part XI, and provides that "damage caused by the failure of a State Party or international organization to carry out its responsibilities under this Part shall entail liability," joint and several where they act together — but then installs the due-diligence shield: a State "shall not however be liable for damage caused by any failure to comply with this Part by a person whom it has sponsored... if the State Party has taken all necessary and appropriate measures to secure effective compliance" [4]. Below the State, the contractor "shall have responsibility or liability for any damage arising out of wrongful acts in the conduct of its operations," the Authority likewise for wrongful acts in exercising its powers, contributory fault apportioned, and "liability in every case shall be for the actual amount of damage" [4]. The Enterprise and the Authority are statutorily firewalled from each other's obligations, and no member of the Authority is liable "by reason only of its membership" for the Enterprise's acts [4]. Liability, in the Convention's design, attaches to conduct and stops at actual damage; it is insurance logic, not punishment.

What happens when responsibility is claimed outside these channels — under the general law Article 304 preserves — is this paper's closing case. Cameroon had asked the Court to hold Nigeria responsible for its occupation and administration of Bakassi and the Lake Chad areas; Nigeria counter-claimed in mirror image. The Court rejected both (operative paragraphs 325 V(D) and (E)), and a separate opinion explains the refusal's logic. The claim was, first, structurally unprecedented: a land boundary claim, a maritime delimitation claim "and a State responsibility claim, all presented at the same time" — "unprecedentedly excessive and indeed unique," the product of a case that "took over eight years" and a "marathon hearing time of five weeks" [13]. It was, second, corrosive by design: "Geography and history compel their eternal co-existence. The Court was not created to consciously or unconsciously create eternal disharmony between brother States," and a responsibility judgment "can only engender bad blood" [13]. The doctrinal core was Nigeria counsel Watts's floodgates argument, quoted with approval: Cameroon "cited no case in which a territorial dispute has been resolved in favour of one State, and in which the losing State was then held internationally responsible for its acts of civil administration or maintenance of public order in areas in which... it was found to have had no right... Any other approach would turn every territorial dispute into a State responsibility case, sometimes of enormous magnitude" [13]. And it was, finally, unending: Cameroon had reserved a later assessment of compensation, Nigeria had asked for reparation to be fixed in a further judgment if not agreed within six months — so that even after judgment "the Court may not see an end to this case" [13]. The pattern matches the Convention's own architecture. Where the law wants liability, it builds a channel with limits, shields and actual-damage caps; where a claim asks liability to do the work of politics — to convert losing a boundary into paying for having administered one — the system, treaty and court alike, declines. Responsibility in the law of the sea is real, but it is deliberately small: priced, channelled, and fenced off from the territorial judgments it might otherwise swallow.

Bakassi: the Cameroon–Nigeria judgment and the peninsula that changed flags

Bakassi is where every doctrine in this paper came to collect. The peninsula's paper chain begins on 10 September 1884, when Great Britain and the Kings and Chiefs of Old Calabar concluded a Treaty of Protection: Britain undertook to extend its protection, and the chiefs promised "inter alia to refrain from entering into any agreements or treaties with foreign nations or Powers without the prior approval of the British Government" [13]. Twenty-nine years later, by the Anglo-German Agreements of 11 March and 12 April 1913 — whose object was "the Settlement of the Frontier between Nigeria and the Cameroons, from Yola to the Sea," with the boundary drawn on annexed map sheets TSGS 2240 and GSGS 2700 — Bakassi was placed in German territory [13] [13]. The twentieth century then layered administrations over the cession: the 1946 Order in Council divided the British-mandated territories into Northern and Southern Cameroons, administered from Nigeria; the plebiscites of 11–12 February 1961 sent the Northern Cameroons to Nigeria and the Southern Cameroons to the Republic of Cameroon; and the two independent states began their own maritime work with the Yaoundé I Declaration of 14 August 1970 (delimitation in three stages, the maritime boundary first), Yaoundé II of 4 April 1971 (a line to the 3-nautical-mile limit, points 1 to 12 on Admiralty Chart No. 3433), and the Maroua Declaration of 1 June 1975, extending the line from point 12 to point G [13]. Cameroon's litigation posture was accordingly austere: its claim could be resolved by the application "pure and simple" of the 1913 Agreements and their cartographic annexes, and it asked the Court to confirm the frontier's course, not to delimit afresh; Nigeria accepted the instruments' relevance everywhere except the "Bakassi provisions" themselves [13].

Against the paper, Nigeria arrayed a century of administration. Its three bases were "(i) Long occupation by Nigeria and by Nigerian nationals constituting an historical consolidation of title and confirming the original title of the Kings and Chiefs of Old Calabar, which title vested in Nigeria at the time of independence in 1960; (ii) peaceful possession by Nigeria, acting as sovereign, and an absence of protest by Cameroon; and (iii) manifestations of sovereignty by Nigeria together with acquiescence by Cameroon" — with Minquiers and Ecrehos invoked for the decisive weight of possession evidence [13]. The catalogue was formidable: no German occupation or administration of Bakassi between March 1913 and May 1916, the peninsula "administered as part of Nigeria" throughout; indirect rule through Warrant Chiefs from 1922 and the native-authority system of the 1933 Ordinance; native courts; taxes collected by the Calabar and Eket divisions; a Methodist school at Abana in 1937; a census under the Eket Division in 1953; health centres, electoral wards and parliamentary elections, customs duties, Nigerian passports, oil exploration permits and production agreements — "a considerable amount and volume of evidence," capped by the claim that Cameroon had acquiesced until 1972, when its village-renaming project itself proved "the previous absence of Cameroonian administration" [13]. Cameroon answered on three levels: a treaty title cannot be displaced by effectivités; Britain's post-Mandate administration was exercised as mandatory Power under Article 22 of the League Covenant on behalf of the international community and the Southern Cameroons, not for Nigeria or the chiefs; and "historical consolidation" is merely "the establishment of title by adverse possession, which has traditionally been labelled as 'acquisitive prescription'," whose strict criteria Nigeria's evidence could not satisfy [13].

The Court's findings dismantled the claim in sequence. Great Britain, "under the applicable law at the time," was in a position in 1913 to determine its boundary with Germany (paragraph 209); the Court was "unable to accept that until Nigeria's independence in 1961… the Bakassi Peninsula had remained under the sovereignty of the Kings and Chiefs of Old Calabar" (paragraph 212); and Nigeria had itself accepted Articles XVIII to XXII of the 1913 Agreement as valid and recognised Cameroonian sovereignty (paragraph 214) — a recognition the Court evidenced, with quiet devastation, from "formal requests up until the 1980s submitted by the Nigerian Embassy in Yaounde, or by the Nigerian consular authorities, before going to visit their nationals residing in Bakassi," requests whose acknowledgment of sovereignty did not depend on any visit actually occurring [13] [13]. It followed that the Agreement "was valid and applicable in its entirety," severability and uti possidetis not requiring decision [13]. Consolidation failed on its own terms: at independence "there existed no Nigerian title capable of being confirmed subsequently by 'long occupation,'" Cameroon having succeeded to the 1913 title, and consolidation "cannot in any event vest title to Bakassi in Nigeria, where its 'occupation' of the peninsula is adverse to Cameroon's prior treaty title and where, moreover, the possession has been for a limited period" [13]. The operative result fixed the Bakassi boundary by Articles XVIII to XX of the 1913 Agreement and declared that "sovereignty over the Bakassi Peninsula lies with the Republic of Cameroon" [13].

A peninsula of fishing settlements, administered for generations as Nigerian — taxed, schooled, censused and policed as Nigerian — was thereby adjudged to have been Cameroonian since a signature in 1913, and the flag followed the paper. The judgment's austere logic is this paper in miniature: maps and treaties are the title deeds of territory, administration is evidence that counts only when it agrees with them, and a community's lived allegiance, however long and however documented, is a fact about people rather than about sovereignty. Whether that hierarchy is the law's wisdom or its original sin is the question Judges Koroma and Al-Khasawneh forced into the open — and the one Bakassi's residents, whose state changed around them without their moving, embody permanently.

The Deep Seabed and the Ocean Commons

The deep seabed regime: pioneer investment and the Enterprise

Part XI declared the deep seabed the common heritage of mankind; Resolutions I and II of the Final Act built the bridge by which the states already spending money there could cross into the new regime without losing it — and by which mankind's own miner, the Enterprise, would be carried across with them. The declared purposes are twinned: "to ensure that the Enterprise will be provided with the funds, technology and expertise necessary to enable it to keep pace with the States and other entities" mining the Area, and to protect investments made compatibly with Part XI before the Convention entered into force [4]. Resolution I created the Preparatory Commission — convened by the Secretary-General once 50 states had signed or acceded, composed of signatory states (and Namibia, represented by the UN Council for Namibia) — to draft the Authority's rules and budget, study its headquarters, handle the pioneer regime, study the problems of developing land-based producer states (including a compensation fund), and, through a special commission, take "all measures necessary for the early entry into effective operation of the Enterprise" [4].

The pioneer investor was defined by sunk cost. The first class comprised France, India, Japan and the Soviet Union — or their state enterprises and controlled entities — provided they had "expended, before 1 January 1983, an amount equivalent to at least $US 30 million... in pioneer activities and... no less than 10 per cent of that amount in the location, survey and evaluation" of the claimed area; the second, four consortia of nationals from Belgium, Canada, the Federal Republic of Germany, Italy, Japan, the Netherlands, the United Kingdom and the United States meeting the same test; the third, any developing state meeting it before 1 January 1985 [4]. Pioneer activities meant the identification, discovery and systematic evaluation of polymetallic nodules — accretions containing "manganese, nickel, cobalt and copper" — including recovery of nodules to design and test exploitation equipment [4] [4]. Registration cost $250,000, carried an annual fixed fee of $1 million from allocation, and obliged expenditures proportioned to what "a bona fide operator who intends to bring that area into commercial production within a reasonable time" would spend [4].

The regime's heart is the parallel system, applied at the application stage. Every application had to "cover a total area... sufficiently large and of sufficient estimated commercial value to allow two mining operations," divided "into two parts of equal estimated commercial value," with all the applicant's data on both parts — mapping, testing, nodule density, metal content — surrendered to the Commission, which within 45 days designated one part "to be reserved... for the conduct of activities in the Area by the Authority through the Enterprise or in association with developing States," allocating the other to the investor [4]. The pioneer's own half, the pioneer area, could not exceed 150,000 square kilometres and shrank by schedule: 20 per cent relinquished by the end of year three, a further 10 per cent by year five, and a further 20 per cent after eight years or production authorisation [4]. No investor could hold more than one pioneer area. Overlapping claims — inevitable where consortia had surveyed the same nodule fields — were to be deconflicted by certifying states before application, failing which negotiation, failing which binding UNCITRAL arbitration to run between May 1983 and December 1984, the tribunal weighing dated priority factors: when coordinates were deposited, the continuity and extent of past activities, when at-sea work began, and expenditure in constant dollars [4] [4]. From registration the investor held "the exclusive right to carry out pioneer activities in the pioneer area" [4].

Conversion into the Convention system was automatic but conditional: within six months of entry into force, the registered pioneer applied for approval of a plan of work for exploration and exploitation, which "the Authority shall approve" if compliant — but no plan could be approved unless the certifying state was a Party, and for the consortia, unless all component states were Parties, non-ratification within six months of notification terminating pioneer status absent a three-quarters Council postponement [4]. In production, pioneers took priority over all applicants except the Enterprise, "entitled to production authorizations for two mine sites"; authorisations issued within 30 days of a notice to commence within five years, with a single non-renewable extension of up to five years for economic non-viability, and the Article 151 production ceiling administered as a queue in which delayed pioneers kept priority and simultaneous applicants could apportion tonnage among themselves [4]. The Enterprise's pace-keeping was written as obligations on the pioneers themselves: each registered investor had to explore the reserved area at the Commission's request (costs reimbursed with 10 per cent annual interest), "provide training at all levels for personnel designated by the Commission," and undertake to perform the Convention's technology-transfer obligations; certifying states had to "ensure that the necessary funds are made available to the Enterprise in a timely manner" [4]. The Authority was bound in turn to "recognize and honour the rights and obligations arising from this resolution" [4].

Read as design, Resolution II is the most candidly redistributive instrument in this paper: private and state capital could keep half of what it had found — surveyed, valued and data-stripped — while the other half was banked for a collective miner that did not yet exist, its exploration prepaid by the very investors it would one day compete with. The map of the seabed was to be drawn twice over, once for the market and once for mankind, from the same coordinates.

Protection and preservation of the marine environment

Part XII is the Convention's second constitution, layered over the map of zones: where the rest of UNCLOS allocates rights by distance from baselines, Part XII imposes duties that follow activities wherever they occur. States "shall take, individually or jointly as appropriate, all measures consistent with this Convention that are necessary to prevent, reduce and control pollution of the marine environment from any source, using for this purpose the best practicable means at their disposal and in accordance with their capabilities," covering "all sources" — land-based releases of toxic and persistent substances, vessel pollution, seabed installations, and other devices — and including measures "to protect and preserve rare or fragile ecosystems as well as the habitat of depleted, threatened or endangered species" [4]. Two boundary-crossing rules give the part its distinctive shape. The no-harm rule: states must

“ensure that activities under their jurisdiction or control are so conducted as not to cause damage by pollution to other States and their environment, and that pollution… does not spread beyond the areas where they exercise sovereign rights” [4]. And the anti-displacement rule: in taking measures, states “shall act so as not to transfer, directly or indirectly, damage or hazards from one area to another or transform one type of pollution into another” [4] — the treaty expressly forbidding the oldest trick in environmental management, exporting the harm across a line. A companion provision reaches pollution “resulting from the use of technologies… or the intentional or accidental introduction of species, alien or new” [4].

The part’s regulatory technique is incorporation rather than prescription. For vessel-source pollution, states acting “through the competent international organization or general diplomatic conference” set the international rules; flag States must adopt laws that “at least have the same effect as that of generally accepted international rules and standards”; coastal States in the EEZ may legislate only “conforming to and giving effect to” those standards; and territorial-sea laws must “not hamper innocent passage” [4]. Unilateral ambition is channelled, not excluded: a State may impose particular requirements as a condition of port entry, with publicity, and for a “particular, clearly defined area” of its EEZ with recognised technical justification it may seek special mandatory measures — but the organisation must determine that conditions correspond, the measures bind foreign vessels only 15 months after submission, and they “shall not require foreign vessels to observe design, construction, manning or equipment standards other than generally accepted international rules and standards” [4]. Around the rules sits a cooperation machinery with operational content: immediate notification of States “likely to be affected” when the marine environment “is in imminent danger of being damaged or has been damaged,” joint contingency plans, shared research and scientific criteria, and structured scientific and technical assistance to developing States — training, participation, equipment, and help with environmental assessments [4].

What the duties bite on, in practice, was settled in 2016. The South China Sea tribunal judged features “on the basis of [their] natural capacity, without external additions or modifications,” and held — in Schoenbaum’s summary of the Merits Award — that “[l]and reclamation and construction activities on sensitive marine features will virtually always constitute marine pollution and destruction of marine ecosystems in violation of the provisions of Part XII of UNCLOS on protection of the marine environment” [7]. The significance doubles back on this report’s central theme: the same construction programme that tried to manufacture geography — to upgrade reefs into zone-generating islands — was legally legible only as environmental destruction, because Part XII’s duties attach to the natural feature and its ecosystem, not to the flag planted on the fill. “Protection of the marine environment will be enhanced since states cannot gain rights by land reclamation activities on small maritime features” [7]. The zones allocate the sea on paper; Part XII is the Convention’s insistence that the paper does not exhaust what a state owes there.

Conservation of the living resources of the high seas

High-seas fishing enters the Convention as a freedom already in chains. Article 116 grants all states the right for their nationals to fish on the high seas “subject to: (a) their treaty obligations; (b) the rights and duties as well as the interests of coastal States provided for, inter alia, in article 63, paragraph 2, and articles 64 to 67; and (c) the provisions of this section” — the straddling-stock, highly-migratory, anadromous and catadromous regimes of the EEZ chapters reaching out beyond 200 miles to qualify the freedom where stocks themselves ignore the line [4]. Article 117 states the core duty in the flag-state idiom: “All States have the duty to take, or to cooperate with other States in taking, such measures for their respective nationals as may be necessary for the conservation of the living resources of the high seas” [4]. Jurisdiction runs over one’s own nationals because, by definition, no one else’s may: on the high seas there is no coastal sovereign, so conservation is built from flag duties plus negotiation. Article 118 supplies the negotiation: states whose nationals “exploit identical living resources, or different living resources in the same area, shall enter into negotiations with a view to taking the measures necessary for the conservation of the living resources concerned,” and “shall, as appropriate, cooperate to establish subregional or regional fisheries organizations to this end” [4].

The standard those negotiations must serve is Article 119’s. In determining the allowable catch, states shall take measures “designed, on the best scientific evidence available to the States concerned, to maintain or restore populations of harvested species at levels which can produce the maximum sustainable yield, as qualified by relevant environmental and economic factors, including the special requirements of developing States, and taking into account fishing patterns, the interdependence of stocks and any generally recommended international minimum standards” [4]. The ecosystem clause follows: states must consider “the effects on species associated with or dependent upon harvested species with a view to maintaining or restoring populations of such associated or dependent species above levels at which their reproduction may become seriously threatened” [4]. Information is the regime’s currency

— scientific information, catch and fishing-effort statistics "shall be contributed and exchanged on a regular basis through competent international organizations… with participation by all States concerned" — and its fairness rule is formal equality: conservation measures "and their implementation do not discriminate in form or in fact against the fishermen of any State" [4]. Article 65's marine-mammal regime is expressly extended to the high seas [4].

The regime's modesty is historical. Its 1958 ancestor, the Geneva Convention on Fishing and Conservation of the Living Resources of the High Seas, "was not a success. Only 35 States have ratified this Convention," against 56 for the High Seas Convention and 53 for the Continental Shelf Convention [3]. Meanwhile jurisdiction moved: fishery zones "had progressively extended from 12M in the 1950s to 200M by the middle of the 1970s," driven in part by "the three 'Cod Wars' between the United Kingdom and Iceland between 1958 and 1976" [3]. The 200-mile enclosure absorbed most of the world's fish stocks into EEZs, where a coastal state can legislate and enforce; the high-seas section governs the remainder — the straddling and discrete high-seas stocks — with the weakest instruments in the fisheries toolbox: duties addressed to flags, negotiations among exploiters, organisations by consent, science by contribution. There is no inspectorate; even the enforcement-adjacent rule visible in this Part is corrective — a ship stopped or arrested outside the territorial sea "in circumstances which do not justify the exercise of the right of hot pursuit… shall be compensated for any loss or damage" [4].

The Secretary-General's current assessment measures the gap between that design and the ocean it governs: "Marine ecosystems remain under constant threat from the impacts of human activities, including overfishing, pollution, ocean warming and ocean acidification. These impacts are felt most acutely by coastal communities, especially SIDS and LDCs, with resulting risks to food security and threats to local economies," and "urgent actions are needed to address the wide range of threats" [18]. Cumulative impacts — coastal development, overfishing, pollution, record warming and sea-level rise — "have combined to threaten food security, local economies and the very existence of SIDS and some coastal populations" [18]. Section 2's architecture explains both the persistence and the shortfall. It made conservation a legal duty rather than a courtesy — nationals bound, negotiations obligatory, science and non-discrimination written in — while leaving every operative act (setting the catch, policing the fleet) to the states whose nationals do the fishing. On the high seas, the map has no owner; the regime this paper describes is what law looks like when it must persuade the only authorities present: the flags under which the fishers sail.

Climate change, the ocean, and sea-level rise

Every line in this paper is measured from a coastline, and the coastline has begun to move. The Secretary-General's report to the seventy-ninth session records the physical state of play: "Ocean heat content reached its highest level in the 65-year observational record and the global mean sea level reached a record high since satellite observations began in 1993," while "global mean sea-level rise in the past decade (2014-2023) was more than twice the rate during the first decade of the satellite record (1993-2002), and is virtually certain to continue to rise until at least 2100" [18]. Sea-level rise "continued to pose an existential threat to some small islands and low-lying coasts," driving "coastal habitat contraction, geographical shifts of species, loss of biodiversity and ecosystem functionality, and encroachment on coastal settlements and infrastructure" — and climate change "also exacerbated the spread of invasive alien species" [18]. For a legal order whose baselines are the low-water line as marked on charts, whose island entitlements turn on land "above water at high tide," and whose smallest states are atolls, a rising ocean is not an environmental footnote but a challenge to the geometry of title itself.

The institutions are responding on three tracks. First, codification: "the International Law Commission continued its work on sea-level rise in relation to international law, in particular on issues related to the law of the sea, statehood, and the protection of persons affected by sea-level rise," and the General Assembly convened a high-level plenary meeting on the existential threats of sea-level rise on 25 September 2024 [18]. The ILC's triad names the three questions this paper has circled throughout: do maritime zones freeze or retreat with the shore; can a state whose territory submerges persist; and what is owed to the people displaced. Second, adjudication: in its advisory opinion in Case No. 31 of 21 May 2024, ITLOS "found that anthropogenic greenhouse gas (GHG) emissions into the atmosphere and excess energy absorbed by the ocean constituted 'pollution of the marine environment' under the Convention and that States have the specific obligation to take all necessary measures to prevent, reduce and control this type of pollution," adding that where the marine environment has been degraded, the duty to protect and preserve "may call for measures to restore marine habitats and ecosystems," both for the ocean's carbon-sink function and for resilience as adaptation [18]. The classification matters: by routing the climate problem through Part XII's pollution obligations, the Tribunal made the atmosphere's excess a breach-able duty of the sea's own constitution rather than an external regime's concern.

Third, policy integration — partial and underfunded. The first global stocktake under the Paris Agreement invited parties to scale up ocean-based mitigation; "56 per cent of the 148 new or updated nationally determined contributions (NDCs) integrated coastal and marine nature-based solutions," and of the 158 parties with adaptation components, 30 per cent identified ocean ecosystems as a priority sector [18]. The IMO's revised strategy sets "a collective ambition to achieve net-zero GHG emissions for international shipping by 2050" with adoption of alternative zero and near-zero fuels by 2030 — though UNCTAD found the past two decades' largest fisheries-sector CO2 reductions came from efficiency, smarter navigation and lighter gear, the fuel and vessel alternatives remaining "still in the prototype or product development phase" [18]. The finance gap is the starkest number in the report: adaptation costs for fisheries and aquaculture in developing countries "could reach USD 4.8 billion per year by 2030," while adaptation finance flows to the sector "have averaged only USD 0.22 billion per year from 2017 to 2021" — a twenty-fold shortfall between the assessed need and the money [18].

The report's conclusions place the Convention at the centre of whatever response coheres: "Thirty years after its entry into force, the Convention remains the keystone for global ocean governance," the BBNJ Agreement's adoption demonstrates "the fundamental importance of the Convention and its implementing agreements," and "the high level of engagement on the advisory opinion on climate change by the Tribunal illustrates the relevance of the institutions established by the Convention in addressing such challenges" [18]. Against that, the same conclusions record a "significant increase in the percentage of overfished stocks" and the triple planetary crisis — climate change, pollution and biodiversity loss — compounding: warming and sea-level records "have combined to threaten food security, local economies and the very existence of SIDS and some coastal populations" [18]. The paper that began with an island that never was ends by contemplating islands that may cease to be — and a legal system deciding, in real time, whether the lines it drew from their shores will be the last thing about them to disappear, preserved on charts like Sandy Island in reverse: not a phantom land invented by cartographers, but a real one the ocean reclaims while the map insists.

Piracy: definition, seizure, and the duty to cooperate

Piracy is the one crime the Convention lets every state police everywhere the law otherwise runs out — and its definition is built from exclusions. Under Article 101, piracy consists of "any illegal acts of violence or detention, or any act of depredation, committed for private ends by the crew or the passengers of a private ship or a private aircraft," directed "on the high seas, against another ship or aircraft, or against persons or property on board," or against targets "in a place outside the jurisdiction of any State" — together with voluntary participation in a known pirate ship's operation and any act "of inciting or of intentionally facilitating" the same [4]. Four gates must all be passed: private ends (which excludes state action and, on the classic reading, politically motivated seizure), a private ship (though acts by a warship or government ship "whose crew has mutinied and taken control" are assimilated to private acts), a second victim ship, and the high-seas location. Identical violence inside a territorial sea is therefore not piracy under the Convention at all, but armed robbery under the coastal State's law — the definition polices the jurisdictional vacuum beyond national waters, not maritime crime in general. A ship becomes a pirate ship by intention as much as by act: it qualifies "if it is intended by the persons in dominant control to be used for the purpose of committing" an Article 101 act, or if it has been so used and "remains under the control of the persons guilty of that act," while its nationality, retained or lost, is determined by the law of the State that granted it [4].

Against that narrow definition stands an unusually broad enforcement right, hedged by liability. "On the high seas, or in any other place outside the jurisdiction of any State, every State may seize a pirate ship or aircraft, or a ship or aircraft taken by piracy and under the control of pirates, and arrest the persons and seize the property on board. The courts of the State which carried out the seizure may decide upon the penalties to be imposed," subject to the rights of third parties acting in good faith [4]. This is universal jurisdiction in its oldest form: no nexus of flag, nationality, or territory is required between the seizing State and the pirates. But the right is disciplined from both ends. Only "warships or military aircraft, or other ships or aircraft clearly marked and identifiable as being on government service and authorized to that effect" may carry out a seizure [4] — private counter-piracy is itself outside the scheme. And "[w]here the seizure of a ship or aircraft on suspicion of piracy has been effected without adequate grounds, the State making the seizure shall be liable to the State the nationality of which is possessed by the ship or aircraft for any loss or damage caused by the seizure" [4]. The surrounding articles generalise the cooperative posture without generalising the jurisdiction: all States "shall cooperate in the suppression of illicit traffic in narcotic drugs and psychotropic substances" on the high seas, and likewise for unauthorised broadcasting — where prosecution may be brought by the flag State, the State of registry, the State of the person's nationality, any State where the transmissions can be received, or any State whose authorised communications suffer interference [4]. Piracy's re-

gime is thus the template and the exception at once: the high seas are a commons whose order depends on duties to cooperate, but only for piracy did the Convention trust every State's warships to enforce the peace of a place that belongs to no one.

What's in a Name?

Naming disputes and the politics of geographical names

Every boundary in this paper is finally fixed to words: a named peak, a named river mouth, a named island. Names are the map's addressing system — in the United States topographic series alone, "almost 2 million natural and man-made features are identified," and those names "form a primary reference system essential for the communication of cartographic information" — which is why the USGS calls them, without embarrassment, "part of the Nation's living heritage" [19]. The grid beneath them is industrial in scale: more than 54,000 maps at 1:24,000 in the 7.5-minute quadrangle series covering the 48 contiguous states and Hawaii, each sheet carrying its quota of "place names, buildings, and swamps" alongside the contours [19]. A name is how a coordinate becomes a place that institutions can argue about.

Standardisation was invented because unstandardised names broke the machinery. In the United States, "the surge in mapping and scientific activities after the Civil War left the accuracy and spelling of a large number of names in doubt," a "serious problem to mapmakers and scientists who require nonconflicting nomenclature"; the answer, in 1890, was the U.S. Board on Geographic Names, "the central authority to deal with naming conflicts," an interagency body chaired by the Department of the Interior that standardises "the spelling and application of geographic names on maps and documents published by the U.S. Government" [19]. Every national names authority since — and the UN's geographical-names standardisation apparatus above them — replicates that 1890 move: one spelling, one application, decided by a board, binding on the state's own paper.

The content of the names is itself a political archive. The oldest stratum on U.S. maps comes from Native American languages — "Adirondack, Chippewa, Chesapeake, Shenandoah, Choctaw, Yukon, and the names of 28 States" — overwritten in successive settlement layers: English (New London, Yorktown, Lancaster), French (Fond du Lac, Baton Rouge, St. Louis), Spanish (Rio Grande, San Francisco, Guadalupe) [19]. The descriptive vernacular records who hoped and who feared: Stone Mountain and the Big Muddy River beside "Last Chance, Hells Canyon, Liberty, Thief Lake, Enterprise, Rattlesnake Creek, Dread and Terror Ridge, and Paradise Flats" — names that "evoke the dreams, fears, and color of the frontier" [19]. To rename is to re-found; to retain an indigenous name is to concede a prior layer of possession. That is why naming disputes are never about spelling.

Cross-border practice shows the working compromise. Dunlap's study of the Russian Arctic straits — a law-of-the-sea text, not a gazetteer — states its convention plainly: "Geographical names (except for the seas) are in transliterated Russian, as are most of the useful maps and charts of the area. When an English name differs significantly from the Russian in spelling or is quite well known, it is included parenthetically on the first entry" [8]. Endonym first, exonym in brackets where usage demands it: the standard editorial settlement of a dispute that has no tribunal.

Where sovereignty itself is contested, naming becomes an act of administration — a claim performed in print. The Asian Rim briefing's bibliography preserves a characteristic move: China's Committee of Place Names publishing "Islands in the South China Sea" in the People's Daily of 25 April 1983 — an official list assigning standard Chinese names to the features of a disputed archipelago, asserting through nomenclature the jurisdiction the names presuppose [6]. A gazetteer entry costs nothing to issue and everything to withdraw; rival claimants answer with rival lists, and charts thereafter carry the dispute in their labels even where their lines agree.

The failure mode is legal as much as political: a name in an instrument may not resolve to anything on the ground. In the Cameroon–Nigeria judgment, the 1946 Order in Council routed the boundary over a "prominent peak" — and the Court, transposing the instrument's own data (3.83 miles on a true bearing of 100 degrees from the sources of the M'Fi), found that the only peak the arithmetic identifies is Mount Kombon, 1,658 metres, which "can in no circumstances be regarded as marking that frontier" and does not lie on a crest line as prescribed; neither Tonn Hill, Itang Hill nor Tamnyar Peak satisfied the criteria either, and the Court, "unable to designate a specific peak," fell back on the identifiable crest line of which the unnamed peak must form part [13]. The description named a summit that the terrain declined to supply — the toponymic cousin of the phantom island, a word with coordinates and no referent.

Names, then, are boundaries in miniature: standardised by boards, layered by settlement, wielded by committees of place names as inexpensive sovereignty, and capable — like Bermeja's, invoked in an oil dispute two chapters ago — of carrying legal weight long after anyone has checked whether the named thing exists. The projections that follow distort shapes and distances; the names distort something subtler — whose the shape is said to be.

Every Map Is a Lie: Projections and Distortion

A square equal-area projection with low angular distortion

Every flat world map lies in at least one measurable way; the design question is which lie to choose and how visibly to tell it. The square equal-area projection presented by Matthew Petroff chooses its terms precisely: preserve area exactly, keep angular distortion low and distributed, eliminate the cusps that disfigure its predecessors, and — the engineering constraint that decides its practical fate — keep the mathematics closed-form.

The construction is polyhedral. "The projection is based on a procedure of mapping each octant of a spherical octahedron onto a Euclidean equilateral triangle": each octant splits into two symmetric right triangles — hexadecants, one-sixteenth of the sphere — each hexadecant subdivides into three sub-triangles sized so that "the area ratios between each Euclidean sub-triangle and the Euclidean hexadecant match the area ratios between each spherical sub-triangle and the spherical hexadecant"; each spherical sub-triangle is then mapped onto its Euclidean counterpart "in an equal-area manner," and the flattened hexadecants are "squished and rearranged into a square, in a quincuncial arrangement" [20]. The orientation follows the quincuncial logic: the north pole sits at the origin, the south pole is "split between the four corners," the prime meridian runs from the origin to the bottom centre where it meets the equator and then along the bottom edge to both lower corners, and the ±90° and 180° meridians are interrupted below the equator [20]. The interruptions are the price of the square; the design spends them in the southern oceans.

The comparison set is the two existing equal-area quincuncial projections: Collignon's (1865) and Gringorten's square equal-area map of 1972. On Tissot-indicatrix angular distortion, the new projection's distortion "is more spread out than the Gringorten projection, which concentrates angular distortion near the equator" — Gringorten retains "slightly lower average distortion," so the contest is distribution against mean — while on cusps the verdict is unambiguous: "the new projection presented in this work has considerably smaller cusps at the equator than the other two projections," and since "all three projections are equal-area, comparison of area distortions is moot" [20]. A cusp is where a projection's lie becomes visible as a kink in a coastline; minimising cusps is minimising the moments where the map admits what it is doing.

The computational profile explains the design's intended habitat. The forward mapping, after eliminating trivial identities, requires "12 trigonometric function evaluations and 6 inverse trigonometric function evaluations," reducible by composition identities to eight trigonometric evaluations — heavier than the Collignon quincuncial's single evaluation, but the equations are closed-form in both directions, "which allow it to be implemented as a GPU shader for real-time visualizations, unlike the iterative methods required for the Gringorten projection" [20]. The paper's conclusion generalises the point: the new projection "combines the benefits of the Collignon quincuncial projection with that of the Gringorten projection. It has closed-form forward and inverse solutions, unlike the Gringorten projection, while maintaining relatively low angular distortion when compared to the Collignon quincuncial projection; it also lacks any significant visible cusps, which is an improvement upon both existing equal-area quincuncial projections," and its vertex-oriented great-circle construction "can also be used to produce other map projections, such as equal-area star projections," with the analysis code published [20].

The strangest consequence is that the projection's best customers may not be cartographers. Panoramas and sky maps perform "an analogous operation" — projecting the view from the sphere's centre rather than its outside — and here the square equal-area form beats the incumbent equirectangular format, which "has both extreme area distortions and extreme angle distortions": equal area "avoids the storage of redundant information across multiple pixels," low angular distortion "reduces the occurrence of sampling artifacts in rectilinear views," and the square shape itself is an optimisation target because "current GPU texture size limitations are set with regard to square textures, [so] a square projection maximizes the amount of image information that can be stored in a single texture" — decisive for panoramic video — while seamless tiling can suppress codec edge artefacts [20]. The paper demonstrates with the reprocessed Haslam 408 MHz all-sky map in Galactic coordinates, the radio sky's synchrotron emission "brightest along the Galactic plane" rendered in the same square frame as a holiday panorama [20].

Set against this paper's whole argument, the square projection is the honest end of the projection family: it cannot be conformal and equal-area at once — no flat map can — so it declares its invariant (area), quantifies its error (Tissot distortion, spread rather than hidden), hides its cuts in empty ocean, and publishes its equations. The distortions examined next are what happens when maps are not so scrupulous: flexion, skewness and angular error compounding until the straight line on the chart and the distance on the Earth part company.

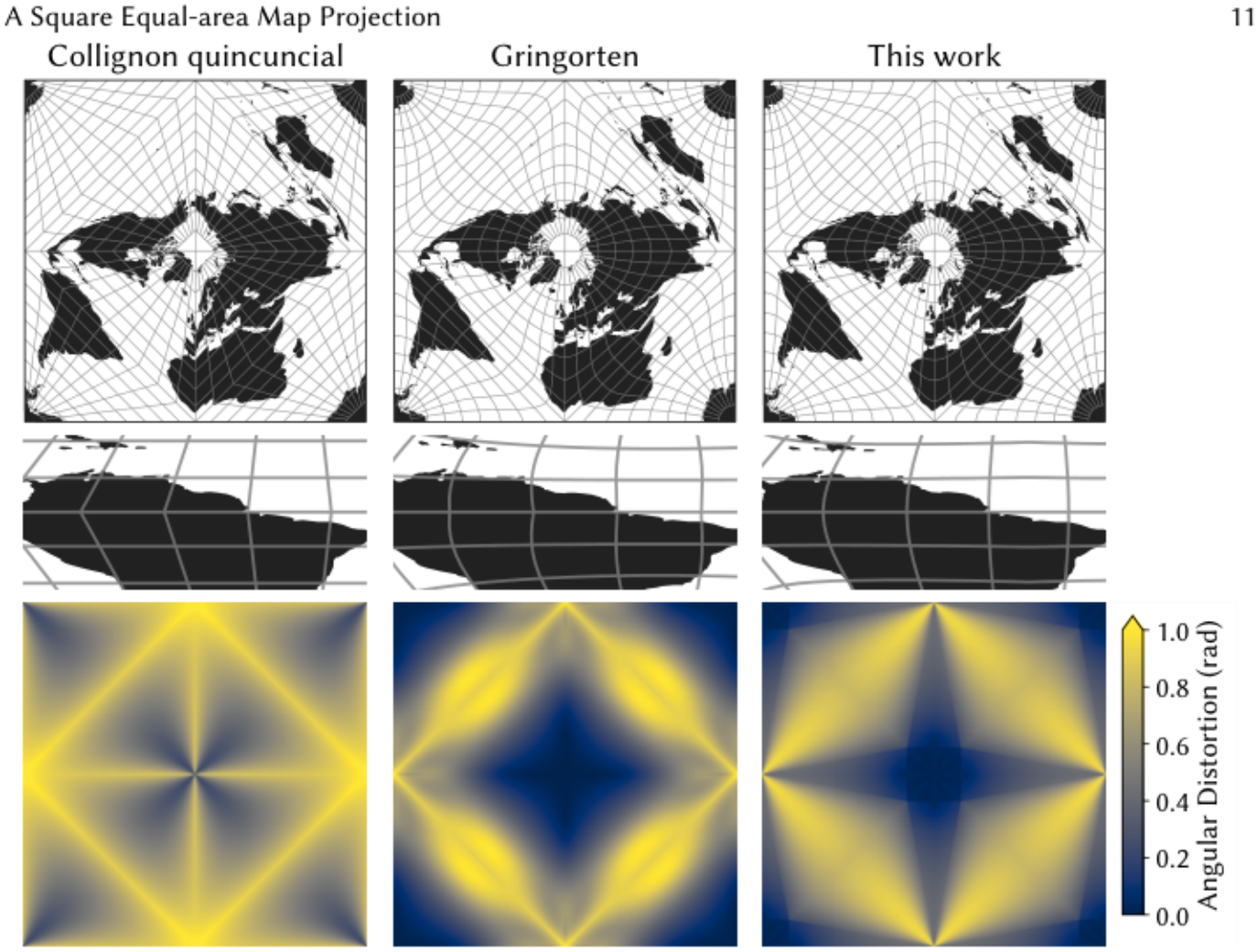


Exhibit. Three square equal-area projections compared: the Collignon quincuncial, Gringorten, and the new projection, with angular-distortion maps. Source: S0001.

The gnomonic cube

The gnomonic cube is the limiting case of the interrupted projection: project the sphere gnomonically onto the six faces of a cube, then unfold the cube into a cross. Goldberg and Gott include it in their numerical survey as the final, multiply-interrupted group — "a projection with multiple interrupts, the gnomonic cube, which is defined piecemeal" — presented as a cross in which "5 edges are included in the map proper" [21]. Its interest is diagnostic. Within any single face the gnomonic projection has a famous virtue: it is "locally flexionfree," because great circles — the geodesics of the sphere — project as straight lines. The cube map as a whole enjoys no such property: "it is clear that geodesics will not trace out straight lines in the gnomonic cube map because they bend when they cross an edge between faces" [21]. Every defect of the map lives exactly at its seams.

Their method measures precisely that. Goldberg and Gott sample 30,000 random points on the globe, follow a random geodesic from each in small numerical steps ($d\tau \simeq 10^{-5}$ rad), and compute the metric and Christoffel symbols directly to obtain flexion and skewness, alongside distance errors for random point pairs and six summary measures: ellipticity (E), area (A), flexion (F), skewness (S), distance (D), and boundary cuts (B) — the average number of map boundary cuts crossed by the shortest geodesic joining two random points, estimated across projections at B = 0.686 [21] [21]. For the cube, "[m]ost of these segments will not hit a connected edge between faces, but those which do will bend and have a small additional acceleration/deceleration (skewness). We find that the average flexion over this whole world projection is 0.115, all of which is attributable to the bends at edges between faces. Likewise, we

find an average skewness of 0.87," of which the boundary-blind method already accounts for 0.84 — "[t]hus, only 0.03 comes from edge crossings" [21]. The cube is thus nearly honest twice over: its flexion is wholly seam-made and quantifiable to the last 0.115, and its skewness is mostly intrinsic to the gnomonic faces rather than the unfolding.

Why include such a construction in a ranking of practical world maps? Because without a boundary term, the scoring can be gamed by interruption. A fan of many narrow gores "would approximate the Azimuthal equidistant projection in distance errors" and score arbitrarily well on E, A, F, and S, "but if a boundary cut term B is included this term would blow up and save us from picking the bad subdivided map" [21]. The gnomonic cube anchors that term empirically — it is the projection whose cuts are impossible to ignore, weighted ×32 in the authors' median normalisation to reflect the census of Snyder's 105 projections [21]. In the event, the overall contest is won elsewhere — the Kavrayskiy VII posts the lowest summed squared normalised errors (4.792), with the Eckert IV the best equal-area map and the Lagrange the best conformal — but the cube's role is the one this report keeps meeting in other materials: a construction that is exact almost everywhere and wrong only where its pieces join, so that the quality of the whole is decided entirely by what happens at the boundaries between its parts.

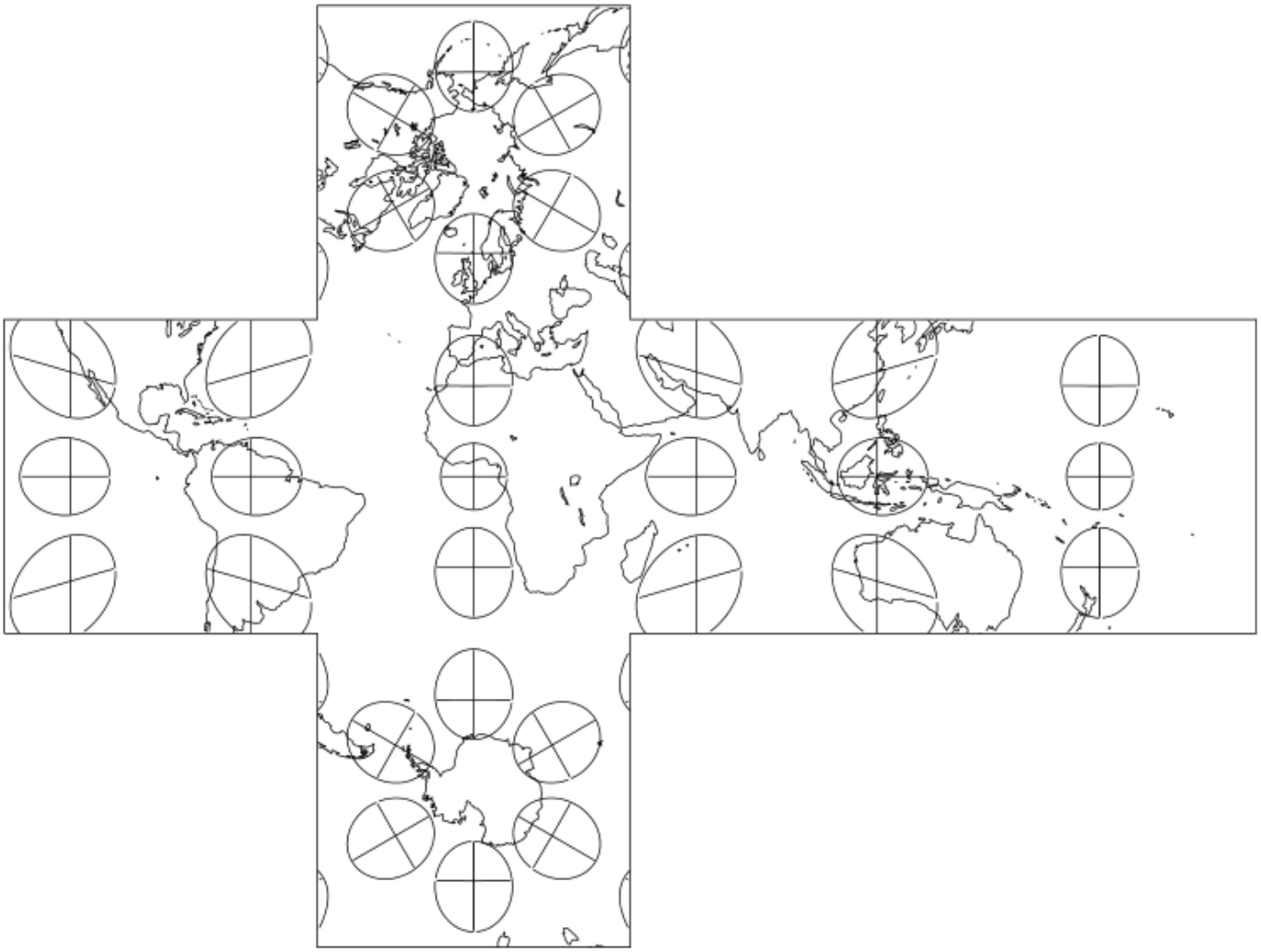


Exhibit. Tissot indicatrices on a cross-shaped world projection, showing how local distortion varies across the map. Source: S0002.

Large-scale distortions: flexion, skewness, and angular distortion

The classical instrument for map error is local. Tissot's indicatrix plants "a small circle of tiny radius (say 0.1 degree of arc in radius)" at a location and projects it: "This always produces an ellipse," specified by major axis, minor axis and orientation angle — exactly the three independent components of the metric tensor, so the ellipse "essentially carries the information on the metric tensor for the map" [21]. Its readings are diagnostic: for conformal projections like Mercator the ellipses remain circles of varying size (scale changes, shape preserved); for equal-area projections like the Mollweide they keep constant area and vary in shape (anisotropy made visible); compromise projections like the Winkel Tripel, the National Geographic's choice, minimise both error families jointly [21]. Goldberg and

Gott's foundational point is that this is not the whole of distortion: "the Tissot ellipse does not carry all of the information related to distortions," because the metric is infinitesimal and continents are not [21].

Their two additional quantities are defined by a truck. Imagine it driving a geodesic — a great circle — "at unit angular speed (say one radian per day)"; on a perfect map its image would move "in a straight line on the map with constant speed," acceleration zero. In reality the image accelerates, and the acceleration splits in two: the perpendicular component "causes the truck to turn without changing its speed on the map. This causes flexion, or bending, of geodesics," quantified as $f = d\alpha/d\tau$, the angular rate at which the velocity vector rotates per radian travelled; the parallel component speeds the image up or slows it down along its path — skewness, "lopsidedness and an asymmetric stretching of large structures" [21]. The terminology is imported from gravitational lensing, and the scale-dependence is the point: "Flexion and skewness only express themselves on large scales. They are not noticeable on infinitesimal scales where the metric contains all the information one needs, but become noticeable on finite scales, with their importance growing with the size of the scale being examined... important on continental scales and larger in a world map" [21]. A projection can therefore be locally impeccable and continentally bent — the error class that decides whether a coastline looks right.

The Hammond Atlas face experiment, which Goldberg and Gott revive, makes the taxonomy visible: draw on the globe a face with a triangular nose, a straight geodesic mouth, and eyes of concentric circles. "In the gnomonic, the mouth was straight, but the eye circles were not circular and were not concentric (what we would call skewness). The Mercator projection had the mouth smiling (what we would call flexion) and although the eyes were circular they were no longer concentric" [21]. The gnomonic is the existence proof that flexion alone can be annihilated — it "shows all great circles as straight lines" — "however it does exhibit anisotropy, scale changes, and skewness, and at best can show only one hemisphere of the globe" [21]. Hammond's own optimal conformal continent maps, built on the Chebyshev criterion of constant scale factor along each continent's boundary, scored better on both qualities precisely because small scale-factor gradients keep flexion and skewness small, "although of course not zero" [21].

A worked school-level investigation of the equidistant cylindrical projection shows the micro-mechanics feeding the macro-measures: the indicatrix is stretched horizontally by the scale factor k (a secant-of-latitude relation), so "the spheres closer to polar areas experience the most horizontal flexion and become elliptical," a point M maps to M′ with its angle θ decreased and $|OM| \neq |OM'|$ — the distance error being the difference of the two — and in Snyder's generalised formulas both k and the bent coordinate depend on "the central meridian λ0 and standard parallel φ1, which are arbitrarily selected": part of every cylindrical map's distortion is a free choice, not a fate [22].

The survey's scoring (six measures — ellipticity, area, flexion, skewness, distance, and the boundary-cut term the gnomonic cube chapter showed to be indispensable) is then normalised following Laskowski against the equirectangular, a choice with a perceptual dividend: area errors end up weighted more heavily than ellipticity errors, which is "fortunate since errors in areas appear subjectively to be more noticeable" — the Lagrange, with area error 0.73, "has fallen out of use because of its large area errors in the polar regions," while the Eckert VI's ellipticity error of the same 0.73 leaves it "popular" in Snyder's judgement; any normalised term above 1.50 means "noticeable trouble" [21]. The summed ranking puts Kavrayskiy VII first (4.792), then Gall Stereographic (5.758), Eckert IV (5.855) — the best equal-area entry — the equirectangular itself (6.0), Lambert Azimuthal two-hemisphere (6.118), Breisemeister (6.179), Miller (6.350), Mollweide (6.388), Eckert VI (6.460), Hammer (6.497) and Lagrange (6.560), the best conformal [21]. On distance errors alone, a different winner: the Gott-Mugnolo Azimuthal, "round" and cut-free [21].

The authors salt their own table: "the detailed rankings one should take with a grain of salt... as any ranking scheme is dependent on the normalizations" — compress Kavrayskiy VII horizontally by 1.25 and it scores slightly better while looking worse, Africa "a bit too thin"; renormalise against the Winkel Tripel and Breisemeister (4.360) takes the crown instead [21]. And the measures can be traded against each other fraudulently — "squashing the Mercator vertically to improve flexion at the expense of greatly increasing ellipticity errors" — which is why the sextet must be read together [21]. That is the mature form of this paper's title-claim: a flat map is wrong in at least six independently measurable ways, each gameable in isolation, and honesty consists in publishing all six. The final section takes the most consequential of them — distance — down to the level of the individual straight line.



Exhibit. A Mercator projection cutout of the continental United States, from a comparative study of projection distortion. Source: S0002.

Distortion in distance and the straight-line versus real distance problem

Of all a map's promises, the straight line is the one users trust most: rule a line between two points, measure it against the scale bar, and read off the distance. It is also the promise no flat map can keep. This closing analysis separates the two failures folded into that gesture — the shortest route on the Earth is generally not straight on the map, and the straight route on the map is generally not the length the scale bar claims — using the equidistant cylindrical projection as the worked example and Mercator as the cautionary exhibit.

The cylindrical construction shows where the error enters. Building the projection on a Cartesian grid with meridian scale factor h set to 1 ("horizontal translations will not cause distortion in distance"), the parallel scale factor k follows from the geometry of the tangent cylinder: the radius of the parallel at latitude φ is r = R cos φ against the cylinder's base radius R, giving k = sec φ — exact on the chosen standard parallel (here the equator, with the central meridian at Greenwich), and growing without bound toward the poles [22]. Every east–west distance off the standard parallel is therefore inflated by a known, latitude-dependent factor; the map's ruler is honest only along one line.

The investigation quantifies the consequence for an arbitrary pair of points. With coordinate differences λ and φ, the true separation computes as $|OM| = \sqrt{(\varphi^2 + \lambda^2 \sec^2\varphi)}$, while the straight segment measured on the projection gives $|OM'| = \sqrt{(\lambda^2 + \varphi^2)}$ — the scale factor silently dropped — so "the maximum distortion in distance is equal to $||OM| - |OM'||$," a quantity that grows with latitude and with the east–west component of the journey; the same analysis bounds the projection's maximum angular distortion at ω = 0.785 radians, attained in the limiting case, with horizontal distortion increasing "without bound at polar areas" [22]. The remedy the investigation reaches for is telling: abandon the map's own geometry and compute through the sphere — a chord-based formula, D = R·arcsin(·) applied to the straight-line chord d between the points, "is able to generate a more accurate result for the real distance between P1 and P2 than the formulas of the equidistant cylindrical projections" [22]. Its conclusion is the section's thesis in miniature: the projection "fails to demonstrate the actual route between two arbitrary points on the surface of the Earth as curves, thus distances become less accurately preserved," and "the straight-line distance which equidistant cylindrical projection illustrates differs from the real distance on the surface of the sphere… the projection must contain distortion in distance" [22]. To measure truly, you must leave the map.

Mercator exhibits the same disease in its most consequential historical form, and Goldberg and Gott's example deserves its fame. The 49th parallel — the northern border of the continental United States — is drawn straight on Mercator, "but really it is a small circle that is concave to the north": drive it and "you would have to turn the steering wheel slightly to the left" continuously, while the actual great-circle route between two points on that parallel, in Washington State and Minnesota, "is a straight line which goes entirely through Canada" and, extended, "passes

south of the northern part of Maine, so the continental United States is bend downward like a frown in the Mercator Map" [21]. Hence their operational summary: "Flexion or bending is manifest on the map as a bending of geodesics on the map, and skewness or lopsidedness is manifest on the map as the midpoint of a geodesic line segment on the map not being at the midpoint of that segment" — the continental centre in Kansas appearing in the map's lower half because the scale factor grows northward [21]. The Mercator skewness is computable in closed form: along a meridian the image accelerates as the scale factor grows, and for a north–south vector "the skewness at 45° is 1, showing a lopsidedness toward the north," positive throughout the northern hemisphere and negative in the southern, its average over all azimuths following by a cos θ integration [21]. Underneath the examples sits the general machinery: the Christoffel symbols computed from the map's metric tensor are precisely the coefficients of the geodesic equation — "Equation (28) describes the bending of straight lines on a particular [projection]" — though computing them analytically "requires that we have an analytic form for the map inversion," which is why the authors publish numerical code instead [21].

This is where the paper's two halves join. The boundary chapters showed lines drawn by law across water and land, their authority resting on charts; the phantom chapters showed features that survived because no one sailed to check; the projection chapters end by showing that even a perfectly surveyed coastline, honestly projected, yields straight lines that are not shortest paths and measured lengths that are not distances — with the errors not random but structured, latitude-dependent, and computable to the last radian by anyone who does the differential geometry. The map is wrong, finally, in the way a language is wrong: not by lying about particular facts, but because its grammar — flat, straight, uniformly scaled — cannot say certain true things about a round world. Everything before this section was an instance. This is the theorem.



Exhibit. Great-circle and rhumb-line tracks between Halifax and Lerwick plotted on a Mercator chart; only the rhumb line is straight. Source: S0012.

Plotting positions on Mercator charts: bearings, distances, and the nautical mile

Boundary work sometimes requires the oldest plotting technique of all: fixing a position by a bearing and a distance from a known charted point. The TALOS Manual permits it with an immediate caution — the method "is only accur-

ate (within the limitations of scale) over quite short distances because of the problems of distance measurement on the Mercator projection, or because of the difficulty of projecting a line of constant bearing on non-Mercator projections that depict meridians as converging lines" [3]. Each of its two ingredients carries its own error budget.

The bearing is measured "either with reference to a nearby meridian using a protractor or specially designed parallel ruler, or it may be made with reference to the compass roses, which will be found located at convenient points on the chart" — with "great care" needed that the compass circle used is "referenced to true and not magnetic north" [3]. Bearings are given in whole-circle notation, 0° to 360° clockwise from true north; the quadrantal form (N36E) is "rarely used today." And whatever the format, a bearing "taken off a chart, or determined by compass bearings... can only have been read off to within about a quarter of a degree (15′)" [3]. The distance is then "measured either with reference to the latitude scale (at the same parallel at which the measurement is being made), or to the distance scale on non-Mercator charts," and "set off using dividers" [3] — the latitude-scale rule being the practical compensation for Mercator's stretching, since the chart's own scale is not constant. The manual notes the natural scale relation "does not hold true for charts based on certain projections e.g. the Mercator projection," which is why delimitation charts are chosen at scales fit to their zone — "from 1:100,000 to 1:1,000,000" for EEZ and continental-shelf lines, "of the order of 1:50,000 to 1:100,000" within the territorial sea — and why plotting error is reckoned at "[approximately] 0.2 mm on paper, which is equivalent to the width of a drawn ink line" [3].

Two further cautions complete the method. The reference point must be beyond argument: a position is described as so many degrees and such a distance "from a specific feature... either defined by precise geographical coordinates or clearly identifiable as an unambiguous point on the chart," such as "a lighthouse or a beacon; if it cannot be identified as an unambiguous point the accuracy of the position is immediately in question. That might occur if the feature were a rounded headland" [3]. And the distances, being short, "are most likely to be given in nautical miles (probably without regard to the difference between a minute of latitude and the International Nautical Mile)" [3] — the sea's traditional unit quietly absorbing a geodetic distinction. Finally, the line drawn is a loxodrome, "a true straight line on a Mercator chart" of constant azimuth, which "projected back onto the reference ellipsoid... will generally differ from the geodesic curve," the difference being "significant, depending upon the length and direction of the line and its latitude" [3]. Bearing-and-distance plotting survives in boundary practice, then, only inside tight tolerances — short legs, unambiguous marks, quarter-degree bearings — and the manual's electronic-chart note points past it: ENCs built from digital data "permit a great increase in accuracy, with no plotting" at all [3].

Mapping Transformed: From Survey to Satellite

The USGS and the mapping of America

The United States is the rare country whose territory was rendered legible by a single national programme. The U.S. Geological Survey "produced its first topographic map in 1879, the same year it was established," having been "[i]nitially charged by Congress with the 'classification of the public lands'" — and "[m]ost of the early USGS mapping activities took place in the vast, largely uninhabited Western United States," where "[m]any locations could be reached only by mule pack train" [19] [19]. The instrument of that era was the planetable — "essentially a portable drawing board on a tripod with a sighting device" — carried to "the area's best vantage point" by topographers who plotted what they could see; "[p]lanetable surveying remained the dominant USGS mapping technique until the 1940's, when it gave way to the airplane and the age of photogrammetry" [19]. Aerial photography had been pioneered in the 1930s, when "the USGS assisted the Tennessee Valley Authority in mapping its area of responsibility" — "the first full-scale test of the use of aerial photographs in creating maps" — and was then massively accelerated by wartime intelligence needs [19].

The mature system is a lattice. The Survey's longstanding goal of "complete, large-scale topographic map coverage of the United States" produced "a series of more than 54,000 maps that cover in detail the entire area of the 48 contiguous States and Hawaii," at 1:24,000 — the 7.5-minute quadrangles, "because each map covers a foursided area of 7.5 minutes of latitude and 7.5 minutes of longitude," with the country "systematically divided into precisely measured quadrangles" whose sheets join seamlessly [19]. Alaska, too large and sparse for that mesh, is mapped at 1:63,360 — one inch to the mile — in 2,700 fifteen-minute quadrangles, and complete national coverage also exists at 1:100,000 and 1:250,000 [19]. What distinguishes the product is the contour line: topographic maps' defining feature "is the use of contour lines to portray the shape and elevation of the land," rendering "the three-dimensional ups and downs of the terrain on a two-dimensional surface," alongside named natural and manmade features from rivers and vegetation to roads, boundaries and buildings [19].

Production industrialised without abandoning the ground. A pair of aerial photographs of the same terrain from different flight positions, "viewed through an instrument called a stereoscope, produc[es] a three-dimensional view of the terrain from which a cartographer can draw a topographic map"; under the National Aerial Photography Program it takes "10 precisely positioned NAPP aerial photographs to provide the stereoscopic coverage needed for each 7.5-minute quadrangle map," flown north–south at a consistent altitude for a uniform 1:40,000, with cameras that "can cost more than $250,000" and seasonal timing — leaf-off, in hardwood country, "so that terrain features are more clearly visible" [19]. Field surveys then establish the control framework: horizontal points fixing latitude and longitude, scale and orientation, and vertical points fixing the elevations from which contours are placed — and the whole process "can take 5 years from the identification of a mapping requirement to the printing" of one quadrangle [19] [19]. Accuracy is codified rather than assumed: the National Map Accuracy Standards, issued in 1941 for all federal map producers, require that "at least 90 percent of horizontal points tested on a 7.5-minute, 1:24,000-scale map must be accurate to within one-fiftieth of an inch on the map (40 feet on the ground)," and 90 percent of tested elevations "to within one-half the map's contour interval" — five feet on a ten-foot-interval sheet [19]. The booklet closes where this report's earlier sections began: with a "mapping revolution" in which digital data for computerised mapping and analysis becomes a new cartographic product in its own right — the quadrangle's successor being not a better sheet but a database [19].

What is a topographic map? The changing landscape of topographic mapping

A topographic map is defined by a single device: "The feature that most distinguishes topographic maps from maps of other types is the use of contour lines to portray the shape and elevation of the land. Topographic maps render the three-dimensional ups and downs of the terrain on a twodimensional surface" [19]. Around that device it assembles the full inventory of a landscape: "both natural and manmade features. They show and name works of nature including mountains, valleys, plains, lakes, rivers, and vegetation. They also identify the principal works of man, such as roads, boundaries, transmission lines, and major buildings" [19]. The USGS booklet that remains the standard popular account calls the map, on paper or screen, "the best tool available to catalog and view the arrangement of things on the Earth's surface," and lists the constituency the definition serves: "engineering, energy exploration, natural resource conservation, environmental management, public works design, commercial and residential planning, and outdoor activities like hiking, camping, and fishing" [19]. A topographic map is thus not a picture of terrain but a general-purpose spatial ledger — elevation, hydrography, culture and boundaries in one registered frame.

Two structural facts govern everything such a ledger can say, and the TALOS manual states both. The first is generalisation: because "map scale is not 1:1… any depiction of features is denser than the real world," compilation requires what the IHO's S-32 defines as "the omission of less important detail when compiling a chart. Its purpose is to avoid overloading charts where space is limited" [3]. The floor on detail is physiological and mechanical: eye acuity is "approximately 0.02mm," printed lines bottom out near "approximately 0.1mm at the minimum," and so "there is a minimum size in order to illustrate a feature on a paper map," a minimum that "is governed by the scale of map; the smaller the scale the larger the minimum size" — features are routinely drawn larger than truth to remain legible, nautical charts more so because they are read "in a challenging environment" [3]. Every topographic map is therefore a curated selection wearing the costume of completeness; the selection rules, not the terrain, decide what a small feature's fate is.

The second fact is reliability, and it is the phantom-island problem in professional dress. Charts are "compiled from the results of hydrographic surveys that are very time consuming operations," undertaken by "only a limited number of maritime States," so "large areas of the world's ocean floor have never been properly surveyed, and surprisingly significant areas of coastal waters have not been surveyed in the detail required for today's shipping or to support the determination of maritime limits and boundaries" [3]. The consequences are itemised: "Geographical positions may be based upon inaccurate, imperfect or inadequate observations"; soft low-water lines of "mud or sand… are most likely to have changed since the surveys were undertaken," especially under strong currents or storm coasts; charts printed from original plates may leave it unclear "which contour line or lines represent the low-water line"; and "the publication date of a chart is not an indication of the age of the source material from which it is compiled" — an old chart may carry modern work, a new one may reprint antique surveys [3]. The map's authority is thus stratified by vintage in ways its face does not disclose — the exact epistemic condition that kept Sandy Island alive for 136 years.

It is into this mature, standardised, quietly uncertain artefact that the digital transition arrived. The USGS booklet, written at the transition's height, announces it without hedging: "A mapping revolution is underway. New technolo-

gies are altering the production and use of traditional maps. Even more significantly, the information age has introduced a new cartographic product that is changing the face of mapping: digital data for computerized mapping and analysis" [19]. The shift is ontological: the product stops being the sheet and becomes the database from which sheets are rendered — "the computer is extending mapping beyond its traditional boundaries," with applications running from home display to analyses that "stretch the definition of cartography" [19]. The Survey positioned itself as the standards-setter for that transition, "the principal agency developing standards and coordinating other matters related to Federal digital cartographic data" [19]. What a topographic map is therefore changed under its users' feet: from a printed, generalised, survey-vintaged ledger of a landscape to a queryable model of one — with generalisation and reliability, the two old governors, reappearing in digital dress as resolution and lineage. The sections that follow trace that transformation through flight, photogrammetry and the database itself.


8086
×8353
8200
Cave
Harvey Meadow
TRAIL
PACK
8300
Sewage Disposal
WT
Ranger Sta
FOOT
TRAIL
8000
Bright Angel Spring
7035
8135
Angel
8339
Heliport
North Rim
Roaring
Tunnel
7200
Spring
7000
7800
8200
Campground
Springs
6400
Footbrid
Ca
Cliff Dweller Spring
FOOT
NORTH
×6897
G R A N D
C A N Y O
6000
TRAIL
×8226
7200
6200
7000
6400
8000
8086
8161
Bright Ange Point
The
5000
6000
N A T I O N A L
P A
×8020
5000
6800
7600
8066×
Oza Butte
4800
6800
Transept
Widforss Point
Upper Upper Ribbon Falls

Exhibit. Detail of a USGS topographic quadrangle: the North Rim of the Grand Canyon, with contour lines, springs, trails, and survey markers. Source: S0023.

Mapping in the age of flight

Flight changed not just the speed of mapping but its epistemology: the topographer stopped drawing what he could see from a summit and started measuring what a camera recorded from altitude. The discipline has a name and a definition — "Photogrammetry is the science of obtaining reliable information by measuring and interpreting photographs" — and its arrival "led to a revolution in mapmaking," significantly increasing "map coverage and enhanced map standardization" alike [19]. The stereoscopic pair is the core instrument: two photographs of the same ground from different positions along the flight line, fused by the viewer's two eyes into a measurable three-dimensional model — the Villanueva, New Mexico pair of 1984, shot at 1:24,000, is the booklet's worked example [19].

Compilation mechanised the fusion. "Upon completion of the field survey, the map manuscript is compiled using stereoscopic plotting instruments. Overlapping aerial photographs are placed in a special projector connected to a separate tracing table. The projected photographs are viewed through an optical system that causes the left eye to see one photograph and the right eye to see another. The result is a three-dimensional impression of the terrain" — and as the operator moves a reference mark through that model, "map features and contour lines are traced as they appear in the stereomodel," the tracing transmitted to the table as the manuscript [19]. The contour line, the topographic map's defining device, is thus no longer sketched from a vantage point but extracted from a photographic solid.

Yet the age of flight did not abolish the ground; it reassigned it. "Technology has reduced the requirement for mapping work in the field... Nevertheless, the field survey still plays an important role in making and revising topographic maps," because the photograph is ambiguous in ways only presence resolves [19]. Field personnel work the photographs themselves, marking and verifying: "a worker in the field can indicate the difference between a perennial stream and one that dries up at certain times of the year" — a distinction with graphic consequences, the perennial stream drawn solid, the intermittent one dash-dot or lighter — and "people who know the local area well, such as fishermen or farmers, are excellent sources of such information" [19]. The same ground pass verifies names and boundaries, "looking at courthouse records and talking to local residents. It can even include a visit to the local cemetery to check the spelling of a feature that has been named after a person buried there" [19]. Successive editions of the Key Largo, Florida sheet — its content changing materially between printings — are the booklet's standing exhibit for why revision cycles exist at all [19].

Most remarkably, the printed map encoded its own confidence. The colour separations carried an honesty layer: "The green layer shows areas of woodland, and the brown layer shows topographic features, including contour lines. The purple layer shows features that are added from aerial photographs and other sources, but are not field checked" [19]. Purple was the ED mark — existence doubtful — of the topographic series: a visible, printed distinction between what had been verified on the ground and what had only been seen from the air. The phantom-island chapters of this paper are a decades-long lesson in what happens when that distinction is lost in a database migration; the flight-era USGS had institutionalised it in ink.

The products diversified with the viewpoint. Detail remained proportionate to scale — at 1:24,000, developed areas show "features like schools, churches, cemeteries, campgrounds, ski lifts, and even fence lines," generalised or omitted at smaller scales — and the catalogue widened beyond the topographic sheet to "topographic-bathymetric maps, photoimage maps, satellite image maps, geologic maps, land use and land cover maps, and hydrologic maps," all public, uncopyrighted, sold at the cost of reproduction and distribution [19]. Flight, in sum, industrialised the survey without severing it from verification: photographs proposed, field parties disposed, and the printing plate confessed which was which. The digital revolution that follows kept the workflow and — as the Sandy Island obituary that opens this paper attests — mislaid the confession.

The digital mapping revolution and the National Digital Cartographic Data Base

The USGS booklet ends its survey of the paper programme by announcing its dissolution. "A mapping revolution is underway," and its decisive product is not a better sheet but "a new cartographic product that is changing the face of mapping: digital data for computerized mapping and analysis" — data that lets a map be displayed on a home computer at the basic end and, at the advanced end, stretches "the definition of cartography" itself [19]. The institutional vehicle is the National Digital Cartographic Data Base: the Survey, "the principal agency developing standards

and coordinating other matters related to Federal digital cartographic data," established the NDCDB "to distribute digital data that meet these standards for use in map production and in automated systems," its contents a framework "of reference for other data about the Earth and its resources" [19]. The database has two primitive types: digital line graphs, "the digital representation of information typically found on a topographic map (point locations, lines, and area outlines)," and digital elevation models, "matrices of elevations for ground points spaced at regular distances" [19].

Getting the paper archive into that form was itself an industrial project. "Most of the USGS's digital map data are collected from existing topographic maps. The task is monumental" — digitising "resembles the original map scribing process in that it requires that each feature on each map separate be located, classified, and traced," across "10 or more different layers — roads, contours, boundaries, surface cover, and manmade features" — by hand with a cursor or automatically with scanners, followed by attribute coding, edge-matching against adjoining files, and accuracy control [19]. The payoff compounds across government: "Nationwide DLG coverage is complete for transportation and hydrographic features found on 1:100,000-scale maps," and "[t]he 1:100,000-scale data served as the base for the Bureau of the Census Topologically Integrated Geographically Encoded Reference files — the digital representation of the Nation used in the 1990 census" [19]. Production economics invert as well: computer-assisted compilation promises to "shorten the 4 to 5 years it takes to produce a map by traditional methods," while demand shifts decisively — "[g]overnment agencies and private businesses now require digital mapping information for their computer-based systems" [19].

The consumer of the database is the geographic information system, which "make[s] it possible to combine layers of digital data from different sources and to manipulate and analyze how the different layers relate to each other" [19]. The booklet's examples already span the modern GIS repertoire: the Department of Agriculture combining DLG data with scanned photographs for soil conservation; police linking record systems to geography to analyse crime patterns; emergency planners merging road types, fire-station locations and fault lines to calculate response times "both under normal conditions and following transportation blockages caused by an earthquake"; and marketers merging sales data with map data [19]. Its closing predictions — personal maps on home computers, dashboard mapping to avoid traffic, news chosen "by touching a map on the screen" — read as a fair sketch of the following three decades [19]. For this report's argument the transition matters beyond convenience: the paper map's errors were frozen at print and discoverable only by re-survey, whereas a database can be corrected continuously — and can also propagate a phantom, as Sandy Island's digital afterlife showed, at the speed of copying. The revolution changed not only how maps are made, but how their mistakes live and die.

Electronic navigational charts and working in GIS

The electronic chart splits into two kinds that share a screen and nothing else. A Raster Navigational Chart is "a digital database of a scanned and passive image of a paper chart," issued under IHO specification S-61; displayed on ECDIS it appears "as a facsimile of the paper chart," with metadata for geo-referencing, automatic updating and day/night display — "[a]s such the chart content cannot be analysed by a computer program to trigger alarms and warnings automatically as is the case with a vector chart" [3]. An Electronic Navigational Chart is the opposite construction: "a database of individual geo-referenced objects extracted from a Hydrographic Office's records," built to the S-57 transfer standard, displayable "as a seamless chart-like image on the ECDIS screen at user selected scales," where the system "continuously monitors the ENC data content (rather than the display) to provide warnings of impending dangers in relation to a vessel's position and its movement" [3]. The raster chart is a picture of a map; the ENC is the map as data — points, lines and areas held as numerical expressions that software can query, alarm on, and recompute [3].

The law of carriage tracks exactly that distinction. SOLAS Chapter V defines the nautical chart to include "a specially compiled database from which such a map or book is derived," accepts a type-approved ECDIS as meeting the carriage requirement, and demands that charts and publications be "adequate and up to date"; the requirement is fulfilled by official up-to-date paper charts, or by ECDIS with an appropriate backup and up-to-date ENCs — while "ECDIS with non-ENC database or ENCs displayed on non-ECDIS platform do not satisfy the carriage requirement," and uncertified systems are merely Electronic Chart Systems whose private databases "have no official status" [3] [3]. For boundary work the same database logic is what the Convention itself anticipates: baselines and limits must be shown "on charts of a scale or scales adequate for ascertaining their position," or replaced by "a list of geographical co-ordinates of points, specifying the geodetic datum," with due publicity and deposit with the UN Secretary-General under Articles 16, 47, 75 and 84 [3].

Working in GIS inherits both the power and the paper chart's old cautions. The TALOS Manual insists the four fundamentals — "[c]hart projection; [c]hart scale; [h]orizontal datum; [v]ertical datum" — retain "such an important influence in the depiction of maritime boundaries" that mixing charts "with dissimilar characteristics and datums" is the standing hazard [3] [3]. And the database is only as current as its surveys: hydrographic surveys are "time consuming and done area by area," so "[h]ydrographic data for some areas may be dated and may not meet modern standards," and "information shown on a chart can be different from the reality" [3]. What the ENC changes for delimitation is nonetheless fundamental — where ENCs are built from digital data rather than digitised from paper, the manual notes elsewhere, they "permit a great increase in accuracy, with no plotting," and dedicated software can compute limits directly from the data [3]. The chart has become a database with a picture attached; boundary work now happens in the database, and its errors are the database's errors — datum mismatches and stale surveys — rather than the draughtsman's.

Satellite services and the world height datum: Sentinel-2 requirements and global gravity models

Every previous chapter assumed a floor: coordinates presuppose a datum, heights presuppose a zero. This final section examines the floor itself — and finds it, like everything else in this paper, plural, conventional, and under repair.

The zero of height is the geoid: "that special equipotential or geopotential surface which coincides with, but is not exactly equivalent to, the mean sea level (MSL) surface of the oceans in an average sense" — the surface the oceans would take under gravity and rotation alone, conceptually extended under the continents, differing "from a best fitting ellipsoid by vertical distances that are up to one hundred metres or so" [3]. It is "the reference surface for heights used in mapping," its heights "commonly known as heights above mean sea level" — but its practical realisation, accepting mean sea level at tide gauges, "carries with it some inherent errors due to the existence of sea surface topography (SST) that may reach well over one metre, and which varies with location and time" [3]. Each nation's vertical datum is thus a local treaty with the ocean, struck at its own gauges.

The unification project follows directly. "The proliferation of vertical datums around the world and the difficulty in transferring heights between these datums has led to renewed efforts by the IAG towards the unification of these datums and the creation of a World Height System (WHS)" — led first by Inter-Commission Project 1.2, "Vertical Reference Frames," and continued as Theme 1, "Unified Global Height System," of the Global Geodetic Observing System [3]. The unresolved problems are the paper's themes in geodetic dress: realising a single reference surface for physical heights; relating tide-gauge records to it; separating "sea level changes and vertical crustal movements at tide gauges" — the datum itself is moving, the same instability the sea-level-rise chapter met in legal form; and connecting the height system to the terrestrial horizontal frame [3]. The recommended construction is satellite-first: heights and the datum surface should come from "a global gravity field model that is independent of inconsistencies in local and regional data… one of the satellite-only global geopotential models (GGMs) such as those produced from a combination of GRACE and GOCE satellite data only," augmented with local gravity data to reach centimetre-level geoid accuracy [3]. The system's anatomy is dual — a geometrical component from the ITRS and reference ellipsoid (ellipsoidal heights h and their rates dh/dt) and a physical component from a conventional geoid potential W0 (geopotential numbers C and their rates dC/dt) — realised by a station network as an International Vertical Reference Frame, its zero surface defining the World Height Datum, against which every local datum's offset becomes a measurable potential difference [3]. Adoption has begun: "countries such as New Zealand have already adopted geoid-based vertical datums. Canada and the USA have decided to do the same in 2013 and 2022, respectively," with GOCE-based models providing "1-2 cm geoid accuracy with 100 km spatial resolution" [3].

Underneath the datum sits the service infrastructure of modern geodesy. The International Association of Geodesy "has established services for all the major satellite geodesy techniques": the GNSS service (IGS), laser ranging (ILRS), very-long-baseline interferometry (IVS), DORIS (IDS), the Earth rotation service (IERS) and the gravity field service (IGFS), generating "precise satellite orbits, ground station coordinates, Earth rotation and orientation values, gravity field quantities and atmospheric parameters" — the enhanced capabilities of space geodesy having made frame definition and maintenance "the primary mission of geodesy," with the law-of-the-sea contribution running through "the definition of national and international datums and the geoid, and to GNSS-based precise positioning" [1].

The Sentinel-2 mission requirements show what this infrastructure is operated for. The GMES services specified in 2007 include GSE RESPOND, applying geographic information across "the entire humanitarian intervention cycle"

for users from the European Commission's humanitarian directorates to UN OCHA, UNHCR and peacekeeping operations and the Red Cross — eight services from basic and crisis/damage mapping to a dedicated "Refugee and Internally Displaced People (IDP) Support Mapping Service" detailing camps, water sources and health resources, with background mapping from archive "within 6 hours (typically 2 hours)" and crisis mapping within 24 hours on a daily cycle, at resolutions of 1–2.5 metres for camp and basic mapping and 2.5–10 metres for crisis work [23]. The land services run in parallel — Urban Atlas sprawl monitoring, inland water quality, impervious-area and irrigation mapping — built on classification of SPOT and Landsat-class imagery, very-high-resolution sensors reserved for small features [23].

So the paper ends beneath the map: heights referred to a conventional potential surface realised by satellite gravity missions and a global station network; positions referred to frames maintained by six international services; imagery delivered to responders within hours at metre resolution. None of it abolishes the argument of the preceding seventy-three sections — it relocates it. The datum is a convention that must be chosen and maintained, the geoid a model that must be computed, the frame a service that must be funded; where any of them lapses, the old failure modes — inherited error, silent propagation, the confident depiction of what was never checked — re-enter through the new machinery. The map is wrong, in the end, whenever its keepers stop paying the maintenance on its truths.

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